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Exploring the Barriers and Facilitators to Active School Transport Among School-Going Children (Aged 5-19) Globally: A Scoping Review

Dissertation Essay

Master of Public Health

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Date of Submission: 18/08/2025

Word Count:
Reflection Statement – 485
Abstract – 490
Main Text – 10,055

Acknowledgments

I would like to extend my gratitude to my supervisors Prasanthi Attwood (PA) and Aldina Mesic (AM) for their guidance and counsel throughout the duration of this research project. I would also like to thank Alejandra Piraguata (AP), Luisa Bartsch (LB), and all other members of The George Institute for Global Health UK for their wisdom, assistance, and support throughout my time with them. Last but certainly not least, I would like to express my gratitude to my personal tutor Sami Altalib (SA), my friends, and especially my family, for all the academic and emotional support they have given me.

Over the course of the research period, I attended seven virtual meetings with my supervisors to discuss both progress and address pertinent questions. Furthermore, this report was formally reviewed on three separate occasions. I hereby acknowledge that the following procedures relevant to this work were contributed (to) by third parties:

- Refinement of the search strategy (PA and AM).
- Determination of the inclusion/exclusion criteria (PA).
- Screening of search result titles and abstracts (LB).
- And resolution of screening conflicts (AP).

All other aspects of this work were conducted and completed solely by myself under authorisation of the Imperial College London School of Public Health towards the completion of the programme code: *PUBH70045 – Research Project 2024-2025*.

Signed,



Reflection Statement

I recall early in the first term when we were first introduced to the hierarchy of evidence in public health. I had been interested science for much of my life, and yet, I had never heard of it. In fact, before my time at Imperial, I was unfamiliar with the very idea of 'determinants of health'. Around the same time, one lecturer contrasted the heroics of clinical practice with public health's 'after-hours', comparing medics to Superman, who saves lives by day, and public health practitioners to Batman, who protects the people by night. It was an amusing remark, but more than that, it was meaningful. By that time, we had only briefly touched on systematic reviews and behavioural research, but I was already determined to undertake a literature review for my research project; an opportunity that would allow me to put into practice the aspects of public health I had come to enjoy the most.

I have since spent a lot of time developing specific methodological expertise. I learned of the purpose of scoping reviews and sought to build the skills they require: comprehensive planning and protocol development; critical reflection and adaptation; consistent charting and structured reporting; multi-levelled synthesis; and technical proficiency in various research software. My steepest challenge was indeed the sheer volume of papers, many among which were characterised by definitional inconsistencies that convoluted screening, coding, and data charting. Progress was slow at fist; but with supervisory guidance and continued persistence, I devised a workflow that allowed me to maintain a meticulous documentation of my decisions, time-box screening, and regularly revisit my inclusion/criteria logic.

The search process also brought back into view several in-lecture discussions about who and what tends to be represented in research. Much of the available evidence in high-impact journals and major databases was written both by and for the Global North, effectively sidelining the Global South. I became more attentive to the politics of evidence, the risks of epistemic injustice, and the ways in which author characteristics and institutional status can shape global agendas and determine dominant frameworks. Though I could not bypass these structural imbalances, I sought to mitigate against them by broadening my search terms and acknowledging downstream effects of publication and language biases, as well as the need for decolonisation in health research.

My choices throughout the project were informed by a variety of domains of knowledge, including behavioural science, public health policy, and epidemiology. This interdisciplinary approach required me to evaluate my methods, identify what was effective, and identify how to refine my work. From an academic point of view, this work offered a comprehensive overview of the barriers and facilitators to active school transport, highlighted key priorities for further research, and served as a foundation

for forthcoming UNICEF guidelines; but for me, it has equipped me with practical skills and valuable research experience that I will carry forward into my future work.

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List of Abbreviations

AST	Active School Transport
SRTS	Safe Routes to School
MVPA	Moderate to Vigorous Physical Activity
BMI	Body Mass Index
EYHS	European Youth Heart Study
US	United States
UK	United Kingdom
HICs	High-Income Countries
MICs	Middle-Income Countries
UMICs	Upper-Middle-Income Countries
LMICs	Lower-Middle-Income Countries
LICs	Low-Income Countries
SES	Socio-Economic Status
SDGs	Sustainable Development Goals
GAPPA	Global Action Plan on Physical Activity
WHO	World Health Organisation
AFRO	WHO African Region
AMRO	WHO Region of the Americas
EMRO	WHO Eastern Mediterranean Region
EURO	WHO European Region
SEARO	WHO South-East Asian Region
WPRO	WHO Western Pacific Region
UNICEF	United Nations Children's Emergency Fund
SEM	Socio-Ecological Model
COM-B	Capability, Opportunity, Motivation Behavioural Framework
TDF	Theoretical Domains Framework
BCW	Behaviour Change Wheel Framework
APEASE	Acceptability, Practicability, Effectiveness, Affordability, Side-effects, and Equity Criteria
PICO	Population, Intervention, Comparison, Outcome Framework
PCC	Population, Context, Concept Framework
OSF	Open Science Framework
PRISMA	Preferred Reporting Items for Systematic revies and Meta-Analyses
ScR	Scoping Review
MeSHs	Medical Subject Headings
WoS	Web of Science
TRID	Transport Research International Documentation
GIS	Geographic Information System
ORs	Odds Ratios
PRs	Prevalence Ratios
SDs	Standard Deviations

Abstract

Background: Active school transport (AST) – defined as non-motorised means of travel for the journey to and/or from school – offers significant health, social, and environmental benefits for children around the globe. Despite a large and growing evidence base, research investigating the global decline of AST remains siloed, with limited research addressing influences across socio-ecological levels. For the operational needs of the United Nations Children's Emergency Fund (UNICEF) and other global actors, consolidated evidence is urgently needed in order to develop intervention strategies that are both effective and theoretically sound.

Aims and Objectives: This review aimed to map the global evidence on the documented barriers and facilitators to AST experienced by children aged 5-19, to identify gaps in the current literature, and to identify policy levers through which to implement behavioural change. Specifically, it sought to: (1) systematically search the available evidence; (2) provide a comprehensive, nuanced, and disaggregated map of behavioural influences across socio-ecological model (SEM); (3) compare and describe determinant patterns across US-equivalent school-levels and World Bank income-strata; (4) identify all available implementation functions and policy categories to inform intervention design; and (5) identify gaps in evidence and recommend areas of priority for further research and systematic revies.

Methods: Descriptive and exploratory in nature, a scoping review design was selected for its capacity to provide broad, comprehensive maps of the evidence, clarify concepts, and catalogue study designs all while identifying inadequacies in the evidence base. Six electronic databases and grey literature sources were systematically searched for papers published between 2010 and 2025. A convergent-integrated mixed-method approach was used to synthesise results, and the SEM, Capability, Opportunity, Motivation (COM-B) behavioural framework, Behaviour Change Wheel (BCW), Theoretical Domains Framework (TDF), and Behaviour Change Wheel (BCW) were sequentially employed to achieve objectives 2-5.

Results: Of the 121 papers included in review, most were of a quantitative, analytic cross-sectional design. The majority of included papers also studied populations from high-income (HICs) and upper-middle (UMICs), with few studying populations from lower-middle (LMICs) countries. Notably, no papers that studies LICs were included in review. The most frequently reported behavioural influences were at the community-level (i.e. long home-school distances, traffic risks, land-use mix, and quality of pedestrian infrastructure), followed by individual-level factors (such as reflective and automatic motivation, perceptions of safety, capability, and socio-economic status (SES)). Institutional and policy-level influences were less frequently reported, but carried notable potential for intervention design. The research gaps that were identified included under-representation of the Global South,

comparative neglect of facilitators when compared to barriers, and limited attention to intersectional and potentially confounding or effect-modifying variables such as sex.

Conclusions: Effective interventions will require integrated implementations that combine environmental restructuring, enablement, education, and incentivisation with supportive policies and service provision. Findings highlight the importance of framing AST as not only a health issue, but an equity also. For UNICEF, this report offers a critical evidence base and practical roadmap to inform well-designed guidelines that would serve to promote AST worldwide.

Section 1: Introduction

1.1 Background

1.1.1 Defining Active School Transport

Active school transport (AST) – defined as walking, cycling, kick scooting, or otherwise using non-motorised modes of travel for the journey to and/or from school – rests at the nexus of child health, urban planning, and climate action (1). It is often the first independent journey that adolescents undertake, serving as a tangible indicator of whether the environments they inhabit support safe, healthy, and equitable mobility (1). Despite its importance, however, there is growing concern that opportunities for active mobility among school-aged children are diminishing, contributing to global trends in physical inactivity and environmental degradation (2).

1.1.2 Global Trends of Active School Transport

Worldwide, physical inactivity among the youth constitutes a serious public health issue (2). According to the World Health Organisation (WHO), more than 80% of adolescents fail to meet the recommended 60 minutes of moderate to vigorous physical activity (MVPA) per day, with little evidence of improvement over the past two decades (3). AST once supplied a dependable share of this activity, serving both as a routine and an incidental source of exercise for children that allowed for common obstacles such as time-constraints to be overcome (1).

Yet, comparative surveillance studies, such as those synthesised in the *Active Healthy Kids Global Alliance Global Matrices 1.0 to 4.0* (2014-2022), report an overall picture of decline in the proportion of school commutes made by active means (4). Behavioural aspects of children's physical activity, such as active transport, were persistently graded at "C-" or lower (and often "D" or "F") in majority of the participating countries spanning six continents, reflecting a consistent pattern of insufficient AST worldwide (5). While these findings reveal a pervasive problem, the evidence also indicates that the decline is unequally distributed, with variations by country, region, and likely also by urban-rural and socio-economic context (5).

In the United States (US), for example, the proportion of children aged 5-18 walking or cycling to school decreased from 40.7% (95% confidence interval (CI): 37.9-43.5) in 1969 to 12.9% (CI: 11.8-13.9) in 2001 – a trend mirrored, albeit less extreme, in other high-income countries (HICs) such as Australia, Canada, England, and Spain in recent decades (6–9). In both middle (MICs) and low-income countries (LICs), rates of AST generally remain higher than in HICs, though there is some evidence of decline in countries experiencing rapid urbanisation and motorisation (10). Withal, most MICs and LICs still lack robust longitudinal data on these trends (10). Overall, these patterns point to a global retreat from everyday physical activity among the youth, driven by a combination of urban

sprawl, traffic-related dangers, and socio-cultural shifts in perceptions of safety and convenience (1,4).

1.1.3 Co-Benefits of Active School Transport

This decline of AST represents both a critical and significant opportunity-cost to the promotion of a wide spectrum of physical (11–14), mental health (9,15), and cognitive benefits (16,17). A robust and growing body of evidence consistently supports the positive effects of routine active commuting on musculoskeletal development, cardiovascular fitness, and weight regulation in children and adolescents, thereby preventing obesity and reducing long-term risk of metabolic disorders such as diabetes (14). In one notable example, the European Youth Heart Study (EYHS) by Ekelund *et al.* (2007) report that children who cycled to school are five times as likely to qualify in the top quartile of cardiovascular fitness compared to those who walked or used passive means (11). In another study, Mendoza *et al.* (2011) report that active commuting is associated with lower Body Mass Index (BMI) Z-scores and body fat percentages, though effect sizes were modest (12). Beyond physical health, AST has been associated with improved mental well-being through reduced stress, enhanced mood, and increased self-esteem (9), alongside greater independence and social interaction (15). Cognitive performance and academic engagement are also positively associated (16), potentially mediated by improved sleep quality (17) and circadian rhythm regulation (13). Importantly, these benefits accrue across socio-economic status (SES) groups (18); and as such, may help mitigate against disparities in access to structured physical activities.

On a broader scale, AST also contributes to a host of societal co-benefits (1,19). Transport produces one-fifth of global energy-related CO_2 emissions – three-quarters of which originate from road vehicles – and motorised school-runs compound congestion and school-local air-quality burdens during peak hours (20,21). Conversely, safe AST practices enhance road safety (22), which is critical given that road-traffic injuries remain the leading cause of death among children and young adults aged 5-29 worldwide (19). The dual health-and-climate dividend of AST position it as a key lever for advancing several Sustainable Development Goals (SDGs), including SDG 3.4 *(reducing premature mortality from non-communicable diseases)*, SDG 3.6 *(halving global deaths and injuries from road-traffic accidents)*, SDG 11.2 *(providing access to safe, affordable, accessible and sustainable transport systems for all)*, and SDG 11.6 *(reducing the adverse environmental impact of cities)* (23,24).

Moreover, AST supports the WHO's *Global Action Plan on Physical Activity 2018-2030* (GAPPA) targets, which call for a 15% relative reduction in physical inactivity by 2030 (25). Likewise, the United Nations Children's Emergency Fund (UNICEF)'s *Guidance for Safe and Healthy Journeys to School During COVID-19 and Beyond* emphasises active mobility for building child-friendly and

resilient environments, urging governments to "build back better" post-pandemic by expanding opportunities for walking and cycling around schools (26).

1.1.4 Equity Disparities in Active School Transport Research

The urgency of promoting AST is further highlighted by several equity considerations (18,27–29). Evidence suggests that females are typically less likely than males to cycle or walk independently, citing both perceived and objective safety risks that are reinforced both by parental safety concerns and social norms that restrict daughters' independent mobility (27). Children in MICs and LICs or in rural and peri-urban contexts often face longer home-school distances, hazardous road environments, and inadequacies in pedestrian and institutional infrastructure – yet these settings remain largely under-represented in existing literature (28). Children with physical or social disabilities face additional challenges, but their experiences, too, are seldom documented (29). Addressing these disparities is essential, especially in order to align with UNICEF's mandate to prioritise the most marginalised children (30).

1.1.5 Barriers and Facilitators to Active School Transport

To deconstruct the interconnected constellation of factors which influence AST-behaviours, this scoping review adopts several complementary conceptual models. The Socio-Ecological Model (SEM) (Appendix A) disaggregates individual (known as intrapersonal), interpersonal, institutional (and organisational), community-environmental, and public policy levels of influence (31), while the Capability, Opportunity, Motivation (COM-B) behavioural framework (Appendix B) enables a unique approach to understanding determinant influences in a manner that translates into behaviour change (32). In simpler terms, the SEM helps *identify* levels of behavioural-influence, while the COM-B supports a granular understanding of *mechanisms* for change. Mapping the documented barriers and facilitators experienced by children onto these frameworks would enable stakeholders with interest and/or power to identify modifiable levers – ranging parental perceptions of safety to standards of street-design – and to gauge which among them are pertinent to different socio-economic settings.

1.2 Literature Review

1.2.1 Conceptual Fragmentation and Methodological Heterogeneity

The global evidence base on behavioural influences to AST has grown significantly in recent decades. Yet, the available literature is characterised by conceptual fragmentation and methodological heterogeneity. Definitions of AST differ, with some authors classifying multi-modal trips that combine walking with public transport as "active", while others restrict use of the term to journeys made exclusively by foot or bicycle (33,34). Similar inconsistencies are also evident among age-range classifications (34). As one illustrative example, Huang *et al.* (2021) refer to sample participants

aged 11, 13, and 15 as "children" (35), while Maulida *et al.* (2022) describe a comparable cohort (including students aged up to 17) as "adolescents" (36). Despite such overlaps in age-group definitions, a lack of standardised terminology persists, with many studies excluding children nearing the upper age boundary defined by UNICEF (which is 19 years). This lack of both definitional and connotational consensus has not only complicated cross-study comparisons, but also potentially discouraged efforts to map global trends in influencing factors associated with AST across the full spectrum of school-aged children.

1.2.2 Typologies of Existing Evidence

Thus far, at least four broad classes of evidence that examine AST have emerged: population-based observational studies, qualitative investigations into students' and parents' perspectives, intervention trials, and research investigating built-environments (9,37,38). Though their intensifying proliferation demonstrates a continued rise of academic interest, the work remains siloed, with findings rarely integrated across paradigms – thereby limiting both a comprehensive understanding of AST and a transparent identification of its underlying determinants (39). Without deliberate syntheses, the landscape risks repeating narrow and repeated questions rather than building coherent and nuanced evidence maps that can be used to identify actionable behavioural interventions and policy levers in both local and global contexts.

1.2.3 Structural Deficits Across Typologies

Furthermore, recent reviews across the four classes reveal three overarching structural deficits:

(i) First, diverse geographies and income-strata appear to be concerningly under-represented. A 2024 international systematic review by Savolainen *et al.* (2024) reports that over two-thirds of published studies focus on HICs in North America, Europe, and Oceania, leaving the experiences of children in MICs and LICs largely under-researched and under-documented (40). One direct implication of this lack of contextual diversity is that global guidance – which is often determined by the Global North – risks exporting culturally inappropriate, ineffective, and/or potentially inequitable interventions into the Global South. In light of UNICEF's mandate, however, developing robust and effective global guidance will require a deliberate effort to incorporate local contexts, values, and perspectives (41).

(ii) Second, intersectional factors such as sex, disability, and cultural norms are only sporadically addressed in the field, despite clear evidence that they influence children's travel-independence and modes of transport (1,29,42–44). The comparatively few studies that examine sex as either a potential confounder or effect-modifier are typically cross-sectional or longitudinal, intervention-focused, or concerned mostly with the built environment, where researchers seek

primarily to understand how both groups differ in their travel behaviours, and what barriers and/or facilitators each face. Conversely, studies that overlook sex as a variable commonly prioritise other determinants (i.e. environmental or policy factors) or lack a sufficiently sized or diverse sample to support a meaningful analysis.

Focus on disability is far less apparent in the evidence base (29), and when present, it is often within the context of broader mobility or inclusion studies rather than AST-specific research (43). Such studies commonly stress the unique barriers faced by children living with disabilities without consideration of (or comparison to) those without. The exclusion of disability as a factor influencing children's AST-behaviours is likely due to challenges in recruiting adequately representative samples, methodological complexities, and/or a general lack of available data.

Meanwhile, cultural norms are most frequently addressed in trans-national or cross-cultural studies, qualitative research, and intervention studies, where researchers consider understanding how parental and social influences shape travel behaviours as pertinent to contextually effective guideline development or policy design (1,44). Studies are more likely to exclude cultural factors when examining culturally homogenous samples, or when measuring them would likely prove methodologically difficult. Ultimately, the inclusion or exclusion of any such intersectional factors is informed both by the indented purpose of each research endeavour, and the practical considerations of the study's design.

(iii) Third, few studies examine intrapersonal, interpersonal, environmental, and policy-level domains simultaneously, compromising the knowledge base required to design both context-sensitive and multi-level interventions (1,45). While there have been attempts to apply socio-ecological models in research, these efforts have revealed additional blind spots. Intrapersonal-level attitudes (such as convenience or enjoyment) and interpersonal influences (such as parental safety concerns) are frequently addressed, but comparatively distal determinants such as political-will and governmental subsidies remain under-explored (42,46). Without application of systematic attention to these upstream public policy-level factors, recommendations and guidance risk over-emphasising behaviour-change messaging while under-specifying the structural changes required to facilitate positive change.

The practical consequences of these deficits are tangible. Interventions developed both in and by car-oriented, high-income 'suburbs' are exported to dense, lower-income 'megacities' without regard for differences in lived experiences and barrier profiles; infrastructure budgets prioritise protected cycle

lanes in areas where bicycle ownership is minimal; and public health goals aimed at increasing children's MVPAs remain unrealised as implementations fail to address the most significant local challenges.

1.2.4 Absence of a Comprehensive, Contextually Diverse Synthesis

To date (and the best of my knowledge), no scoping review has comprehensively mapped the barriers and facilitators to AST across income-strata and age-based demographic groups. Existing syntheses focus either on a single determinant (such as safety perceptions) or present aggregated findings without disaggregating by context, thereby obscuring important variations that are critical for informing global guidance. As such, a scoping review that captures both the full range of determinants and the nuanced, lived experiences of barriers and facilitators across diverse contexts would make a valuable and timely contribution to the field.

1.2.5 Limitations in Theoretical Integration and Analytical Depth

Moreover, prior systematic and narrative reviews were often descriptive and atheoretical. Few drew on socio-ecological or behavioural theory to critically examine how determinants interact across all levels of the socio-ecological system, and even fewer assessed the quality or consistency of evidence when findings were conflicting (42,47). Of particular concern, a 2018 systematic review by Larouche *et al.* screened 6,318 papers and found that *none* of the 27 included papers employed structural equation modelling to investigate causal pathways or applied established behavioural frameworks such as the COM-B (1). The absence of a comprehensive overview of the research landscape could complicate efforts to identify areas of emerging consensus, catch on where theoretical development has stalled, and recognise populations or determinant categories that remain largely under-studied.

1.2.6 Research Question

Because AST is shaped by a complex interplay of a wide range of factors, there is an undeniable need for an inclusive review that draws on evidence from published studies, grey literature, and policy documents across diverse geographies and income-strata. By bridging the strengths of quantitative research's positivist rigor with the contextual insights offered by relativist qualitative approaches, a pragmatic mixed-method review could offer a more nuanced understanding of AST among children around the globe. A scoping review, unconstrained by the narrowness of the Population, Intervention, Comparison, Outcome (PICO) criteria of a systematic review, is uniquely suited to provide such a map that would capture the breadth of determinants, elucidate research clusters and chasms, and identify intervention and policy-relevant levers for future intervention.

The convergence of evidence gaps identified in this literature review with UNICEF's operational needs therefore warrants the question: *"What are the documented barriers and facilitators to active school transport among school-going children (aged 5-19) globally?"*

1.3 Aims and Objectives

The aim of this project is hence to identify and synthesise the documented barriers and facilitators to AST among school-going children aged 5-19, creating comprehensive evidence maps to inform UNICEF's forthcoming global guidelines on active mobility (with a secondary aim to elucidate a clearer view of the current knowledge base).

The objectives are as follows:

(i) Perform a systematic search on peer-reviewed (and grey literature) on the documented barriers and facilitators to AST among school-going children aged 5-19.
(ii) Map the documented barriers and facilitators within the SEM.
(iii) Compare and describe determinant patterns across homogenised school-levels and World Bank income-strata.
(iv) Identify all available implementation and policy mechanisms through which to which to implement positive behaviour change using the COM-B (and additive frameworks).
(v) Identify gaps in evidence and recommend areas of priority for further research, intervention design, and guidance development relevant to key stakeholders.

Section 2: Methodology

The protocol for this study will be registered on the Open Science Framework (OSF), a free, open-source platform designed to support researchers in managing scoping reviews throughout the research process. The review was conducted in accordance with the Preferred Reporting Items for Systematic reviews and Meta-Analyses extension for Scoping Reviews (PRISMA-ScR) guidelines (Appendix C) (48). Ethical approval was not required, as all data utilised in this study was obtained from published sources or otherwise publicly available documentation. Moreover, concept groups used in the literature search were developed using the Population, Context, Concept (PCC) framework (49), where:

Population	=	School-going children (aged 5-19).
Context	=	Barriers and facilitators to AST across all SEM levels.
Concept	=	Non-motorised travel to and/or from school specifically.

2.1 Search Strategy

Six electronic databases – namely OVID Embase, OVID Medline, PubMed, Scopus, Web of Science (WoS), and Transport Research International Documentation (TRID)) – were systematically searched to retrieve papers between the 1st of January, 2010, and the 27th of May, 2025. A fifteen-year period was chosen because it is a widely accepted timeframe that supports a balance between comprehensiveness and relevance (50). The search concepts described above were agreed upon *a priori* and further informed by deconstruction of the research question. The search strategy was first developed for Embase and subsequently adapted for use in other databases, with guidance from experienced researchers (PA and AM, part of the George Institute supervisory team). Following every modification, searches were piloted severally, and abstracts were reviewed for appropriateness and accuracy. The final search strategy was then validated by an experienced librarian from my institution.

Several scoping and systematic reviews with similar concept groups were examined to identify relevant keywords. Emtree in Embase was then exhaustively searched to capture all pertinent subject headings. In other databases, Medical Subject Headings (MeSHs) and equivalent controlled verbiage were thoroughly explored. A combination of search terms (keywords and subject headings) were then combined with Boolean operators to relate the three concept groups (Table 1). Search results were limited to papers of the English-language, focused on human studies exclusively. A supplementary manual search of the reference lists of included papers and grey literature was additionally performed; the latter included governmental reports and policy documents, program evaluations, and external dissertations.

Table 1: *First Iteration Search Strategy for OVID Embase. Individual subject headings for each concept group were combined using Boolean "OR" and integrated with corresponding keywords using "AND". Multi-purpose fields were utilised to search across title, abstract, subject heading, and author fields. Syntax was applied as appropriate to account for homonyms, plurals, and continuing phrases. To further refine the search given the broadness of included terms, proximity searches and common phrases were also employed. The complete search strategy across the six databases is described in Appendix D.*

Search Strategy		
Population (P): School-aged children aged 5-19	Subject headings (Emtree)	Child/
		Adolescent/
		Minors/
		Juvenile/
		Teenager/
		exp Parent/
		exp Teacher/
	Keywords	("School-Age*" or "School-Going" or Child* or Schoolchild* or Adolescent* or Teen* or Kid or Kids or Minor or Minors or Youth* or Parent* or Guardian* or Teacher*).mp.
Context (C): Barriers and facilitators to AST across SEM levels.	Subject headings (Emtree)	Social determinants of health/
		Health behavior/
		Behavior change/
		Motivation/
		Extrinsic motivation/
		Intrinsic motivation/
		Incentive/
		Social incentive/
		Perverse incentive/
		Disincentive/
	Keywords	("Health Behavio?r" or "Mobility Behavio?r" or "Behavio?r Change" or "Health Literac*" or "Self-Efficac*" or "Peer-Pressure" or "Peer-Influence" or ((Cultural or Systemic or Underlying or Social) adj1 (Belief* or Determinant* or Barrier* or Facilitat* or Challenge* or Reason or Reasons)) or ((Positive or Good or Negative or Bad) adj1 (Experience* or Opinion* or Belief* or Perception*)) or Barrier* or Facilitat* or Reason or Reasons or Policy or Constrain* or Obstruct* or Restrain* or Restrict* or Enabler* or Obstacle* or Influence* or Determinant* or Motivator* or Limit* or Challenge or Attitude* or Belief* or Perception* or Motivation* or Stigma* or Trust* or Access* or Availab* or Guideline*).mp.
Concept (C): Non-motorised travel to and/or from school specifically.	Keywords	((("Non-Motori?ed Transport*" or "Activ* Travel*" or "Active Mobility" or "Physical Mobility" or "Soft Mobility" or "Cycle Path*" or "Cycle Zone*" or "Cycle Lane*" or "Cycling Path*" or "Cycling Zone*" or "Cycling Lane*" or "Bike Path*" or "Bike Zone*" or "Bike Lane*" or "Bicycle Path*" or "Bicycle Zone*" or "Bicycle Lane*" or "Walk* Path*" or "Walk* Zone*" or "Walk* Lane*" or "Foot Path*" or "Foot Zone*" or "Foot Lane*" or "Pedestrian Path*" or "Pedestrian Zone*" or "Pedestrian Lane*" or "Home-to-School Travel*" or "Home-from-School Travel" or "School-to-Home Travel" or "School-from-Home Travel" or "Home-to-and-from School Travel") adj10 (School* or Highschool* or Kindergarten*)) or (("Non-Motori?ed Transport*" or "Activ* Travel*" or "Active Mobility" or "Physical Mobility" or "Soft Mobility" or "Cycle Path*" or "Cycle Zone*" or "Cycle Lane*" or "Cycling Path*" or "Cycling Zone*" or "Cycling Lane*" or "Bike Path*" or "Bike Zone*" or "Bike Lane*" or "Bicycle Path*" or "Bicycle Zone*" or "Bicycle Lane*" or "Walk* Path*" or "Walk* Zone*" or "Walk* Lane*" or "Foot Path*" or "Foot Zone*" or "Foot Lane*" or "Pedestrian Path*" or "Pedestrian Zone*" or "Pedestrian Lane*" or "Home-to-School Travel*" or "Home-from-School Travel" or "School-to-Home Travel" or "School-from-Home Travel" or "Home-to-and-from School Travel") adj10 ("High School*" or "Secondary School*" or "Middle School*" or "Primary School*" or "Elementary School*")) or ((Commut* or Walk* or Sidewalk or Footpath or Bike or Bicycle or Cycle or Cycling or Trip or Trips or Travel* or Route* or Journey* or "Drop-off" or "Drop-offs" or "Pick-up" or "Pick-ups") adj10 (School* or Highschool* or Kindergarten*)) or ((Commut* or Walk* or Sidewalk or Footpath or Bike or Bicycle or Cycle or Cycling or Trip or Trips or Travel* or Route* or Journey* or "Drop-off" or "Drop-offs" or "Pick-up" or "Pick-ups") adj10 ("High School*" or "Secondary School*" or "Middle School*" or "Primary School*" or "Elementary School*"))).mp.

EndNote20 (a reference management application) and Rayyan (a web-based software) were used to identify and eliminate duplicate studies in order to assist independent reviewers during the screening process. EndNote was selected for its robust and customisable deduplication capabilities, while Rayyan was used for its enhanced sensitivity and accuracy in the manual removal of papers with differing fields.

2.2 Design

The review employed a convergent integrated mixed-method design, wherein quantitative, qualitative, and mixed-method evidence were first qualitized and subsequently integrated simultaneously (Figure 1). For the purposes of this work, 'integration synthesis' was defined as refining, comparing, and contrasting re-emerging themes of barriers and facilitators across included papers. Due to the uneven distribution of included papers across school-levels and income-strata (and small sample sizes within each sub-group), findings were reported in both absolute terms and as a proportion of papers within each sub-group to enable a more accurate and transparent representation of the data. Brief reminders of the calculation method were provided at the beginning of each major result sub-section to enhance readability.

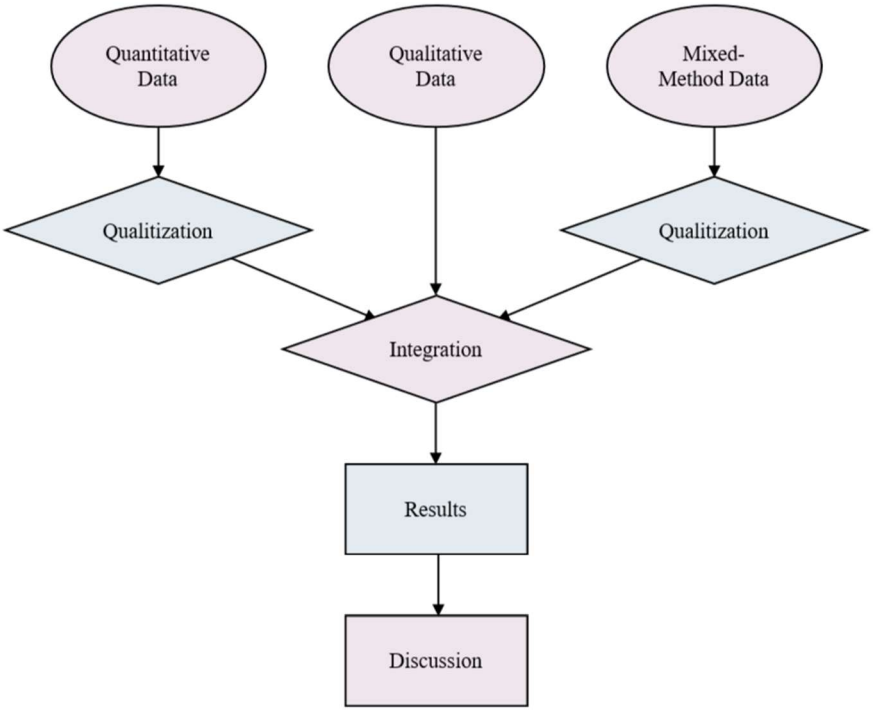


Figure 1: *Overview of the Convergent-Integrated Mixed-Method Scoping Review Design. Qualitization is the process of converting quantitative data into qualitative form such that quantitative findings can be integrated alongside qualitative data where a unified qualitative interpretive approach is ideal.*

2.3 Study Selection

2.3.1 Eligibility Criteria

Titles, abstracts, and full-text papers retrieved from searches and included for review were those that met the following inclusion criteria:

(1) Focuses on children between the ages of 5 and 19.
(2) Focuses on travel to and/or from school (or any appropriate synonym).
(3) Focuses on barriers and/or facilitators to AST experienced by children.
(4) Is any primary or secondary data source which is NOT a clinical or health intervention trial (including policy-related papers that revolve around these).
(5) Pertains to any combination of LICs, MICs, or HICs.
(6) Pertains to any combination of rural, peri-urban, or urban contexts.
(7) Addresses non-motorised modes of transport among school-going children.
(8) Includes any combination of child, parent, or guardian perspectives on perceived barriers and facilitators to AST experienced by children across SEM levels.

Search results were excluded if they:

(1) Did not focus on children between the ages of 5 and 19.
(2) Had no mention of active travel (or any appropriate synonym) to and/or from school.
(3) Had no mention of barriers or facilitators (or any appropriate synonym) to AST across any SEM level.
(4) Were a clinical or health intervention study.
(5) Were of a conference or editorial publication type.
(6) Had an exclusive focus on motorised modes of transport.
(7) Had a focus on vehicle safety technology rather than AST.
(8) Were a duplicate.

The criteria described above were determined through deconstruction of both the research question and objectives, as informed by UNICEF's needs. Clinical and health intervention studies and conference papers were excluded so as to maintain clarity of purpose and manageability of the review.

2.3.2 Defining Barriers and Facilitators

While there is currently no single, universally accepted definition of either a 'barrier' or 'facilitator' in behavioural research, there is a clear and consistent conceptual understanding of how and when these terms are to be used. In the context of this work, barriers were understood to be variables that impede(d) or prevent(ed) the adoption, engagement, or maintenance of AST among children aged 5-19. Conversely, facilitators were taken to refer to factors that enable(d), support(ed), or promote(d)

participation in the desired positive health-behaviour. As these terms have become standard in the field, they were adopted to ensure the interpretability and comparability of findings across papers by all relevant stakeholders (i.e. researchers, practitioners, and policymakers), as well as to best serve the design of well-informed guidelines. All pertinent synonyms (e.g. belief, social determinant, obstacle, or enabler) were included as search terms to reflect the full range of common-use vernacular.

2.3.3 Defining School Levels

School-levels were classified based on the US educational framework to accommodate for variations in global school systems (51). Herein, 'kindergarten' corresponded to students aged 5-6, 'elementary' to ~6-12, 'middle' to ~12-14, and 'high' to ~14-19.

2.3.4 Selection Process

The extracted studies were either included or excluded as part of a three-step process based on title/abstract and full-text screening. All titles and abstracts were screened via two independent reviewers (myself and LB, who were 97.2% aligned). Abstracts that passed the initial screen then progressed to full-text review to ensure compliance with the eligibility criteria. Conflicts were resolved through independent assessment by a third reviewer (AP), followed by discussion until consensus was reached. In no cases were authors or identified papers contacted to request additional information where eligibility was unclear due to time constraints.

2.4 Data Charting

Data charting was performed by myself in Microsoft Excel. The information extracted across papers included bibliographic details (i.e. author name(s), title, year of publication, journal, and DOI/URL); geographical details (inter-country region, country, WHO region, and World Bank income-strata); paper design details (methodological orientation, paper-type, data-collection tools, and analytical approach); population characteristics (age, sex, US-equivalent school-level, and sample size); mode of transport; identified barriers and facilitators; and policy or practice implications. This charting framework was piloted to best suit relevant and extractable information across papers, and behavioural factors were assigned by themes identified via qualitization and integration.

2.5 Data Mapping

Following data synthesis, the SEM was used to map barriers and facilitators to AST in a way that recognised behaviours were shaped not only by individual choices, but also by broader social, environmental, and policy contexts. Similarly, the COM-B was employed to frame barriers in a way that elucidated both why they existed and how they might be addressed. The COM-B was used alongside commonly accompanying models – namely the Theoretical Domains Framework (TDF) (Appendix E) (52) and Behaviour Change Wheel (BCW) (Appendix F) (32) – to identify all possible intervention functions and actionable policy categories through which to implement change.

Section 3: Results

Results are presented in chronological order, in line with objectives (i) to (iv). This section is divided into four main sub-sections: 3.1 summarises the paper selection process; 3.2 describes the characteristics of included papers; 3.3 maps the identified barriers and facilitators across the SEM; and 3.4 elucidates all available intervention functions and actionable policy categories using the COM-B and associated frameworks.

3.1 Paper Selection Process

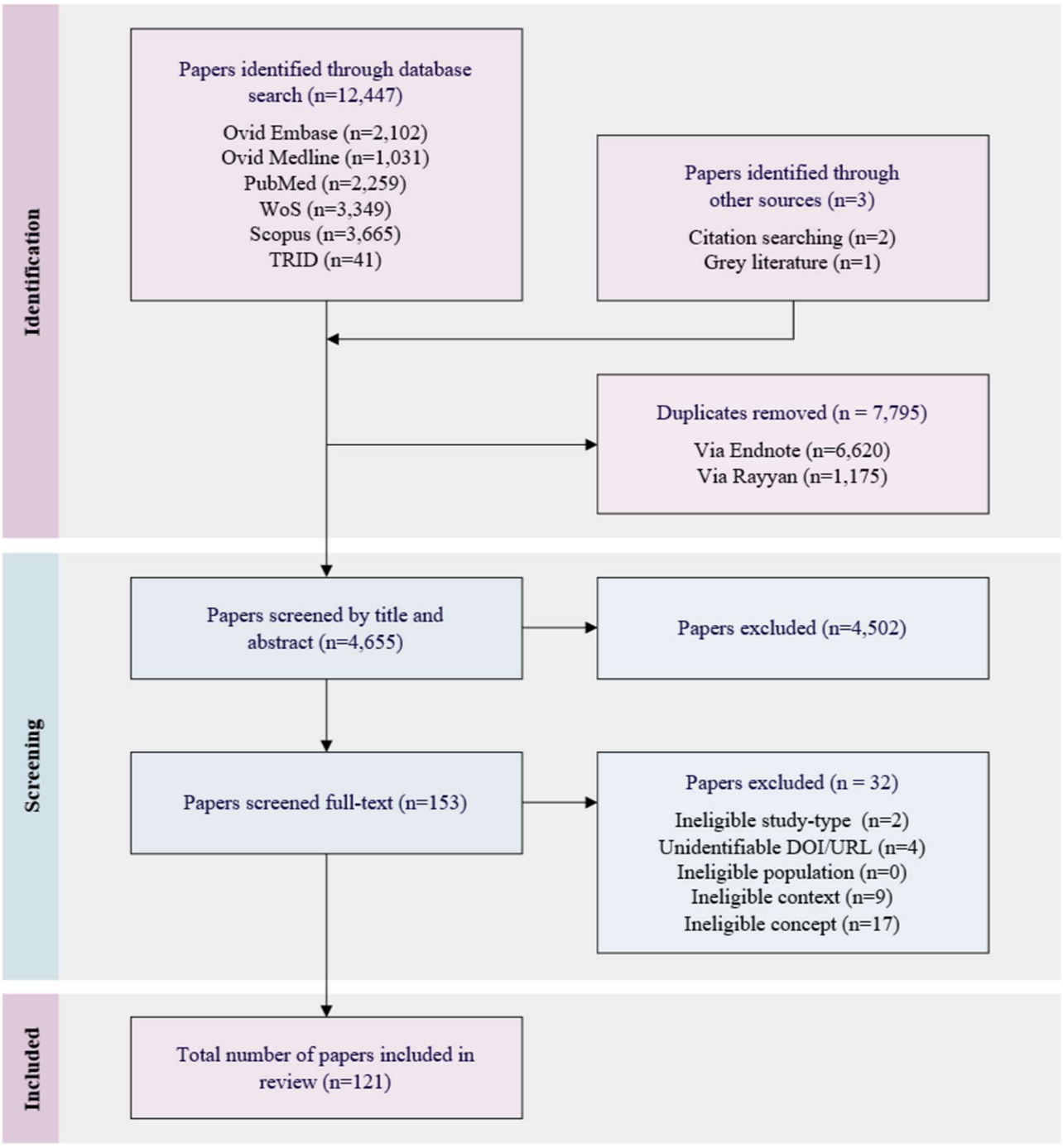


Figure 2: *PPRISMA Flowchart Overview of the Paper Selection Process. A total of 12,329 papers were excluded.*

A total of 12,447 papers were identified from search of the six electronic databases (Figure 2). Three papers were identified through alternative means, including 2 via citation searching, and 1 via grey literature. EndNote20 was first used to exclude 6,620 duplicates, and Rayyan aided in the removal of an additional 1,175 hidden by differing fields. 4,655 papers were subsequently screened for eligibility by title and abstract, leaving 153 papers for full-text review. A further 32 papers were then excluded for the reasons recorded in Figure 2.

3.2 Characteristics of Included Papers

The resultant 121 included papers comprised book chapters, studies, an implementation evaluation, and literature reviews of various methodological approaches and sample populations. A summary of the included paper characteristics is described in Table 2.

Table 2: *Overview of Included Paper Characteristics. Sum of percentages for school-level characteristic is >100 due to significant overlaps between age-ranges of study populations in included papers. Full table of charted items – including identified barriers and facilitators – is described in Appendix G.*

Results Section	Paper Characteristic	Category		Number of papers n, (% of Included Papers)	
3.2.1	WHO Region of Focus	Africa (AFRO)			4, (3.31)
		Americas (AMRO)			38, (31.40)
		South-East Asia (SEARO)			2, (1.65)
		Europe (EURO)			31, (25.62)
		Eastern Mediterranean (EMRO)			2, (1.65)
		Western Pacific (WPRO)			34, (28.10)
		Multi-Regional			5, (4.13)
		Unspecified			5, (4.13)
	World Bank Income Strata	High-Income Country (HIC)			79 (65.29)
		Upper-Middle-Income Country (UMIC)			30, (24.79)
		Lower-Middle-Income Country (LMIC)			5, (4.13)
		Low-Income Country (LIC)			0, (0.00)
		Multi-Countries			3, (2.48)
		Unspecified			4, (3.31)
3.2.2	Methodological Approach	Quantitative			93, (76.86)
		Qualitative			15, (12.40)
		Mixed Methods			13, (10.74)
	Paper-Type	Book Chapter	*Descriptive*	*3, (2.48)*	4, (3.31)
			Analytic	*1, (0.83)*	
		Cross-Sectional	*Descriptive*	*5, (4.13)*	95, (78.51)
			Analytic	*90, (74.38)*	
		Case-Study	*Descriptive*	*2, (1.65)*	4, (3.31)
			Analytic	*2, (1.65)*	
		Case-Control			1, (0.83)
		Descriptive Exploratory			4, (3.31)
		Implementation Evaluation			1, (0.83)
		Longitudinal	*Descriptive*	*0, (0.00)*	4, (3.31)
			Analytic	*4, (3.31)*	
		Literature Review	*Systematic*	*5, (4.13)*	8, (6.61)
			Narrative	*3, (2.48)*	
3.2.3	School-Level (Age ranges)	Kindergarten (5-6)			27, (22.31)
		Elementary School (~6-12)			89, (73.55)
		Middle School (~12-14)			70, (57.85)
		High School (~14-18+)			41, (33.88)

3.2.1 Mapping Geographies

Twenty-one papers were based on sample populations originating from the US; 14 from China and New Zealand; 9 from Spain; 8 from Brazil and Canada; 6 from Australia; and 4 from Greece, Malaysia, Portugal, and England (United Kingdom (UK)). Three studied children from the Czech Republic, Ireland, India, and the Netherlands; 2 from Belgium, Denmark, Germany, Iran, and South Africa; and 1 from Bangladesh, Cameroon, Chile, Columbia, Ecuador, Ghana, Israel, Italy, Mexico, Nigeria, Saudi Arabia, Slovenia, Switzerland, and Vietnam (Figure 3). The geographical representation illustrated by included papers indicated a skew towards the WHO regions of the Americas (AMRO) (n=38, 31% of included countries), the Western Pacific (WPRO) (n=34, 28%), and Europe (EURO) (n=31, 26%), particularly towards HICs (n=79, 65%) and upper-MICs (UMICs) (n=30, 25%). Notably, no papers that examined LICs were eligible for review.

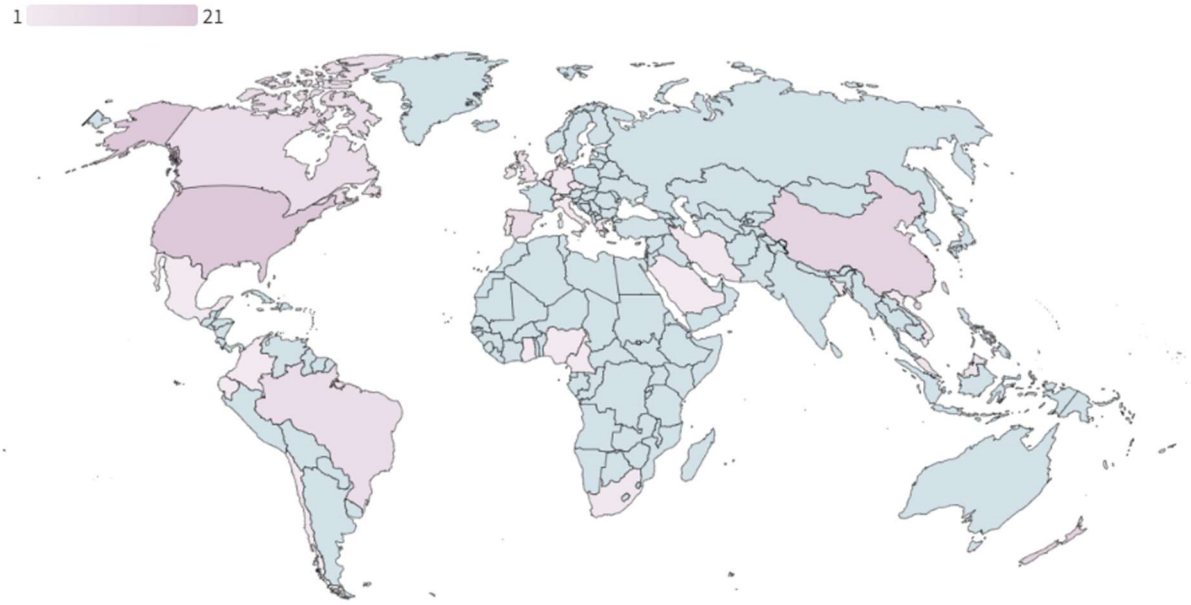


Figure 3: *Overview of Included Papers by Geographical Distribution. Pink shades represent countries from which sample populations included in review originate. Darker pink shades indicate countries featured in more papers. Blue shades represent countries from which no sample populations were included in review. Figure was made in app.flourish.studio.*

3.2.2 Mapping Paper Designs

Methodological Orientations and Paper-Types

The majority of included papers employed an analytic approach (n=103, 85% of included papers), while the remainder adopted a more descriptive orientation. Analytic studies sought to examine relationships, predictors, or correlates of AST, while descriptive papers aimed primarily at characterising (and sometimes quantifying) the prevalence of travel behaviours, perceptions, and/or contextual factors, without testing for associations.

In terms of methodological orientation, a disproportionate number of studies were quantitative (n=93, 77%), followed by qualitative (n=15, 12%) and mixed-method designs (n=13, 11%) (Table 3). The most common paper-type were cross-sectional studies (n=95, 79%), particularly among analytic

studies (n=90, 87% of analytic studies). Smaller portions of studies were longitudinal (n=4), descriptive-exploratory (n=4), case-study (n=4), and case-control (n=1) designs, and non-study contributions were literature reviews (n=8), book chapters (n=4), and a single implementation evaluation (n=1). En masse, the variety of included papers captured both empirically measurable outcomes and contextually relevant experiences across school-levels and income-strata.

Table 3: *Overview of Included papers by Methodological Approach. 'D' is indicative of descriptive studies, while 'A' represents analytic ones. 'S' represents systematic reviews, and 'N' represents narrative ones. Sum of percentages in each row is =100, denotating the full catalogue of each paper-type.*

		n, (% of Paper-Type)					
Paper-Type		**Quantitative**		**Qualitative**		**Mixed Methods**	
Book Chapter		2, (50.00)		0, (0.00)		2, (50.00)	
Cross Sectional	*D*	*2, (40.00)*	84, (88.42)	*2, (40.00)*	3, (3.16)	*1, (20.00)*	8, (8.42)
	A	*82, (91.11)*		*1, (1.11)*		*7, (7.78)*	
Case Study	*D*	*0, (0.00)*	2, (50.00)	*2, (100.00)*	2, (50.00)	*0, (0.00)*	0, (0.00)
	A	*2, (100.00)*		*0, (0.00)*		*0, (0.00)*	
Case Control		1, (100.00)		0, (0.00)		0, (0.00)	
Descriptive Exploratory		0, (0.00)		4, (100.00)		0, (0.00)	
Implementation Evaluation		0, (0.00)		0, (0.00)		1, (100.00)	
Longitudinal	*D*	*0, (0.00)*	4, (100.00)	*0, (0.00)*	0, (0.00)	*0, (0.00)*	0, (0.00)
	A	*4, (100.00)*		*0, (0.00)*		*0, (0.00)*	
Literature Review	*S*	*0, (0.00)*	0, (0.00)	*3, (60.00)*	6, (75.00)	*2, (40.00)*	2, (25.00)
	N	*0, (0.00)*		*3, (100)*		*0, (0.00)*	

Data-Collection Tools

The methods of data-collection identified in review were diverse, reflecting the purposes of each research endeavour. Eighty-five (or 79% of the sum of) quantitative or mixed-method studies relied on structured questionnaires or surveys administered to children, their parents, both, or administrative school-personnel. These were used to gather information on school travel mode and frequency, status of travel independence, socio-demographic characteristics, and perceptions of social and built environments. While some tools were indeed validated scales, others were adapted or developed for specific contexts. A subset of quantitative or mixed-method studies (n=21, 19%) also incorporated objective or secondary data sources, such as Geographic Information System (GIS)-based environmental audits, administrative school records, or computational mapping tools such as Google Maps to estimate route distances or assess potential exposures to traffic and infrastructural variables of interest. Notably, these objective measures were often used to corroborate self-reported data, therein promoting overall validity.

On the other hand, qualitative studies utilised a broader range of data-collection methods, including semi-structured interviews, focus group discussions, and open-ended questionnaires. Two (or 13%) of these adopted atypical approaches, employing student-generated drawings or mapping activities to capture rich, interpretivistic data via a simulated experience of the journey to school. When used in

mixed-method research, qualitative data-collection tools were typically combined with questionnaires, surveys, or objective measures, enabling methodological triangulation while enhancing the pragmatic depth of findings.

Analytical Approaches

Analytic studies, particularly those of quantitative or mixed-method designs, predominantly applied a variety of regression-based analytical approaches to assess the relationships between AST and wide array of underlying behavioural predictors (n=69, 64% of the sum of quantitative and mixed-method papers). These comprised binary, multilevel, and multivariate logistic regressions; generalised linear and additive mixed models; Poisson and gamma regressions; proportional odds models; and structural equation modelling. Moreover, 3 studies utilised machine-learning algorithms (such as random forest and support vector machines) where large or complex datasets were imminent. In both quantitative and mixed-method papers, outcomes were commonly reported using Odds Ratios (ORs), Prevalence Ratios (PRs), or regression coefficients (e.g. β), alongside 95% CIs to convey the strength and direction of associations. Statistical significance was typically assessed using p-values, chi-squared (χ^2) tests, or distributional tests such as the Shapiro-Wilk test, and descriptive statistics were used to complement these. Frequencies, percentages, and proportions were used to summarise categorical variables, while means and standard deviations (SDs) described continuous ones.

In contrast, qualitative studies adopted a variety of interpretive strategies. Thematic analysis was the most dominant analytical approach (n=10, 67% of qualitative papers), in some instances guided by established theoretical models such as the Ecological Model, Theory of Planned Behaviour, or Prototype Willingness Model. Other analytical strategies included reflexive thematic analysis, visual content analysis, and narrative synthesis. The mixed-method portion of studies typically presented both quantitative-arm and qualitative-arm findings either in parallel or via an integrated approach, drawing on statistical modelling to elucidate behavioural patterns and qualitative data to contextualise the lived experiences underlying these trends.

3.2.3 Mapping School-Levels

The sample populations of the included studies adequately represented the four levels of the US educational framework (Figure 4).

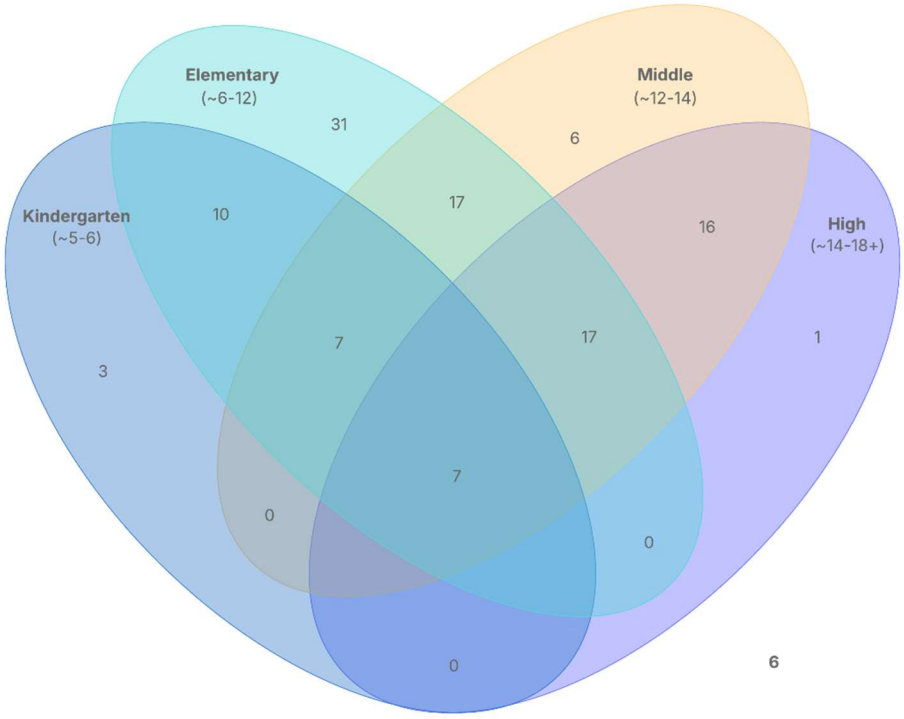


Figure 4: *Overview of Included Papers by United States (US)-Equivalent School-Levels. 41 papers addressed a single school-level, while 58 addressed multiple levels. 6 papers were not identifiable as any particular school-level – these were literature reviews. 27 papers addressed kindergarten; 89 addressed elementary; 70 addressed middle; and 41 addressed high-school. Youngest age identified in kindergarten level was 5; highest in high-school was 19. Figure was made in lucid.app.*

Forty-one papers focused exclusively on a single school-level. Of these, 31 focussed on elementary-school students, 6 on middle-school students, 3 on kindergarteners, and 1 on high-schoolers. Forty-three papers included participants from two adjacent school-levels: 17 spanning elementary and middle-school levels, 16 spanning middle and high-school, and 10 spanning kindergarten and elementary-school. Moreover, 24 papers included participants across three school-levels: 7 spanning kindergarten through middle-school, and 17 spanning elementary through high-school. Seven papers included participants across all four levels, and no papers addressed non-adjacent level intersections. An additional 6 papers were not classifiable, as authors did not specify age-ranges pertaining to their study populations. Three among these were book chapters; the remaining 3 were systematic reviews.

3.3 Mapping Barriers and Facilitators to Active School Transport Using the SEM

The documented barriers and facilitators to AST among children across school-levels and World Bank income-strata were systematically mapped into intrapersonal, interpersonal, institutional, community-environmental, and public policy-level influences using the SEM (Appendix A). Use of the convergent-integrated mixed-method design identified and consolidated re-emerging themes such that closely related but variable factors were grouped together into broader, well-defined categories (refer to Figure 1).

3.3.1 Intrapersonal Factors

First, 60 papers (50% of included papers) identified intrapersonal-level influences affecting children's participation in AST. These factors were internal to the individual, most relating to SES, logistical considerations, and psycho-social motivations, with variations observed across school-levels and income-strata (Table 4).

Table 4: *Map of the Identified Intrapersonal-Level Barriers and Facilitators to Children's Active School Transport by US-Equivalent School-Level and World Bank Income-Strata. Proportions are relative to the total number of papers pertinent to each school-level, such that '3, (11.11)' indicates 3 (or 11.11%) of all papers pertinent to kindergarten reported being of higher socio-economic status as a barrier to AST. Proportions pertinent to LMICs should be interpreted with caution due to small numbers of included papers. Homogenisation in school-levels resulted in age-range overlap, so the sum of each integer does not equate the true number of papers assessing HICs. Full relative statistics are presented in the table; Quantitative values are detailed in-text only for the most frequently reported barrier and facilitator to illustrate the link.*

		School-Level n, (% of Level)											
		Kindergarten (K)			Elementary (E)			Middle (M)			High (H)		
		HICs	UMICs	LIMCs	HICs	UMICs	LMICs	HICs	UMICs	LMICs	HICs	UMICs	LMICs
Barriers	Higher household income and socio-economic status (SES)	2, (7.41)	1, (3.70)	0, (0.00)	4, (4.49)	8, (8.99)	1, (1.12)	3, (4.29)	8, (11.43)	1, (1.43)	3, (4.29)	5, (7.14)	0, (0.00)
	Carriage of (heavy) backpacks	0, (0.00)	0, (0.00)	0, (0.00)	2, (2.24)	0, (0.00)	0, (0.00)	0, (0.00)	0, (0.00)	0, (0.00)	1, (2.44)	0, (0.00)	0, (0.00)
	Time constraints	1, (3.70)	1, (3.70)	0, (0.00)	3, (3.37)	1, (1.12)	0, (0.00)	3, (4.29)	1, (1.43)	0, (0.00)	2, (4.88)	1, (2.44)	0, (0.00)
	Pre-existing sedentary behaviour	1, (3.70)	0, (0.00)	0, (0.00)	1, (1.12)	0, (0.00)	0, (0.00)	1, (1.43)	1, (1.43)	0, (0.00)	1, (2.44)	1, (2.44)	0, (0.00)
	Obesity	1, (3.70)	0, (0.00)	0, (0.00)	2, (2.24)	0, (0.00)	0, (0.00)	1, (1.43)	0, (0.00)	1, (1.43)	0, (0.00)	0, (0.00)	1, (2.44)
	Low self-efficacy	0, (0.00)	0, (0.00)	0, (0.00)	1, (1.12)	1, (1.12)	1, (1.12)	2, (2.86)	0, (0.00)	1, (1.43)	1, (2.44)	0, (0.00)	1, (2.44)
	Female sex	0, (0.00)	1, (3.70)	0, (0.00)	3, (3.37)	2, (2.24)	1, (1.12)	4, (5.71)	2, (2.86)	1, (1.43)	1, (2.44)	2, (4.83)	0, (0.00)
	Limited access to bicycles	0, (0.00)	0, (0.00)	0, (0.00)	1, (1.12)	0, (0.00)	0, (0.00)	2, (2.86)	0, (0.00)	0, (0.00)	2, (4.88)	0, (0.00)	0, (0.00)

Table 4 *(Continued)*

Facilitators	Perceived safety, comfort, and convenience	4, (14.81)	0, (0.00)	0, (0.00)	14, (15.73)	4, (4.49)	0, (0.00)	7, (10.00)	2, (2.86)	0, (0.00)	3, (7.32)	2, (4.88)	0, (0.00)
	Lower household income and SES	1, (3.70)	0, (0.00)	0, (0.00)	0, (0.00)	1, (1.12)	0, (0.00)	0, (0.00)	2, (2.86)	0, (0.00)	0, (0.00)	2, (4.83)	0, (0.00)
	Perceived health benefit	1, (3.70)	0, (0.00)	0, (0.00)	2, (2.24)	2, (2.24)	0, (0.00)	2, (2.86)	1, (1.43)	0, (0.00)	0, (0.00)	1, (2.44)	0, (0.00)
	Perceived opportunity for exercise	0, (0.00)	0, (0.00)	0, (0.00)	0, (0.00)	1, (1.12)	0, (0.00)	1, (1.43)	1, (1.43)	0, (0.00)	0, (0.00)	0, (0.00)	0, (0.00)
	Perceived enjoyment	1, (3.70)	0, (0.00)	0, (0.00)	3, (3.37)	1, (1.12)	0, (0.00)	1, (1.43)	0, (0.00)	0, (0.00)	0, (0.00)	0, (0.00)	0, (0.00)
	Perceived opportunity for independence	1, (3.70)	0, (0.00)	0, (0.00)	1, (1.12)	0, (0.00)	0, (0.00)	0, (0.00)	0, (0.00)	0, (0.00)	0, (0.00)	0, (0.00)	0, (0.00)
	Older age	2, (7.41)	0, (0.00)	0, (0.00)	3, (3.37)	0, (0.00)	0, (0.00)	3, (4.29)	0, (0.00)	0, (0.00)	1, (2.44)	0, (0.00)	0, (0.00)

Note for Calculations: *Included in review were: 27 papers pertinent to Kindergarten; 89 to Elementary; 70 to Middle; and 41 to High school; 79 to HICs; 30 to UMICs; and 5 to LMICs See* ***Table 2*****.**

Intrapersonal Barriers

The most frequently cited intrapersonal barriers which emerged in review were related to higher household income and SES, described in 16 papers (47,53–67). These barriers were especially evident among middle (n=12, 17%) and high-school (n=8, 20%) populations, and were most commonly reported in papers pertinent to UMICs (n=9, 30%) and lower-MICs (LMICs) (n=1, 20%), relative to the total number of paper of papers addressing each school-level and income-stratum, respectively. These characteristics often conferred increased access to private vehicles, fostering familial preferences for convenience that were shaped both by interpersonal and community-level influences. In turn, these external elements contributed to students' perceptions of AST as socially undesirable, some students citing feelings of embarrassment or fear of peer-judgement when engaging in AST.

An additional 8 papers referenced students' descriptions of AST as physically demanding (68,69) or time-consuming (62,70–74), especially when compared against the convenience offered by motorised travel. These perceptions were not strictly intrapersonal, but also reflected broader interpersonal, institutional, and community-level influences such as family routines, school schedules, and neighbourhood design.

Nine papers identified low motivation to shift from passive school transport to AST, often reinforced by habitual sedentary behaviours and social norms that favour car-use (55,61,66,69,75–79). Among these, 3 papers specifically noted obesity as a barrier to AST, citing factors such as physical discomfort and social stigma as key deterrents (55,75,79). While underlying causes varied by context, these factors contributed to low self-efficacies, with many students lacking the confidence to navigate areas perceived as unsafe due to various community-level concerns including risks of traffic or crime.

Sex differences were also apparent. In 8 papers, female students expressed a greater reluctance than males to participate in AST, citing internal concerns such as safety and external pressures such as parental restrictions and cultural gender norms (62,65,72,76,80–83). School uniforms were identified as an additional deterrent to cycling among older females, while limited access to bicycles constrained older students of both sexes who were both willing and able to cycle to and/or from school (70,84).

Intrapersonal Facilitators

Likewise, many of the intrapersonal facilitators which emerged in review exhibited points of overlap with social and environmental contexts. The most frequently cited among these related to perceptions of safety, comfort, and convenience, which were often influenced by community-level factors such as the built environment (73,79,81,85–100). These facilitators were reported across school-levels – but especially among kindergarten (n=4, 15%) and elementary-school populations (n=14, 16%) – and in HICs (n=15, 19%) and UMICs (n=5, 17%), but not in LMICs, possibly reflecting that lower-income contexts tend to be more driven by necessity than by choice. This interpretation was supported

by findings in 3 included analytic studies, wherein lower household income, parental education, and SES were associated with increased participation in AST, motivated not only by financial constraints but also by limited access to private vehicles (60,76,101).

Six papers identified children's perceptions of AST as beneficial for health as another instrumental facilitator, particularly for home-school distances under 1-2 miles (78,102–106). Two additional papers reported that students were more likely to walk when they viewed the journey to school as an opportunity for weight-loss or exercise (96,107). In both cases, intrapersonal motivations were reinforced by supportive peers, positive media messaging, and wider social norms, further highlighting the intersection between individual motivations and social influence.

Enjoyment (68,70,89,96,108), independence (70), and a sense of accomplishment (70) were other noted facilitators of AST, whose benefits were most evident when parents granted their children greater autonomy and/or considered their neighbourhoods to be safe.

3.3.2 Interpersonal Factors

Second, 50 papers (41% of included papers) identified interpersonal-level influences affecting children's participation in AST. These factors comprised parent-child dynamics, peer and sibling involvement, and broader familial attitudes towards safety, autonomy, and mobility (Table 5).

Table 5: *Map of the Identified Interpersonal-Level Barriers and Facilitators to Children's Active School Transport by US-Equivalent School-Level and World Bank Income Strata. Proportions are relative to the total number of papers pertinent to each school-level. Proportions pertinent to LMICs should be interpreted with caution due to small numbers of included papers. Homogenisation in school-levels resulted in age-range overlap, so the sum of each integer does not equate the true number of papers assessing HICs. Full statistics are presented in the table; Quantitative values are detailed in-text only for the most frequently reported barrier and facilitator to illustrate the link.*

		School-Level n, (% of Level)											
		Kindergarten			Elementary			Middle			High		
		HICs	UMICs	LIMCs	HICs	UMICs	LMICs	HICs	UMICs	LMICs	HICs	UMICs	LMICs
Barriers	Parental safety and security concerns	3, (11.11)	0, (0.00)	0, (0.00)	9, (10.11)	6, (6.74)	0, (0.00)	8, (11.43)	6, (8.57)	0, (0.00)	5, (12.20)	1, (2.44)	0, (0.00)
	Non-conducive parental work schedules	1, (3.70)	0, (0.00)	0, (0.00)	2, (2.25)	0, (0.00)	0, (0.00)	1, (1.43)	0, (0.00)	0, (0.00)	1, (2.44)	0, (0.00)	0, (0.00)
	Parental commutes to and from work	0, (0.00)	0, (0.00)	0, (0.00)	2, (2.25)	1, (1.12)	0, (0.00)	0, (0.00)	1, (1.43)	0, (0.00)	0, (0.00)	1, (2.44)	0, (0.00)
	Modelling of parental non-active behaviour	1, (3.70)	0, (0.00)	0, (0.00)	3, (3.37)	1, (1.12)	0, (0.00)	1, (1.43)	1, (1.43)	0, (0.00)	1, (2.44)	1, (2.44)	0, (0.00)
	Parental accompaniment	0, (0.00)	0, (0.00)	0, (0.00)	2, (2.25)	0, (0.00)	0, (0.00)	1, (1.43)	0, (0.00)	0, (0.00)	1, (2.44)	0, (0.00)	0, (0.00)
	Parental encouragement and support	1, (3.70)	0, (0.00)	0, (0.00)	4, (14.81)	1, (1.12)	1, (1.12)	5, (7.14)	1, (1.43)	1, (1.43)	1, (2.44)	0, (0.00)	1, (2.44)
	Parental accompaniment	0, (0.00)	0, (0.00)	0, (0.00)	2, (2.25)	0, (0.00)	0, (0.00)	1, (1.43)	1, (1.43)	0, (0.00)	1, (2.44)	1, (2.44)	0, (0.00)
	Peer support and social interaction	1, (3.70)	0, (0.00)	0, (0.00)	4, (14.81)	1, (1.12)	0, (0.00)	4, (5.71)	1, (1.43)	0, (0.00)	2, (4.88)	1, (2.44)	0, (0.00)
	Presence of (an) older sibling(s)	0, (0.00)	0, (0.00)	0, (0.00)	2, (2.25)	0, (0.00)	0, (0.00)	2, (2.86)	0, (0.00)	0, (0.00)	0, (0.00)	0, (0.00)	0, (0.00)

Interpersonal Barriers

Parental concerns for children's safety emerged as the most frequently reported interpersonal barrier, described in 26 papers (47,59,62,67,74,76,83,87,91,92,96,102,109–122). These concerns were particularly evident among guardians of children across elementary (n=15, 17%), middle (n=14, 20%), and high-school populations (n=6, 15%), and were most commonly reported in papers pertinent to HICs (n=16, 20%, %) and UMICs (n=6, 20%), often reflecting both perceived and objective intrapersonal and community-level indicators of potential dangers along the route. Parents of kindergarteners expressed comparatively low concern, likely because they often did not permit their children to travel independently.

Logistical constraints were also reported, with 5 papers recognising parental time pressures – particularly in the case of dual-earner households – to play an inhibitory role (54,66,90,100,123). Tight work schedules, quicker travel times, and a need to drop-off multiple children at different locations made driving a more convenient option, especially in higher-income contexts where capitalist cultural values are more prominent.

A subset of studies also found that modelling of parental transport behaviours influenced children's travel preferences. In 6 papers, parent's prioritisation of efficiency, negative attitudes towards walking or cycling, or otherwise lack of engagement in these attenuated children's likelihood of participating in AST (54,81,113,123–125). This effect was observed across all age-groups, where children's travel habits may remain (at least partially) prone to parental mimicry.

Interpersonal Facilitators

By contrast, parental encouragement and support were identified as particularly robust facilitators of AST, cited in 8 papers (59,77,81,103,117,121,126,127). Parents who expressed affirmative confidence in their children's abilities or reflectively promoted AST as beneficial were linked to a higher likelihood of students' active commuting. In one prominent example, an included analytic study by Ross *et al.* (2018) found that positive parental attitudes were strongly associated with increased use of AST among elementary and middle-school children in a HIC ($\beta=0.535$; $p<0.001$) (103). Comparable influences were notable among elementary-school populations (n=6, 22%) and were most commonly reported in HICs (n=5, 6%) and LMICs (n=1, 20%), where emotional support continues to play a formative role.

Four papers identified parental accompaniment as a facilitator to AST (74,76,84,124), though 2 papers noted it could also function as a barrier, with the barrier function more pronounced among males than females (128,129). Contrastingly, positive peer influence was more consistently characterised, reported as a facilitator in 10 papers (62,84,86,93,99,102,105,117,121,130). Children who no longer required adult supervision were more likely to walk or cycle when joined by friends or classmates,

spurred on by various intrapersonal factors including perceived enjoyment, safety, and psycho-social motivation.

In 2 papers, the presence of older siblings attending as escorts served to mitigate against parental availability constraints and safety concerns, particularly in households with inflexible work schedules and limited access to private transport (64,83).

3.3.3 Institutional Factors

Third, 18 papers (15% of included papers) reported institutional and organisational-level influences affecting children's participation in AST. These factors constituted organisational policies, practices, and infrastructure elements pertaining to schools and related organisations (Table 6).

Table 6: *Map of the Identified Institutional-Level Barriers and Facilitators to Children's Active School Transport by US-Equivalent School-Level and World Bank Income-Strata. Proportions are relative to the total number of papers pertinent to each school-level. Proportions pertinent to all institutional factors should be interpreted with caution due to small numbers of included papers. Homogenisation in school-levels resulted in age-range overlap, so the sum of each integer does not equate the true number of papers assessing HICs. Full statistics are presented in the table; Quantitative values are detailed in-text only for the most frequently reported barrier and facilitator to illustrate the link.*

		School-Level n, (% of Level)											
		Kindergarten			Elementary			Middle			High		
		HICs	UMICs	LIMCs	HICs	UMICs	LMICs	HICs	UMICs	LMICs	HICs	UMICs	LMICs
Barriers	Non-conducive school uniforms	0, (0.00)	0, (0.00)	0, (0.00)	0, (0.00)	0, (0.00)	0, (0.00)	4, (5.71)	0, (0.00)	0, (0.00)	3, (7.32)	0, (0.00)	0, (0.00)
	Lack of secure bicycle storage	0, (0.00)	0, (0.00)	0, (0.00)	1, (1.12)	0, (0.00)	0, (0.00)	1, (1.43)	0, (0.00)	0, (0.00)	1, (1.43)	0, (0.00)	0, (0.00)
	Non-conducive school policies	1, (3.70)	0, (0.00)	0, (0.00)	1, (1.12)	0, (0.00)	0, (0.00)	1, (1.43)	0, (0.00)	0, (0.00)	1, (1.43)	0, (0.00)	0, (0.00)
Facilitators	AST programs	3, (11.11)	1, (3.70)	0, (0.00)	3, (3.37)	1, (1.12)	0, (0.00)	2, (2.86)	0, (0.00)	0, (0.00)	1, (1.43)	0, (0.00)	0, (0.00)
	Supportive school and staff engagement	1, (3.70)	0, (0.00)	0, (0.00)	2, (2.25)	0, (0.00)	0, (0.00)	3, (4.29)	0, (0.00)	0, (0.00)	2, (4.88)	0, (0.00)	0, (0.00)
	Organisational incentives	1, (3.70)	0, (0.00)	0, (0.00)	2, (2.25)	0, (0.00)	0, (0.00)	1, (1.43)	0, (0.00)	0, (0.00)	1, (1.43)	0, (0.00)	0, (0.00)
	Conducive school schedules	1, (3.70)	0, (0.00)	0, (0.00)	1, (1.12)	0, (0.00)	0, (0.00)	0, (0.00)	0, (0.00)	0, (0.00)	0, (0.00)	0, (0.00)	0, (0.00)
	Presence of crossing guards or police along the route	2, (7.41)	0, (0.00)	0, (0.00)	3, (3.37)	0, (0.00)	0, (0.00)	1, (1.43)	0, (0.00)	0, (0.00)	0, (0.00)	0, (0.00)	0, (0.00)

Institutional Barriers

Uniforms deemed unsuitable for cycling were the most frequently reported institutional barrier, described in 4 papers (96,131–133). This issue was identified to play an inhibitory role in HICs exclusively (n=4, 5%), where it primarily affected female middle (n=4, 6%) and high-school populations (n=3, 7%). Female guardians did not support their daughters cycling due to safety and practicality concerns, while male guardians did not discriminate between either sex. Parents and their daughters referenced concerns of skirts getting caught in bicycle chains, gears, or wheels, creating potential for accidents. In-skirt cycling was also associated with risks of exposure and discomfort, raising parental concerns about modesty and their children's self-confidence.

Shortcomings in school-related infrastructure – including limited availability of bicycle racks and the absence of adult supervision during arrival to and dismissal from school – were noted in 1 paper (84). Two papers reported rigid or unfavourable school schedules as additional barriers, especially when in conflict with interpersonal and community-level influences such as work and family schedules or daylight patterns (74,108).

These 2 papers also proposed that non-conducive school policies may indirectly discourage AST-participation, possibly due to an overall lack of alignment between AST-related initiatives and broader health priorities or SDGs (74,108). Further, inconsistent policy implementation across schools or districts may even create inequities in access to environments supportive of AST.

Institutional Facilitators

On the other hand, well-implemented AST-related initiatives were identified as instrumental facilitators in 5 papers (74,94,126,134,135). These facilitators were particularly evident among kindergarten (n=4, 15%) and elementary-school populations (n=4, 5%), and exclusively reported in HICs (n=6, 8%) and UMICs (n=3, 10%). In one notable example, an included analytic study by McDonald *et al.* (2014) reported that education and encouragement programs implemented over a five-year period could lead to a 25% relative increase in AST (135). This was corroborated by Stewarta *et al.* (2018), whose evaluation of state-funded Safe Routes to School (SRTS) projects across four US states demonstrated an overall rise of 5% in AST-participation ensuing implementation (74). Interestingly, the greatest increases in cycling rates were observed in areas with low baseline rates, suggesting SRTS as particularly effective at introducing cycling to communities where previously uncommon. Further evidence from a similar demographic suggested more frequent walking bus trips and the presence of multiple route leaders also promoted AST (74,134).

Health-promoting school cultures (82,84,126) and institutional incentives (84,136) were also noted to have played an encouraging role. Support from principals and administrative parties was associated

with increased engagement with AST-related initiatives, especially when in-tandem with broader health-promotion efforts such as bicycle training and skills development initiatives.

Conducive school schedules were reported as a facilitator in 1 paper (70), and the presence of trustworthy institutional figures such as crossing guards or police along the home-school route was identified in 4 to address individual, interpersonal, and community-level concerns (70,81,86,117).

3.3.4 Community Factors

Fourth, 89 papers (74% of included papers) reported community-level influences affecting children's participation in AST. These comprised both perceived and objective environmental elements (Table 7).

Table 7: *Map of the Identified Community-Level Barriers and Facilitators to Children's Active School Transport by US-Equivalent School-Level and World Bank Income-Strata. Proportions are relative to the total number of papers pertinent to each school-level. Proportions pertinent to LMICs should be interpreted with caution due to small numbers of included papers. Homogenisation in school-levels resulted in age-range overlap, so the sum of each integer does not equate the true number of papers assessing HICs. Full statistics are presented in the table; Quantitative values are detailed in-text only for the most frequently reported barrier and facilitator to illustrate the link.*

		School-Level n, (% of Level)											
		Kindergarten			Elementary			Middle			High		
		HICs	UMICs	LIMCs	HICs	UMICs	LMICs	HICs	UMICs	LMICs	HICs	UMICs	LMICs
Barriers	Long home-school distance	6, (22.22)	0, (0.00)	0, (0.00)	27, (30.34)	4, (4.49)	3, (3.37)	21, (30.00)	4, (5.71)	3, (4.29)	11, (26.83)	4, (9.76)	2, (4.88)
	Inadequate pedestrian infrastructure	3, (11.11)	0, (0.00)	0, (0.00)	18, (20.22)	4, (4.49)	0, (0.00)	14, (20.00)	4, (5.71)	0, (0.00)	10, (24.39)	3, (7.32)	0, (0.00)
	Infrastructural deterrents	0, (0.00)	0, (0.00)	0, (0.00)	1, (1.12)	1, (1.12)	0, (0.00)	0, (0.00)	0, (0.00)	0, (0.00)	0, (0.00)	0, (0.00)	0, (0.00)
	Poor neighbourhood aesthetics	0, (0.00)	0, (0.00)	0, (0.00)	0, (0.00)	2, (2.25)	0, (0.00)	0, (0.00)	0, (0.00)	0, (0.00)	0, (0.00)	0, (0.00)	0, (0.00)
	Perceptions of air pollution	1, (3.70)	0, (0.00)	1, (3.70)	2, (2.25)	0, (0.00)	1, (1.12)	1, (1.43)	1, (1.43)	1, (1.43)	1, (2.44)	0, (0.00)	1, (2.44)
	Traffic hazards	6, (22.22)	0, (0.00)	0, (0.00)	9, (10.11)	8, (8.99)	0, (0.00)	11, (15.71)	4, (5.71)	1, (1.43)	9, (21.95)	2, (4.88)	1, (2.44)
	Risks of crime	3, (11.11)	0, (0.00)	0, (0.00)	6, (6.74)	3, (3.37)	0, (0.00)	5, (7.14)	4, (5.71)	1, (1.43)	3, (7.32)	3, (7.32)	1, (2.44)
	Adverse weather	2, (7.41)	0, (0.00)	0, (0.00)	4, (4.49)	0, (0.00)	0, (0.00)	2, (2.86)	1, (1.43)	0, (0.00)	1, (2.44)	0, (0.00)	0, (0.00)
	Attending schools in higher SES areas	0, (0.00)	0, (0.00)	0, (0.00)	3, (3.37)	0, (0.00)	1, (1.12)	2, (2.86)	0, (0.00)	2, (2.86)	0, (0.00)	0, (0.00)	1, (2.44)
	Attending private school as opposed to public school	0, (0.00)	0, (0.00)	0, (0.00)	0, (0.00)	1, (1.12)	1, (1.12)	0, (0.00)	1, (1.43)	1, (1.43)	0, (0.00)	1, (2.44)	0, (0.00)
	Living in more urbanized areas	0, (0.00)	0, (0.00)	0, (0.00)	2, (2.25)	1, (1.12)	0, (0.00)	2, (2.86)	1, (1.43)	0, (0.00)	2, (4.88)	1, (2.44)	0, (0.00)
	Social and cultural norms	0, (0.00)	0, (0.00)	0, (0.00)	1, (1.12)	0, (0.00)	0, (0.00)	3, (4.29)	0, (0.00)	0, (0.00)	2, (4.88)	0, (0.00)	0, (0.00)
	Residential density	0, (0.00)	0, (0.00)	0, (0.00)	0, (0.00)	1, (1.12)	0, (0.00)	2, (2.86)	1, (1.43)	1, (1.43)	2, (4.88)	1, (2.44)	1, (2.44)
Facilitato	Residential density	0, (0.00)	0, (0.00)	0, (0.00)	2, (2.25)	0, (0.00)	0, (0.00)	1, (1.43)	0, (0.00)	0, (0.00)	0, (0.00)	0, (0.00)	0, (0.00)
	Land-use mix diversity	2, (7.41)	0, (0.00)	0, (0.00)	5, (5.62)	3, (3.37)	0, (0.00)	3, (4.29)	2, (2.86)	1, (1.43)	2, (4.88)	1, (2.44)	1, (2.44)

Table 7 *(Continued)*

	Presence of recreational amenities along the route	0, (0.00)	0, (0.00)	0, (0.00)	1, (1.12)	0, (0.00)	0, (0.00)	0, (0.00)	0, (0.00)	0, (0.00)	0, (0.00)	0, (0.00)	0, (0.00)
	Close proximity to business centres	1, (3.70)	0, (0.00)	0, (0.00)	3, (3.37)	0, (0.00)	0, (0.00)	2, (2.86)	0, (0.00)	0, (0.00)	1, (2.44)	0, (0.00)	0, (0.00)
	Adequate pedestrian infrastructure	3, (11.11)	0, (0.00)	0, (0.00)	10, (11.24)	2, (2.25)	1, (1.12)	8, (11.43)	2, (2.86)	2, (2.86)	6, (14.63)	2, (4.88)	2, (4.88)
	Neighbourhood walkability	3, (11.11)	2, (7.41)	0, (0.00)	11, (12.36)	4, (4.49)	0, (0.00)	3, (4.29)	3, (4.29)	0, (0.00)	2, (4.88)	2, (4.88)	0, (0.00)
	Grid-like street patterns	0, (0.00)	0, (0.00)	0, (0.00)	1, (1.12)	0, (0.00)	0, (0.00)	1, (1.43)	0, (0.00)	0, (0.00)	0, (0.00)	0, (0.00)	0, (0.00)
	Green spaces	1, (3.70)	1, (3.70)	0, (0.00)	1, (1.12)	3, (3.37)	0, (0.00)	0, (0.00)	3, (4.29)	0, (0.00)	0, (0.00)	2, (4.88)	0, (0.00)
	Aesthetic home-schools routes	0, (0.00)	1, (3.70)	0, (0.00)	0, (0.00)	4, (4.49)	0, (0.00)	1, (1.43)	2, (2.86)	1, (1.43)	1, (2.44)	2, (4.88)	1, 2.44)
	((In)formal) community surveillance	2, (7.41)	0, (0.00)	0, (0.00)	3, (3.37)	0, (0.00)	0, (0.00)	0, (0.00)	0, (0.00)	0, (0.00)	0, (0.00)	0, (0.00)	0, (0.00)

Community Barriers

Long distances to school – often associated with urban encroachment and/or rural isolation – emerged as the most frequently reported deterrent to AST, described in 47 papers (47,55,56,60,62,65,67,71,73,74,77,79,81,86,87,90,94–96,98,100,101,103,109,111,116–118,121,123–125,129,137–150). These papers reported greater distances were negatively associated with both walking and cycling to-and-from school across school levels (Kindergarten: n=6, 22%; Elementary: n=34, 89%; Middle: n=28, 70%; High: n=17, 39%) and income-strata (HICs: n=36, 46%; UMICs: n=7, 23%; LMICs: n=4, 80%), but particularly in LMICs, where school sitting was commonly considered inadequate.

Unsafe or inadequate pedestrian and cycling infrastructure constrained both the perceived safety and appeal of active commuting, noted in 30 papers

(67,68,73,76,79,81,87,96–98,100,109,112,115,116,118,119,132,135,139,140,143,145,148,150–155). Common deterrents included stray animals, poor lighting, narrow or damaged sidewalks, hilly landscapes, insufficient or inadequate signage, a lack of (safe) cycling lanes, a high number of intersections, and perceptions of air pollution linked to industrial land-use. 'Road obstacles' such as roundabouts and crosswalks were additionally described, referenced as "problematic" by students in 3 papers (67,140,155).

Perceived and environmental risks further deterred AST in a large subset of papers. Traffic-related hazards characterised by high speeds, large volumes, and dangers of collision were reported in 31 papers, often intensifying interpersonal concerns such as parental fears of children's safety (47,55,62,67,70,74,80,87,88,91,98–100,102,111,113–115,117–119,121,122,131–133,137,140,142,153,156). Crime and perceptions of insecurity were reported in 16 papers, particularly affecting younger children (47,53,55,67,70,91,102,108,111,114,116–118,122,137,157); and adverse weather conditions marked by excessive rain, heat, or snow were identified as barriers in 8 more (70,74,108,115–117,150,158).

The socio-economic context of school sitings also influenced students' travel behaviours. Five papers found that attending schools situated in higher SES areas was associated with a lower likelihood of AST, independent of household SES (65,75,90,103,147). Similarly, students enrolled in private schools tended to travel less by active means when compared to their public or governmental school counterparts (101,159). In both cases, these behaviours reflected social and communal stigmas relating to social roles and identities.

Community Facilitators

While residential density was identified as a barrier in 3 papers (85,109,137), it was also recognised as a facilitator in 2 others (83,160). Nine papers reported comparatively higher rates of AST where neighbourhoods and home-school routes featured higher intersection densities, on-street parking, recreational amenities, food outlets, and continuous, taller buildings with small setbacks (55,87,130,137,139,151,153,160,161). These denser, mixed-use environments (such as those offered by city centres and central business districts) were reported to alleviate some level of interpersonal parental concern and were positively associated with walking but not cycling both to-and-from school.

Improved accessibility and pedestrian infrastructure were also linked to a higher likelihood of AST, cited in 17 studies (76,77,84,87,94,95,97,109,117–119,133,137,144,146,155,161). These facilitators were evident across all school-levels (Kindergarten: n=3, 11%; Elementary: n=13, 15%; Middle: n=12, 17%; High: n=10, 24%) and income-strata (HICs: n=13, 17%; UMICs: n=4, 13%; LMICs: n=2, 40%). Neighbourhood walkability was also reported in 9 papers, marked by high street connectivity and low traffic volumes (67,73,74,100,124,125,134,152,156). Other common features included grid-like street patterns (67,83), access to green spaces, trees, and shaded paths, or other aesthetically pleasing home-school routes (53,87,98,153,155,162). Children residing in these environments enjoyed fewer structural barriers and realities more conducive to active means of travel.

Three papers cited community cohesion as an additional facilitator of AST, wherein trans-neighbourhood trust and informal surveillance enriched perceptions of safety, thereby increasing parental willingness to allow independent travel (86,98,150).

3.3.5 Public Policy Factors

Fifth, 6 papers (5% of included papers) addressed public policy-level influences affecting children's participation in AST. These related to public policies, regulation, and guidelines (Table 8).

Table 8: *Map of the Identified Public Policy-Level Barriers and Facilitators to Children's Active School Transport by US-Equivalent School-Level and World Bank Income Strata. Proportions are relative to the total number of papers pertinent to each school-level. Proportions pertinent to all public policy factors should be interpreted with caution due to small numbers of included papers. Homogenisation in school-levels resulted in age-range overlap, so the sum of each integer does not equate the true number of papers assessing HICs. Full statistics are presented in the table; Quantitative values are detailed in-text only for the most frequently reported barrier and facilitator to illustrate the link.*

		School-Level n, (% of Level)											
		Kindergarten			Elementary			Middle			High		
		HICs	UMICs	LIMCs	HICs	UMICs	LMICs	HICs	UMICs	LMICs	HICs	UMICs	LMICs
Facilitators	AST-related policies	0, (0.00)	0, (0.00)	0, (0.00)	1, (1.12)	0, (0.00)	0, (0.00)	2, (2.86)	0, (0.00)	0, (0.00)	0, (0.00)	0, (0.00)	0, (0.00)
	Tight school bus regulations	0, (0.00)	0, (0.00)	0, (0.00)	1, (1.12)	0, (0.00)	0, (0.00)	0, (0.00)	0, (0.00)	0, (0.00)	0, (0.00)	0, (0.00)	0, (0.00)
	Tight screentime guidelines	0, (0.00)	0, (0.00)	0, (0.00)	1, (1.12)	0, (0.00)	0, (0.00)	2, (2.86)	0, (0.00)	0, (0.00)	1, (2.44)	0, (0.00)	0, (0.00)

Public Policy Facilitators

No public policy barriers were identified among the papers included in review. However, a few public policy-level factors were identified to play a facilitating role. Three papers reported that the presence of AST-related public policies encouraged elementary (n=1, 1%) and middle-school populations (n=2, 3%) in HICs (n=3, 4%) to engage in AST by providing formalised guidance and institutional support for AST-related initiatives (69,82,163).

Another paper referenced how reductions in school bus services prompted children of a similar demographic towards active commuting through paternalist-libertarian nudge-based principles (94).

Likewise, national and regional screen-time guidelines were associated with reduced sedentary behaviours and increased MVPAs in 2 papers (62,79).

3.3.6 Overall Trends in SEM Factors by School-Level and Income-Strata

Overall, proportions of the identified SEM barriers and facilitators were observed to have varied by school-level and income-strata. However, it should be noted that proportions must be interpreted with caution due to small sample sizes. 50% of included papers identified at least one intrapersonal-level influence on children's AST-behaviours. 41% identified ≥ 1 at the interpersonal-level; 15% identified ≥ 1 at the institutional-level; 74% identified ≥ 1 at the community-level; and 5% identified ≥ 1 at the public policy-level.

Intrapersonal (n=8, 21%) and community-level barriers (n=22, 61%) were the most frequently reported deterrents to AST among kindergarten populations, relative to the total number of barriers across all levels of the SEM pertinent to that same school-level (Table 9). The same pattern was observed for facilitators, with a comparable trend extending across elementary through high-school populations. Markedly, public policy-level factors were the least frequently reported across school-levels.

Table 9: *Overall Trends in Socio-Ecological Model Factors by US-equivalent School-Level. Proportions are relative to the total number of papers pertinent to each sub-column (either barriers or facilitators) identified in each school-level. Proportions pertinent to institutional and public policy factors should be interpreted with caution due to small numbers of included papers.*

	School-Level n, (% of Level)							
	Kindergarten		**Elementary**		**Middle**		**High**	
	Barriers	Facilitators	Barriers	Facilitators	Barriers	Facilitators	Barriers	Facilitators
Intrapersonal	8, (20.51)	9, (25.00)	32, (19.39)	32, (27.59)	32, (21.19)	20, (24.39)	20, (20.20)	9, (19.57)
Interpersonal	5, (13.89)	2, (5.56)	26, (15.76)	15, (12.93)	19, (12.58)	16, (19.51)	11, (11.11)	7, (15.22)
Institutional	1, (2.78)	9, (25.00)	2, (1.21)	12, (10.344)	6, (3.97)	7, (8.54)	5, (5.05)	4, (8.70)
Community	22, (61.11)	16, (44.44)	105, (63.64)	54, (46.55)	94, (62.25)	35, (42.69)	63, (63.64)	25, (45.35)
Public Policy	0, (0.00)	0, (0.00)	0, (0.00)	3, (2.59)	0, (0.00)	4, (4.88)	0, (0.00)	1, (2.17)

Similar observations were reflected in the proportions of identified SEM-factors across income-strata (Table 10). Notably, no public policy-level barriers were identified.

Table 10: *Overall Trends in Socio-Ecological Model Factors by World Bank Income-Strata. Proportions are relative to the total number of papers pertinent to each sub-column (either barriers or facilitators) identified in each income-stratum. Proportions pertinent to LMICs and institutional and public policy factors should be interpreted with caution due to small numbers of included papers.*

	Income-Stratum n, (% of Level)					
	HICs		**UMICs**		**LMICs**	
	Barriers	Facilitators	Barriers	Facilitators	Barriers	Facilitators
Intrapersonal	24, (14.12)	30, (27.03)	16, (25.00)	12, (28.57)	4, (35.71)	0, (0.00)
Interpersonal	27, (15.88)	19, (17.12)	10, (15.63)	4, (9.52)	0, (0.00)	1, (20.00)
Institutional	6, (35.29)	16, (14.41)	0, (0.00)	3, (7.14)	0, (0.00)	0, (0.00)
Community	113, (66.47)	42, (37.84)	38, (59.38)	23, (54.76)	10, (71.42)	4, (80.00)
Public Policy	0, (0.00)	4, (3.60)	0, (0.00)	0, (0.00)	0, (0.00)	0, (0.00)

3.4 Mapping Intervention Functions and Policy Categories Using the COM-B

Application of the COM-B (Appendix B) and associated frameworks contextualised predictors of the negative health-behaviour (passive school transport) which emerged in review (Figure 5).

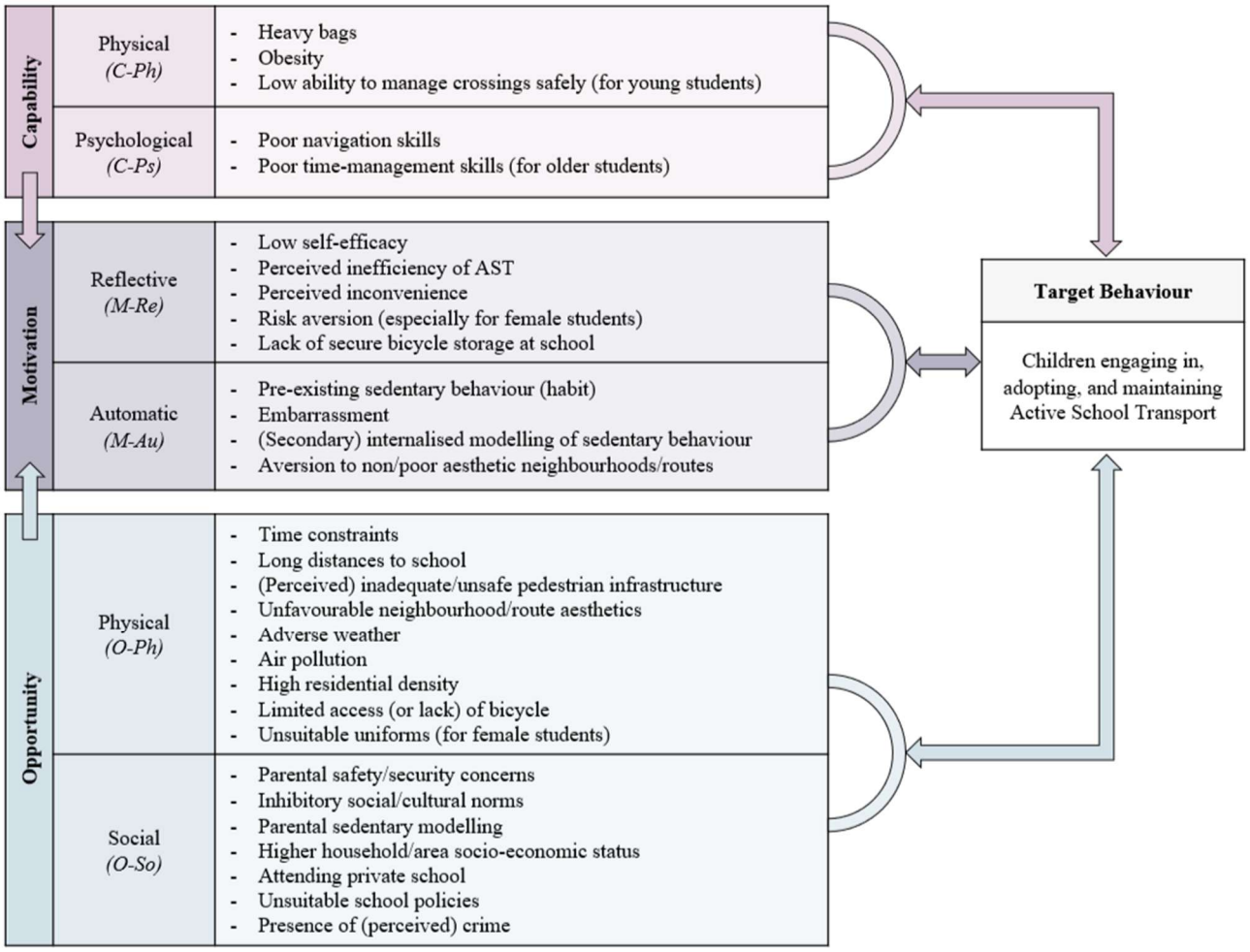


Figure 5: *Overview of the Barriers to Children's Adoption of Active School Transport by Capability, Opportunity, Motivation Behavioural Model Application. C-Ph is defined as physical skills, strength, stamina, or physical health to perform the behaviour; C-Ps as knowledge, cognitive skills, memory, reasoning, or decision-making processes; M-Re as evaluations, plans, intentions, or beliefs about what is good or possible; M-Au as emotional reactions, desires, impulses, habits, or reflex responses; O-Ph as environmental resources and affordances; and O-So as interpersonal influences, cultural norms, and social cues that shape/permit behaviour.*

Use of the TDF (Appendix E) enabled a more granular understanding of underlying determinants by identifying what factors that can change require positive change to promote uptake of the AST (Table 11). Application of the TDF identified skills, knowledge, mental processes, behavioural regulation, goals, beliefs about consequences and capabilities, optimism, social role and identity, emotions, reinforcement, environmental context and resources, and social influence as targetable domains through which to implement positive behavioural change.

Table 11: *Overview of the Factors that Require Change to Promote Uptake of Active School Transport by Theoretical Domains Framework Application. Skills are defined as ability or proficiency acquired through practice; Knowledge as awareness of the existence of a condition, task, or environment; Memory, attention and decision processes as ability to retain information, focus selectively, and choose between two or more alternatives; Behavioural regulation as anything aimed at managing/changing objectively measured actions; Goals as mental representations of outcomes an individual aims to achieve; Intentions as conscious decisions to perform a behaviour in a certain way; Beliefs about consequences as acceptance, truth, reality, or validity about outcomes of a behaviour; Beliefs about capabilities as acceptance of truth, reality, or validity about an individual's ability, talent, or facility that can be put to constructive use; Optimism as confidence that things will occur for the best; Social role and identity as a coherent set of behaviours and displayed personal qualities of an individual in a social setting; Emotions as complex, reactionary patterns involving experiential, behavioural, and psychological elements; Reinforcement as increasing probability of a response by arranging a dependent relationship or contingency; Environmental context and resources as circumstances of a person's situation or environment that discourages development of skills and abilities; and Social influences as interpersonal processes that cause an individual to alter their thoughts, feelings, or behaviours.*

COM-B Subdomain	TDF Domain	Relevance – Addressing Behavioural Barriers
C-Ph	*Skills*	Facilitate improved road-safety and navigational skills.
C-Ps	*Skills*	Facilitate improved time-management skills.
	Knowledge	Facilitate improved road-safety and navigational knowledge.
	Memory, Attention, and Decision Processes (mental processes)	Facilitate attention and decision processes pertaining to navigation.
		Encourage improved time-management decision processes.
	Behavioural Regulation	Address sedentary behaviours.
M-Re	*Goals*	Address capitalist goals regarding maximised efficiency and convenience.
	Intentions	Address intentions to avoid AST due to misconceptions and social/cultural/peer norms and stigmas.
	Beliefs about Consequences	Address negative social/cultural/peer or media narratives.
		Address anticipations of pain/discomfort stemming from school bag loads.
		Address parental and student concerns of potential dangers stemming from traffic/crime/pollution.
		Address (female student and parent) concerns of immodesty and embarrassment.
	Beliefs about Capabilities	Encourage improvement of *C-Ph* self-efficacy about ability to engage in AST and improving road safety and navigational skills and knowledge.
	Optimism	Address lack of optimism due to reflective processes stemming from low-self efficacies and social/cultural/peer/familial norms and stigmas
	Social/Professional Role and Identity	Address social norms relating to social identities in schools in higher SES areas, private schools, and higher-income households.
M-Au	*Emotions*	Address feelings of embarrassment and psycho-social pressures relating to identity.
		Address parental concerns of the safety of their children.
		Address lack of enjoyment of AST due to non/poor aesthetic neighbourhoods/routes.
	Reinforcement	Install incentives to participate in AST-related implementation initiatives.
	Optimism	Address lack of optimism due to automatic processes stemming from negative emotions and parental modelling.
	Social/Professional Role and Identity	None identified.
O-Ph	*Environmental Context and Resources*	Address long home-school distances, unsafe/inadequate pedestrian infrastructure, non/poor aesthetic neighbourhood/routes, proximal industrial land use, residential densities, traffic, crime, air pollution, and non-conducive school and public policies.

Table 11 *(Continued)*

O-So	*Social Influences*	Mitigate against pedestrian crowding, crime, and non-conducive school and public policies.
		Address social/cultural/peer/familial norms and stigmas.

3.4.2 Intervention Functions

Relevant COM-B and TDF domains were then mapped systematically to nine possible BCW intervention functions (Appendix F) (Table 12).

Table 12: *Overview of All Possible Behaviour Change Wheel Intervention Functions. Coloured blocks indicate all possible links between COM-B and intervention functions, while blocks marked 'x' represent contextually relevant links between TDF and intervention functions identified in review. Education is defined as increasing knowledge or understanding; Persuasion as using communication to induce positive or negative perceptions or feelings; Incentivisation as creating expectation(s) of reward(s); Coercion as creating expectation(s) of punishment or cost; Training as imparting skills; Restriction as using rules to reduce opportunity (or opportunities) to engage in the target behaviour; Environmental restructuring as changing physical or social context(s); Modelling as providing (an) example(s) for individuals to aspire to or imitate; and Enablement as increasing means or mitigating barriers to increase capability.*

COM-B Subdomain	**TDF Domain**	**BCW Intervention Functions**								
		Education	**Persuasion**	**Incentivisation**	**Coercion**	**Training**	**Restriction**	**Environmental Restructuring**	**Modelling**	**Enablement**
C-Ph	Skills					x				x
C-Ps	Skills	x				x				
	Knowledge	x				x				
	Mental processes									x
	Behavioural regulation	x								
M-Re	Goals			x	x					
	Intentions		x	x	x					
	Beliefs about consequences	x	x	x	x					
	Beliefs about capabilities	x	x							
	Optimism	x	x	x	x					
	Social identity			x						
M-Au	Emotions			x				x	x	
	Reinforcement			x				x		
	Optimism			x				x	x	x
	Social role and identity									
O-Ph	Environmental context						x	x		x
O-So	Social influences						x	x		x

Education, incentivisation, environmental restructuring, and enablement were identified as the functions that can contribute the most positive change towards the target-behaviour. Others – namely persuasion and coercion – warranted further assessment through APEASE criteria, which considered acceptability, practicability, effectiveness, affordability, side-effects, and equity dimensions of policy implementation (Table 13).

Table 13: *Assessment of Intervention Functions by Acceptability, Practicability, Effectiveness, Affordability, Side-Effects, and Equity Criteria. Assessment was contextualised by findings from the review.*

Intervention Function	APEASE Criteria in the Context of AST Behaviour
Persuasion	Affordable and practicable; can shift parental perceptions of safety and social norms when delivered by trusted local figures and peers; effectiveness is contingent on credible safety assurances and visible uptake, however; would work best in-tandem with enablement or environmental change rather than in isolation.
Coercion	Costly to enforce; not acceptable to stakeholders, practicable, or effective in the context of promoting participation in AST; may increase disparity between advantaged and disadvantaged individuals.

Given the considerations outlined above, persuasion was additionally noted, though primarily as an adjunct.

3.4.3 Policy Categories

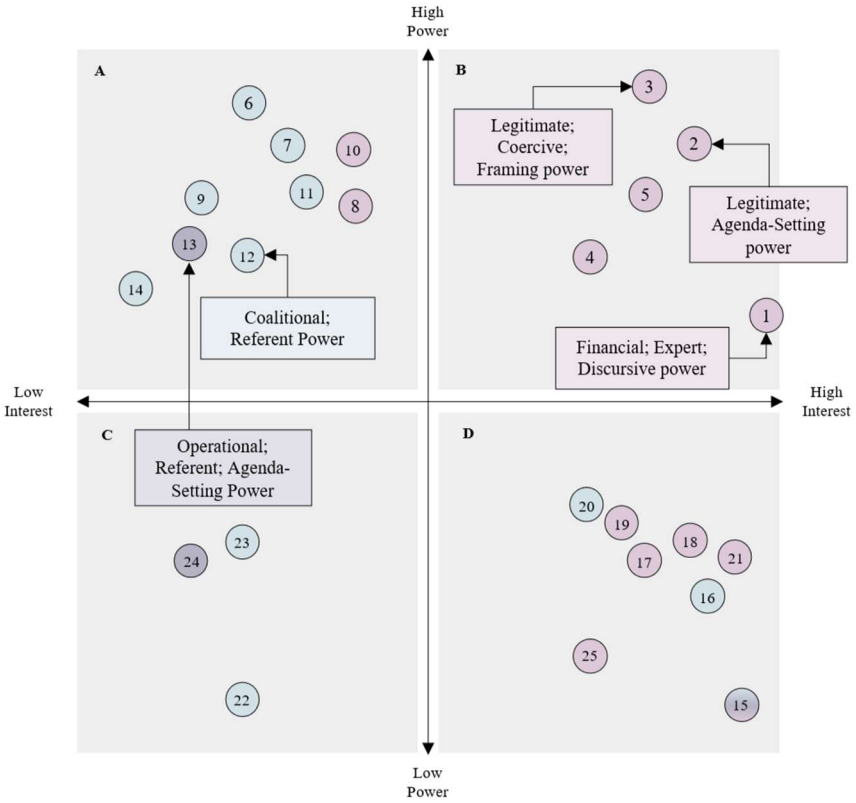
The BCW was then used to match each of the five recognised intervention functions to seven possible policy categories, which were also assessed through the APEASE criteria (Table 15). Application of the BCW identified environmental/social planning, service provision, and regulation as primary policy categories through which to implement behavioural change.

Table 14: *Overview of All Possible Behaviour Change Wheel Policy Interventions. Coloured blocks indicate all possible links between intervention functions and policy categories, while blocks marked 'x' indicate contextually relevant links between the two. Communication/marketing is defined as disseminating information, portrayals, or campaigns (sometimes allowing for interaction) to influence behaviour; Guidelines as creation and dissemination of recommendations or instructions advising on specific behaviours; Fiscal measures as employing financial (dis)incentives to influence behaviour; Regulation as creation and application of rules with sanctions for non-compliance which may be implemented by authorities without legislative power; Legislation as laws; Environmental or social planning as formal planning and designing of physical or social environments supportive or discouraging of certain behaviours: and Service provision as providing resources or services supportive of behavioural change.*

Policy Categories	**Intervention Functions**					**Assessment of whether the policy category meets the APEASE criteria in the context of promoting Active School Transport in Children**
	Education	**Persuasion**	**Incentivisation**	**Environmental Restructuring**	**Enablement**	
Communication / Marketing	x	x	x			Trusted and highly localised campaigns that aim to normalise AST and identify safety and time concerns (and stigmas) that require addressment are acceptable, low-cost, and scalable; especially effective in-tandem with visible changes in environmental/social planning.
Guidelines	x	x	x	x		Effective if well-enforced; strong standards for 'school-safe' routes, drop-off zones, and AST-promotions are practicable and affordable at the community and institutional levels; can mitigate against inequities in-tandem with monitoring/evaluation and support.
Fiscal			x	x	x	Targeted subsidises (e.g. towards bicycles and helmets, cycling training, repair, etc.) are acceptable, equitable, and cost-effective; punitive charges (such as parking fines) risk low acceptability and may marginalise families with safety/access constraints.
Regulation		x	x	x	x	Can be highly effective and safety-enhancing (e.g. via timed-traffic restrictions, speed limits, and parking regulations round schools) but require enforcement capacity and exemptions to temper acceptability and equity.
Legislation	x	x	x	x	x	Can enforce large-scale change but have long lead-times and significant political and administrative costs.
Environmental / Social planning				x	x	Can deliver significant safety and visibility gains (e.g. via restructuring (or implementation) of sidewalks, crossings, bicycle lanes, school-street designs etc. to mitigate against capability and opportunity barriers); can be costly, but benefits are substantial.
Service provision	x	x	x		x	Can deliver immediate reassurance to parents (e.g. via implementation of crossing guards or police, AST-related initiative); acceptable and practicable; high benefit per unit currency when targeted to marginalised populations.

Ergo, key stakeholders can employ education, persuasion, incentivisation, environmental restructuring, and enablement through planning, service provision, and regulation as primary policy categories through which to promote uptake of AST-behaviours. Communication/marketing can serve

as effective, low-cost amplifiers; fiscal measures as conditional amplifiers; and legislation as long-horizon enablers. Given its lack of both legislative and regulatory power, UNICEF's capabilities are limited to issuing guidelines that serve as enabling scaffolds, fostering consistency and accountability but limited in capacity to effect behavioural change without support of additional policy mechanisms. Effective stakeholder management is therefore essential, providing clarity on the positions of those who support, are neutral towards, or oppose the implementation of interventions (Figure 6).



1. UNICEF and United Nation (UN) agencies (e.g. WHO).
2. Ministries of Education.
3. City Mayors and Transport/Urban Planners.
4. School Boards or Headteachers.
5. Road Safety Councils or Agencies.
6. Ministries of Finance or Treasuries.
7. National Road Agencies.
8. Police and Traffic Enforcement.
9. Public Works and Procurement.
10. Development Banks / Donors.
11. Public Transport Operators.
12. Teachers' Unions.
13. Informal Transport Unions.
14. Building Developers.
15. Students.
16. Parents or Guardians.
17. School Councils.
18. Local Non-Governmental Organisations (NGOs) and Advocacy Groups.
19. Public Health Teams.
20. Community Leaders.
21. School Travel Plan Co-ordinators.
22. Local Businesses on home-school route.
23. Media and Influencers.
24. Taxicab Operators.
25. Bicycle / Active Transport Equipment Vendors.

Stakeholder positions are not stagnant.

Figure 6: *Overview of Stakeholder Positions by Interest and Power. Pink circles indicate stakeholders in support of promotion of AST-behaviours, while blue denotates those against. Blue is indicative of those who are neutral. Stakeholders in block A should have their needs anticipated and met; those in B should be managed most thoroughly; those in C should be contacted minimally but regularly; and those in D should be kept completely informed.*

Section 4: Discussion

In line with objectives (i) to (iv), a systematic search on peer-reviewed and grey literature was performed. Identified barriers and facilitators were mapped within the SEM, and determinant patterns were compared and described across homogenised school-levels and World Bank income-strata. Finally, all actionable implementation categories and policy mechanisms through which to implement positive behavioural change were identified. In line with objective (v), this section is divided into five main sub-sections: 4.1 summarises the review's key findings; 4.2 discusses methodological considerations of included papers; 4.3 outlines strengths and limitations of this work; 4.4 makes suggestions for future directions; and 4.5 points to key implications for policy and practice.

4.1 Key Findings

4.1.1 Characteristics of Included Papers

Paper Designs

Of the 121 included papers, 85% were analytic in orientation, most employing parent or child surveys or questionnaires as at least one method of data collection, paired with regression-based analytical approaches. Of these, 87% were cross-sectional by design, 5% were systematic reviews, 4% were longitudinal, 2% were case-studies, 1% were case-control studies, and 1% were book chapters. The remaining 15% of included papers were descriptive in orientation, wherein 28% were cross-sectional, 22% were descriptive exploratory studies, 17% were book chapters, 17% were narrative reviews, 11% were case-studies, and 6% were implementation evaluations.

Geographies and School-Levels

Geographically, the majority of included papers examined populations from HICs (65% of included papers) and UMICs (25%), predominantly within the WHO regions of AMRO (31%), WPRO (28%), and EURO (26%). Notably, no papers that investigated populations from LICs were eligible for review. Among the included papers, most focused on children of both sexes aged between 6 and 14, attending school-levels equivalent to elementary (74%) and middle-school (58%) in the US.

4.1.2 Barriers and Facilitators to AST

Barriers and facilitators identified in the review varied by school-level and income-strata, summarised in Tables 15 and 16 below. The SEM elucidated how influences interacted across multiple levels to shape behaviour, while the COM-B illustrated how barriers in capability and opportunity may have constrained motivations.

Table 15: *Overview of the Identified Barriers and Facilitators to Active School Transport by US-Equivalent School-Level. Coloured blocks correspond to Tables 4 to 8 in section 3.3 such that red is indicative of intrapersonal; orange is indicative or interpersonal; yellow is indicative of institutional; green is indicative of community; and blue is indicative of public policy-level influences.*

		Barriers	Facilitators
Factors of Influence by School-Level	**Kindergarten (K)**	None identified.	None identified.
	Elementary (E)	Infrastructural deterrents; Poor neighbourhood aesthetic.	((In)formal) community surveillance and inter-neighbourhood trust.
			Tight school bus guidelines.
	Middle (M)	None identified.	None identified.
	High (H)	None identified.	Presence of recreational amenities along the route.
	K+E	None identified.	Perceived opportunity for independence.
			Organisational incentives; Conducive school schedules.
	E+M	None identified.	Perceived opportunity for exercise.
			Presence of (an) older sibling(s).
			Grid-like street patterns.
			AST-related public policies.
	M+H	Heavy bags.	None identified.
		Non-conducive school uniforms (female students).	
	K+E+M	None identified.	Perceived enjoyment.
	E+M+H	Low self-efficacy; Limited access to bicycles.	Presence of trustworthy individuals such as crossing guards or police along the route.
		Parental commutes to and from work; Parental accompaniment.	
		Lack of secure bicycle storage.	Tight screentime guidelines.
		Attending schools in higher SES area; Attending private school; Living in more urbanised areas; Residential density.	
	K+E+M+H	Higher household income SES; Time constraints; Pre-existing sedentary behaviour; Obesity; Sex (female).	Perceived safety, comfort, and convenience; Lower household income and SES; Perceived health benefit; Older age.
		Parental safety and security concerns; Non-conducive parental work schedules; Modelling of parental non-active behaviour.	Parental support; Parental accompaniment; Peer support and social interaction.
		Non-conducive school policies.	Institutional AST-related programs; Supportive school and staff engagement.
		Long home-school distance; Inadequate pedestrian infrastructure; Perceptions of air pollution (and proximity to industrial land-use); (Perceived) traffic hazards; (Perceived) risk of crime; Adverse weather; Social and cultural norms.	Residential density; Land-use mix diversity; Close proximity to business centres; Adequate pedestrian infrastructure; Neighbourhood walkability; Aesthetic home-school routes.
	K+M	Heavy bags.	None identified.
	E+H	None identified.	None identified.
	K+H	None identified.	None identified.

Table 16: *Overview of the Identified Barriers and Facilitators to Active School Transport by World Bank Income-Strata. Coloured blocks correspond to Tables 4 to 8 in section 3.3 such that red is indicative of intrapersonal; orange is indicative or interpersonal; yellow is indicative of institutional; green is indicative of community; and blue is indicative of public policy-level influences.*

		Barriers	Facilitators
Factors of Influence by Income-Strata	**HICs (H)**	Heavy bags; Limited access to bicycles.	Perceived opportunity for independence; Older age.
		Non-conducive parental work schedules; Parental accompaniment.	Presence of (an) older sibling(s).
		Non-conducive school uniforms (female students); Lack of secure bicycle storage; Non-conducive school policies.	Supportive school and staff engagement; Organisational incentives; Conducive school schedules; Presence of trustworthy individuals such as crossing guards or police along the route.
		Social and cultural norms.	Residential density; Presence of recreational amenities along the route; Close proximity to business centres; Grid-like street patterns; ((In)formal) community surveillance.
			AST-related public policies; Tight school bus regulations; Tight screentime guidelines.
	UMICs (U)	Poor neighbourhood aesthetics.	None identified.
	LMICs (L)	None identified.	None identified.
	H+U	Time constraints; Pre-existing sedentary behaviour.	Perceived safety, comfort, and convenience; Lower household income and SES; Perceived health benefit; Perceived opportunity for exercise; Perceived enjoyment.
		Parental safety and security concerns; Parental commutes to and from work; Modelling of parental non-active behaviour.	Parental accompaniment; Peer support and social interaction.
		Inadequate pedestrian infrastructure; Infrastructural deterrents; Adverse weather; Living in more urbanized areas.	Institutional AST-related programs.
			Neighborhood walkability; Green spaces.
	U+L	Attending private school.	None identified.
	H+U+L	Higher household income SES; Low self-efficacy; Sex (female).	Parental support.
		Long home-school distance; Perceptions of air pollution (and proximity to industrial land-use); (Perceived) traffic hazards; (Perceived) risk of crime; Residential density.	Land-use mix diversity; Adequate pedestrian infrastructure; Aesthetic home-school routes.
	H+L	Obesity.	None identified.
		Attending schools in higher SES areas.	

Across included papers, barriers to AST were more commonly assessed than facilitators. Yet, both barriers and facilitators were most frequently identified at the community-environmental level (74% of included papers), followed by the intrapersonal-level (50%), with public policy influences (5%)

emerging as the least frequently cited across school-levels and income-strata. Notably, each level of influence both shaped and was shaped by the others: intrapersonal behaviours were affected by interpersonal relationships, which themselves were shaped by upstream institutional and community contexts; institutions – through their effect on social networks and the environments wherein people interact – were influenced by broader community norms and public policy factors; and community norms, in turn, both influenced and were influenced by individual behaviours and institutional practices.

4.1.3 Consistency with Broader Literature

Consistency with Evidence

These findings align well with existing literature. Several literature and systematic reviews consistently emphasise the predominant influence of community-level factors such as the built environment, social cohesion, and neighbourhood walkability on children's AST and physical mobility across HICs, UMICs, and LMICs (164–166). Persistently reported behavioural determinants include long home-school distances, quality and acceptability of pedestrian infrastructure, and traffic safety (42), with personal perceptions, motivations, and attitudes playing critical roles at the intrapersonal-level (167). As one example, a major systematic review by Aranda-Balboa *et al.* (2019) mapped parental barriers chiefly to safety concerns and environmental features (1,164). The findings also reinforce a fundamental principle of the SEM, which proposes that while multiple layers shape behaviour, the most salient barriers and facilitators for children and their families are those closest to their lived experience, likely because they are often considered more tangible, relevant, and actionable (1,168).

While policy-level influences are indeed reported, they are infrequently documented and under-evaluated, particularly in LMIC contexts, where both research and interventions tend to focus more on proximal, modifiable influences (1,42). Such 'macro' factors are more commonly mentioned as recommendations, rather than systematically explored determinants with influence over AST-behaviours (42). Likewise, the disproportionate lack of evidence investigating LICs is also echoed in large-scale reviews, with the global literature predominantly focused on higher-income settings (164).

Moreover, few papers include intersectional factors such as sex, and no papers adopted structural equation modelling to establish causality, or established frameworks such as the SEM or COM-B (1,29,42–44).

4.2 Methodological Considerations of Included Papers

4.2.1 Internal and External Validity

The analysis phase of this scoping review raised several important methodological considerations. All included studies employed observational designs, precluding causal inferences between AST and

any reported behavioural influences. While the studies demonstrated reasonably good external validity (i.e. statistical generalisability), internal validity was likely limited by measurement error. Possibly in mitigation against the risk of random error, most studies recruited sample sizes exceeding 300 participants – a threshold widely regarded as robust for the purposes of behavioural research and trend analyses (169) – thereby reducing variability while enhancing statistical power. Violations in internal validity, however, could not be minimised by such means.

Bias

Potential sources of non-response bias may have led to samples that over-represented children experiencing particular barriers in studies that utilised interviews. By contrast, Neyman's bias may have been introduced by the under-representation of children who experienced said barriers, distorting the true relationship between those barriers and AST. Few studies attempted to limit these biases by use of nation-wide surveys or databases to create sample populations representative of respective countries. Potential for information bias, both non-differential and differential, posed additional concern. The prospective nature of the parent/child interviews may have meant many interviewers were not blinded to participant status, introducing risk of interviewer bias. Though the use of standardised questionnaires may have mitigated against some level of differential bias, reliance on retrospective self-reported data created potential for recall bias which may have been compounded by social-desirability bias. Moreover, the reporting of children's lived experiences by guardians instead of the children themselves may have contributed further to differential misclassification.

Confounding

In analytic studies, internal validity was also threatened by inadequate adjustment for confounders. It was often unclear whether crude and adjusted estimates (such as ORs) were compared to assess the presence of confounding, or the lack thereof. Many papers did not report sufficiently on SES, and as such could have benefitted from more comprehensive data on lifestyle factors to further reduce residual confounding and strengthen validity. Risks of over-adjustment (e.g. via controlling for parental permission) and collider bias (via conditioning on AST-programme participation, which itself depends on neighbourhood risk) may have further attenuated or even reversed associations.

Because the majority of analytic studies were cross-sectional, exposures and outcomes were assessed simultaneously. While this allowed for the elucidation of associations between reported influences and AST-behaviours, it precluded causal inference due to the lack of temporality. Hence, it cannot be determined whether the barriers and facilitators identified in these studies truly influence AST in children or merely co-occurred.

4.2.2 Effect Modification

In addition, few studies explored effect-modification, meaning important heterogeneity by sex, SES, or cultural norms was likely overlooked.

4.2.3 Implications for this Review

Because the purpose of this review was to map breadth rather than estimate pooled effects, the behavioural influences identified in this work should therefore be interpreted as *reported* barriers and facilitators, rather than empirically proven causal factors.

4.3 Methodological Considerations of Review

4.3.1 Strengths

The rigour of this review was maintained through use of a comprehensive search strategy, strict eligibility criteria, and the involvement of at least two independent reviewers, ensuring descriptive validity (i.e. accuracy of data) despite the broad scope of the research question. Interpretative validity may be limited, as I alone conducted the data charting phase; however, this risk was mitigated against through use of a clear, structured charting framework described in Appendix G. In order to strengthen theoretical validity (reliability of data interpretation), thematic findings were discussed at each stage of synthesis by two reviewers (myself and PA), and disagreements were resolved when consensus was reached. Pragmatic validity (applicability and transferability of findings) was enhanced by transparent reporting of all charted items, providing sufficient contextual detail to enable readers to judge the relevance of the findings for themselves. While a formal and critical appraisal of the individual sources of evidence was not pursued, a general appraisal was also included.

4.3.2 Limitations

Design

Despite these efforts, however, this review is not without limitations. The use of a convergent-integrated approach traded some nuance for a more unified, direct combination of quantitative, qualitative, and mixed-method evidence. By integrating findings across methodologies into a single synthesis, the review assumed a degree of commensurability that may not be warranted, potentially dulling context-specific insights, minority perspectives, and contradictory findings. Similarly, though it provided useful structure, the coding frame inevitably foregrounded more frequently reported or easily qualitizable narratives at the expense of more localised, context-specific narratives.

Use of an alternative results-based convergent synthesis might have preserved the methodological and conceptual distinctness of included papers by first analysing quantitative, qualitative, and mixed-method lines separately, before integrating them at a deeper interpretive level via joint displays and meta-inferences; synthesising charted results by this approach could have yielded contextually richer insights across the three data forms. Credibility and transparency might also have benefitted from

sensitivity analyses by study-type, a more explicit mapping (or otherwise detailing) of areas of contradictory findings, and stricter reporting that distinguished between data sources and analytic decisions. While both time constraints and the purpose of review informed my methodological choices, the cost of this complexity was consequential for both interpretation and use.

Internal and External Validity

External validity is also limited. Though global in both intent and purpose, the skew in literature towards urban, HIC settings and school-specific populations has rendered rural and non-school-going children central to UNICEF's work effectively overlooked. Publication bias likely played a notable role, given studies reporting favourable findings or those from the Global North are more likely to become available in online databases, while studies from the Global South or those with unfavourable results are less likely to be published. Language bias only exacerbated this issue, given that studies published in English are more likely to be made available in high-impact journals, and this review included English-only papers. Because the exclusion of non-English papers systematically omitted evidence that would have altered this review's findings, internal validity is also threatened. Withal, these factors compromise the generalisability of the findings, and may at least partially explain the apparent lack of barriers and facilitators reported in lower-income countries.

4.4 Future Directions

The considerations outlined in sections 1.2 and 4.1-4.4 therefore point to seven overarching deficits in the evidence base, which will require further research:

(i) Investigating intersectional factors such as sex, disability, and cultural norms, which are currently under-explored.
(ii) Investigating (and increasing attention to) lower-income contexts (particularly LMICs, LICs, and other marginalised populations), who are currently under-represented.
(iii) Investigating younger (US-equivalent kindergarteners (aged 5-6)) and older (US-equivalent high-school students (~14-19)), who are comparatively under-studied.
(iv) Investigating facilitators of AST, which are comparatively less explored.
(v) Integrating investigation and analyses of multiple SEM domains, to better capture their interactions.
(vi) Utilising robust analytic longitudinal designs, to clarify causal pathways.
(vii) And evaluating multi-level interventions that simultaneously address community and intrapersonal-level influences.

4.5 Implications for Policy and Practice

Findings from this review suggest that the most salient barriers and facilitators to AST – such as those operating at the community and interpersonal-levels – are best addressed through education,

persuasion, incentivisation, environmental restructuring, and/or enablement, supported primarily by environmental and social planning, service provision, and/or regulation. While it is important to outline broad principles that can guide policy and practice, context-specific adaptation remains essential. In HICs, implementation may need to prioritise persuasion and incentivisation to mitigate against longstanding car-dependencies, whereas in lower-income countries, emphases may lie on regulation, enforcement, and infrastructure provision to improve both perceived and objective safety.

For UNICEF, the findings highlight the importance of framing AST as both a health and equity issue, pushing for cross-sectoral governance and accountability, and providing a critical foundation of evidence-based guidance that can be adapted to local cultural, socio-economic, and environmental realities worldwide (Table 17).

Table 17: *Overview of Suggested Directions for Active School Transport (AST)-Promotion and Implications for UNICEF Guidance. Real-world examples are indicative of successful initiatives utilizing the intervention functions and policy categories identified in review.*

Principle	Policy Categories	Real-World Examples	Implications for UNICEF
Create safe and supportive environments.	Service provision; Regulation; Environmental restructuring.	Vision Zero (Sweeden) (170)	Advocate for infrastructure as a children's rights priority; Advocate for integration of AST into urban planning and transport regulation.
Bolster parental and community confidence.	Education; Persuasion; Communication/marketing.	Ciclovía (Bogotá, Columbia) (171)	Support campaigns to address parental safety concerns and cultural norms.
Leverage schools as points for change.	Guidelines; Enablement; Incentivisation.	The Walking School Bus (New Zealand) (172)	Position schools as platforms for scalable, community or institutional-led interventions.
Integrate policies together.	Regulation; Fiscal; Communication/marketing; Legislation.	Cycling education and bicycle subsidies (Netherlands) (173)	Promote multi-sectoral policy packages that can be adapted to local contexts.
Promote equity and local adaptation.	Service provision; Regulation; Guidelines.	Shova Kalula Bicycle Project (South Africa) (174)	Provide flexible options to ensure interventions are both context-specific and equitable.

Section 5: Conclusion

This review aimed to identify and synthesise the documented barriers and facilitators to AST among school-going children (aged 5-19) in order to inform UNICEF's forthcoming global guidance on active mobility. A scoping review was well suited to this purpose because it enabled the provision of broad, comprehensive maps of the available evidence, key concepts, and research gaps across the literature. Descriptive and exploratory in nature, this work transparently reported concepts, catalogued the types and extent of existing research, and discussed inadequacies – especially around AST where the evidence base is large and concepts are insufficiently defined.

Disclosure Statement

No potential conflict of interest was identified.

Funding

This review was not funded by any research or institutional body. As such, the views expressed within this work are solely those of myself.

References

1. Larouche R, Mammen G, Rowe DA, Faulkner G. Effectiveness of active school transport interventions: a systematic review and update. BMC Public Health. 2018 Dec 1;18(1):206.

2. Frohlich KL, Collins PA. Children's right to the city and their independent mobility: why it matters for public health. J Epidemiol Community Health (1978). 2024 Jan;78(1):66–8.

3. Bull FC, Al-Ansari SS, Biddle S, Borodulin K, Buman MP, Cardon G, et al. World Health Organization 2020 guidelines on physical activity and sedentary behaviour. Br J Sports Med. 2020 Dec;54(24):1451–62.

4. Morrison SA, Aubert S, Brazo-Sayavera J, Demchenko I, Reilly JJ, Tremblay MS. Celebrating a decade of the Active Healthy Kids Global Alliance and its Global Matrix initiative. Br J Sports Med. 2025 Feb;59(3):198–9.

5. Tremblay MS, Barnes JD, Demchenko I, Gonzalez SA, Brazo-Sayavera J, Kalinowski J, et al. Active Healthy Kids Global Alliance Global Matrix 4.0—A Resource for Physical Activity Researchers. J Phys Act Health. 2022 Nov 1;19(11):693–9.

6. MCDONALD N. Active Transportation to SchoolTrends Among U.S. Schoolchildren, 1969–2001. Am J Prev Med. 2007 Jun;32(6):509–16.

7. Garrard J. Walking, riding or driving to school: what influences parents' decision making? [Internet]. 2016 Sep [cited 2025 Jul 14]. Available from: https://www.dit.sa.gov.au/__data/assets/pdf_file/0004/513508/Walking_riding_or_driving_to_school-_what_influences_parents_decision_making-Literature_Review_.pdf

8. Haug E, Smith ORF, Bucksch J, Brindley C, Pavelka J, Hamrik Z, et al. 12-Year Trends in Active School Transport across Four European Countries-Findings from the Health Behaviour in School-Aged Children (HBSC) Study. Int J Environ Res Public Health. 2021 Feb 22;18(4).

9. Rothman L, Macpherson AK, Ross T, Buliung RN. The decline in active school transportation (AST): A systematic review of the factors related to AST and changes in school transport over time in North America. Prev Med (Baltim). 2018 Jun;111:314–22.

10. González SA, Aubert S, Barnes JD, Larouche R, Tremblay MS. Profiles of Active Transportation among Children and Adolescents in the Global Matrix 3.0 Initiative: A 49-Country Comparison. Int J Environ Res Public Health. 2020 Aug 18;17(16).

11. Ekelund U, Anderssen SA, Froberg K, Sardinha LB, Andersen LB, Brage S. Independent associations of physical activity and cardiorespiratory fitness with metabolic risk factors in children: the European youth heart study. Diabetologia. 2007 Aug 6;50(9):1832–40.

12. Mendoza JA, Watson K, Nguyen N, Cerin E, Baranowski T, Nicklas TA. Active Commuting to School and Association With Physical Activity and Adiposity Among US Youth. J Phys Act Health. 2011 May;8(4):488–95.

13. Korkutata A, Korkutata M, Lazarus M. The impact of exercise on sleep and sleep disorders. npj Biological Timing and Sleep. 2025 Feb 3;2(1):5.

14. Larouche R, Saunders TJ, John Faulkner GE, Colley R, Tremblay M. Associations Between Active School Transport and Physical Activity, Body Composition, and Cardiovascular Fitness: A Systematic Review of 68 Studies. J Phys Act Health. 2014 Jan;11(1):206–27.

15. Fyhri A, Hjorthol R, Mackett RL, Fotel TN, Kyttä M. Children's active travel and independent mobility in four countries: Development, social contributing trends and measures. Transp Policy (Oxf). 2011 Sep;18(5):703–10.

16. Mendoza JA, Cowan D, Liu Y. Predictors of children's active commuting to school: an observational evaluation in 5 U.S. communities. J Phys Act Health. 2014 May;11(4):729–33.

17. Alnawwar MA, Alraddadi MI, Algethmi RA, Salem GA, Salem MA, Alharbi AA. The Effect of Physical Activity on Sleep Quality and Sleep Disorder: A Systematic Review. Cureus. 2023 Aug 16;

18. Laverty AA, Hone T, Goodman A, Kelly Y, Millett C. Associations of active travel with adiposity among children and socioeconomic differentials: a longitudinal study. BMJ Open. 2021 Jan 12;11(1):e036041.

19. Road traffic injuries [Internet]. World Health Organization (WHO); 2023 [cited 2025 Jul 15]. Available from: https://www.who.int/news-room/fact-sheets/detail/road-traffic-injuries

20. Ritchie H. Cars, planes, trains: where do CO_2 emissions from transport come from? [Internet]. OurWorldinData; 2020 [cited 2025 Jul 14]. Available from: https://ourworldindata.org/co2-emissions-from-transport

21. Tiseo I. Distribution of carbon dioxide emissions produced by the transportation sector worldwide in 2024, by sub sector [Internet]. 2025 [cited 2025 Jul 14]. Available from: https://www.statista.com/statistics/1185535/transport-carbon-dioxide-emissions-breakdown/

22. Road injury prevention - Resources to support schools to promote safe active travel [Internet]. 2016 Mar [cited 2025 Jul 15]. Available from: https://assets.publishing.service.gov.uk/media/5a81a1a2e5274a2e8ab5507a/25_Feb16FINAL_DOCUMENT.pdf

23. United Nations (UN) Department of Economic and Social Affairs. Transforming our world: the 2030 Agenda for Sustainable Development [Internet]. 2015 [cited 2025 Jul 15]. Available from: https://docs.un.org/en/A/RES/70/1

24. The United Nations (UN) Depaertment of Economic and Social Affairs. Child Health Initiative: Safe and healthy journeys to school to reduce emissions and exposure to air pollution, enable active transport, and ensure safe roads [Internet]. [cited 2025 Jul 14]. Available from: https://sdgs.un.org/partnerships/child-health-initiative-safe-and-healthy-journeys-school-reduce-emissions-and-exposure

25. Global Action Plan on Physical Activity 2018-2030 [Internet]. 2018 [cited 2025 Jul 15]. Available from: https://iris.who.int/bitstream/handle/10665/272722/9789241514187-eng.pdf

26. Draisin N, Vincenten J. Guidance for safe and healthy journeys to school during the COVID-19 pandemic and beyond [Internet]. United Nations International Children's Fund (UNICEF); 2020 [cited 2025 Jul 15]. Available from: https://healtheducationresources.unesco.org/library/documents/guidance-safe-and-healthy-journeys-school-during-covid-19-pandemic-and-beyond

27. Goel R, Oyebode O, Foley L, Tatah L, Millett C, Woodcock J. Gender differences in active travel in major cities across the world. Transportation (Amst). 2023 Apr;50(2):733–49.

28. Katoch B, Ghosh I, Chandra S. Safety of children in school zones − A systematic review. Transp Res Part F Traffic Psychol Behav. 2025 Aug;113:554–69.

29. Walker A. Scottish Research Project (SRP) 9: Disabled Children's Travel to School [Internet]. SUSR2258. Sustrans and Transport Scotland Còmdhail Alba; 2024 [cited 2025 Jul 15]. Available from: https://www.sustrans.org.uk/media/arrp3ndg/srp9_disabled-childrens-travel-to-school.pdf

30. The United Nations Children's Emergency Fund. UNICEF mission statement [Internet]. [cited 2025 Jul 14]. Available from: https://www.unicef.org/about/mission-statement

31. Seamon D. Review of The Ecology of Human Development: Experiments by Nature and Design, by U. Bronfenbrenner. JSTOR [Internet]. 1980 [cited 2025 Jul 15];8(2):179–83. Available from: http://www.jstor.org/stable/4602552

32. Michie S, van Stralen MM, West R. The behaviour change wheel: A new method for characterising and designing behaviour change interventions. Implementation Science. 2011 Dec 23;6(1):42.

33. Smith M, Ikeda E, Hawley G, Mavoa S, Hosking J, Egli V, et al. An integrated conceptual model of environmental needs for New Zealand children's active travel to school. J Transp Health. 2020;16.

34. Costa J, Adamakis M, O'Brien W, Martins J. A Scoping Review of Children and Adolescents' Active Travel in Ireland. Int J Environ Res Public Health [Internet]. 2020;17(6). Available from: https://library-search.imperial.ac.uk/primo-

explore/openurl?vid=ICL_VU1&institution=44IMP&?sid=OVID&isbn=&issn=1660-4601&volume=17&issue=6&date=2020&title=International+Journal+of+Environmental+Research+%26+Public+Health+%5BElectronic+Resource%5D&atitle=A+Scoping+Review+of+Children+and+Adolescents%27+Active+Travel+in+Ireland.&aulast=Costa+J&spage=

35. Huang C, Memon AR, Yan J, Lin Y, Chen ST. The Associations of Active Travel to School With Physical Activity and Screen Time Among Adolescents: Do Individual and Parental Characteristics Matter? Front Public Health. 2021 Dec 15;9.

36. Maulida R, Goel R, Labib SM, Oni T, van Sluijs EMF. Environmental correlates of adolescent active travel to school in Asia: An ecological study. Health Place. 2023 May;81:103024.

37. Villa-González E, Barranco-Ruiz Y, Evenson KR, Chillón P. Systematic review of interventions for promoting active school transport. Prev Med (Baltim). 2018 Jun;111:115–34.

38. Lam HY, Jayasinghe S, Ahuja KDK, Hills AP. Active School Commuting in School Children: A Narrative Review of Current Evidence and Future Research Implications. Int J Environ Res Public Health. 2023 Oct 16;20(20):6929.

39. Schuch L, Curtis JW, Curtis A, Hudson C, Wuensch H, Sampsell M, et al. Breaking Out of Surveillance Silos: Integrative Geospatial Data Collection for Child Injury Risk and Active School Transport. Journal of Urban Health. 2016 Feb 14;93(1):36–52.

40. Savolainen E, Lindqvist AK, Mikaelsson K, Nyberg L, Rutberg S. Children's active school transportation: an international scoping review of psychosocial factors. Syst Rev. 2024 Jan 30;13(1):47.

41. UNICEF Strategic Plan 2022–2025 - Renewed Ambition Towards 2030 [Internet]. 2022 Jan [cited 2025 Jul 14]. Available from: https://www.unicef.org/reports/unicef-strategic-plan-2022-2025

42. Boland P, Nowland R, Tellis KD, Adams M, Westwood J, Crook D, et al. Barriers and Facilitators to Cycling to School for Children in the UK: A Systematic Review. Active Travel Studies. 2025 Feb 11;5(1).

43. Comeau É, Chan A, Buliung R, Lewis I, Ross T. Understanding the manifestations and impacts of ableism in school transportation: a scoping review. Transp Rev. 2025 Jul 4;45(4):514–36.

44. Galán A, Ruiz-Apilánez B, Macdonald E. Built environment and active transportation to school in the west: latest evidence and research methods. Discover Cities. 2024 Mar 27;1(1):3.

45. Gardner B, Thomopoulos N, Itani N, Yang YC. Identifying Indicators of School Travel Plan Effectiveness [Internet]. 2025 Jun [cited 2025 Jul 16]. Available from: https://www.surrey.ac.uk/sites/default/files/2025-06/UoS-Innovation-Hub-School-Travel-Plan-Report.pdf

46. Parish N, Bryant B. The future of Home to School Transport [Internet]. 2025 Jan [cited 2025 Jul 17]. Available from: https://www.local.gov.uk/sites/default/files/documents/Home%20to%20school%20transport%20report%20final%202025.pdf

47. Aranda-Balboa MJ, Huertas-Delgado FJ, Herrador-Colmenero M, Cardon G, Chillón P. Parental barriers to active transport to school: a systematic review. Int J Public Health. 2020;65(1):87–98.

48. Tricco AC, Lillie E, Zarin W, O'Brien KK, Colquhoun H, Levac D, et al. PRISMA Extension for Scoping Reviews (PRISMA-ScR): Checklist and Explanation. Ann Intern Med. 2018 Oct 2;169(7):467–73.

49. Aromataris E, Lockwood C, Porritt K, Pilla B, Jordan Z, editors. JBI Manual for Evidence Synthesis. JBI; 2024.

50. Bramer WM, De Jonge GB, Rethlefsen ML, Mast F, Kleijnen J. A systematic approach to searching: an efficient and complete method to develop literature searches. Journal of the Medical Library Association. 2018 Oct 4;106(4).

51. Education system in the USA [Internet]. Unipage; [cited 2025 Jul 19]. Available from: https://www.unipage.net/en/education_system_usa

52. Atkins L, Francis J, Islam R, O'Connor D, Patey A, Ivers N, et al. A guide to using the Theoretical Domains Framework of behaviour change to investigate implementation problems. Implementation Science. 2017 Dec 21;12(1):77.

53. Wang X, Liu Y, Yao Y, Zhou S, Zhu Q, Liu M, et al. Adolescents' environmental perceptions mediate associations between streetscape environments and active school travel. Transp Res D Transp Environ [Internet]. 2023;114. Available from: https://www.scopus.com/inward/record.uri?eid=2-s2.0-85143807591&doi=10.1016%2fj.trd.2022.103549&partnerID=40&md5=f1a692be6c85d5cd5fdd5708ce6ec23b

54. Brand C, Reuter CP, Dias AF, Mota J, Duncan M, Gaya AR, et al. Like Mother, like Son: Physical Activity, Commuting, and Associated Demographic Factors. Sustainability. 2020;12(14).

55. Bosch LSMM, Wells JCK, Lum S, Reid AM. Associations of the objective built environment along the route to school with children's modes of commuting: A multilevel modelling analysis (the SLIC study). PLoS One [Internet]. 2020;15(4):e0231478. Available from: https://library-search.imperial.ac.uk/primo-explore/openurl?vid=ICL_VU1&institution=44IMP&?sid=OVID&isbn=&issn=1932-6203&volume=15&issue=4&date=2020&title=PLoS+ONE+%5BElectronic+Resource%5D&atitle=Associations+of+the+objective+built+environment+along+the+route+to+school+with+children%27s+modes+of+commuting%3A+A+multilevel+modelling+analysis+%28the+SLIC+study%29.&aulast=Bosch+LSMM&spage=e0231478

56. Fallah Zavareh M, Abolhasannejad V, Mamdoohi A, Nordfjærn T. Barriers to children's walking to school in Iranian and Chinese samples. Transp Res Part F Traffic Psychol Behav [Internet]. 2020;73:399–414. Available from: https://www.scopus.com/inward/record.uri?eid=2-s2.0-85089109502&doi=10.1016%2fj.trf.2020.07.008&partnerID=40&md5=14e7c2d546238d14ddc72c2d4fe27a81

57. de Camargo EM, Santos MPM, Ribeiro AGP, Mota J, de Campos W. Interaction of sociodemographic factors in the association between psychosocial factors and active commuting to school. Cad Saude Publica. 2020;36(5).

58. Silva AA de P da, Fermino RC, Souza CA, Lima AV, Rodriguez-Anez CR, Reis RS. Socioeconomic status moderates the association between perceived environment and active commuting to school. Rev Saude Publica [Internet]. 2018;52:93. Available from: https://library-search.imperial.ac.uk/primo-explore/openurl?vid=ICL_VU1&institution=44IMP&?sid=OVID&isbn=&issn=0034-8910&volume=52&issue=&date=2018&title=Revista+de+Saude+Publica&atitle=Socioeconomic+status+moderates+the+association+between+perceived+environment+and+active+commuting+to+school.&aulast=Silva+AAPD&spage=93

59. Simons A, Koekemoer K, van Niekerk A, Govender R. Parental supervision and discomfort with children walking to school in low-income communities in Cape Town, South Africa. Traffic Inj Prev. 2018;19(4):391–8.

60. Yang Y, Hong X, Gurney JG, Wang Y. Active Travel to and From School Among School-Age Children During 1997-2011 and Associated Factors in China. J Phys Act Health [Internet]. 2017;14(9):684–91. Available from: https://library-search.imperial.ac.uk/primo-explore/openurl?vid=ICL_VU1&institution=44IMP&?sid=OVID&isbn=&issn=1543-3080&volume=14&issue=9&date=2017&title=Journal+of+Physical+Activity+%26+Health&atitle=Active+Travel+to+and+From+School+Among+School-Age+Children+During+1997-2011+and+Associated+Factors+in+China.&aulast=Yang+Y&spage=684

61. Lu W, McKyer ELJ, Lee C, Ory MG, Goodson P, Wang S. Children's active commuting to school: an interplay of self-efficacy, social economic disadvantage, and environmental characteristics. Int J Behav Nutr Phys Act [Internet]. 2015;12:29. Available from: https://library-search.imperial.ac.uk/primo-explore/openurl?vid=ICL_VU1&institution=44IMP&?sid=OVID&isbn=&issn=1479-5868&volume=12&issue=&date=2015&title=International+Journal+of+Behavioral+Nutrition+%26+Physical+Activity&atitle=Children%27s+active+commuting+to+school%3A+an+interplay+of+self-efficacy%2C+social+economic+disadvantage%2C+and+environmental+characteristics.&aulast=Lu+W&spage=29

62. Mandic S, Leon de la Barra S, Garcia Bengoechea E, Stevens E, Flaherty C, Moore A, et al. Personal, social and environmental correlates of active transport to school among adolescents in Otago, New Zealand. J Sci Med Sport [Internet]. 2015;18(4):432–7. Available from: https://library-search.imperial.ac.uk/primo-explore/openurl?vid=ICL_VU1&institution=44IMP&?sid=OVID&isbn=&issn=1878-1861&volume=18&issue=4&date=2015&title=Journal+of+Science+%26+Medicine+in+Sport&atitle=Personal%2C+social+and+environmental+correlates+of+active+transport+to+school+among+adolescents+in+Otago%2C+New+Zealand.&aulast=Mandic+S&spage=432

63. Melo EN, Barros M, Reis RS, Hino AAF, Santos CM, Farias Junior JC. Is the environment near school associated with active commuting to school among preschoolers? Revista Brasileira de Cineantropometria e Desempenho Humano. 2013 Apr 30;15(4).

64. Pabayo RA, Gauvin L, Barnett TA, Morency P, Nikiéma B, Séguin L. Understanding the determinants of active transportation to school among children: evidence of environmental injustice from the Quebec Longitudinal Study of Child Development. Health Place. 2012;18(2):163–71.

65. Trang NHHD, Hong TK, Dibley MJ. Active commuting to school among adolescents in Ho Chi Minh City, Vietnam: change and predictors in a longitudinal study, 2004 to 2009. Am J Prev Med [Internet]. 2012;42(2):120–8. Available from: https://library-search.imperial.ac.uk/primo-explore/openurl?vid=ICL_VU1&institution=44IMP&?sid=OVID&isbn=&issn=0749-3797&volume=42&issue=2&date=2012&title=American+Journal+of+Preventive+Medicine&atitle=Active+commuting+to+school+among+adolescents+in+Ho+Chi+Minh+City%2C+Vietnam%3A+change+and+predictors+in+a+longitudinal+study%2C+2004+to+2009.&aulast=Trang+NH&spage=120

66. Seraj S, Sidharthan R, Bhat C, Pendyala R, Goulias K. Parental attitudes toward children walking and bicycling to school. Transp Res Rec [Internet]. 2012;(2323):46–55. Available from: https://www.scopus.com/inward/record.uri?eid=2-s2.0-84872859319&doi=10.3141%2f2323-06&partnerID=40&md5=faa0e531b7278901d61b340ebe93f4d8

67. Lee C, Zhu X. School transportation, health, and equity: The role of built environments. In: Urban Planning in the 21st Century [Internet]. 2013. p. 141–75. Available from: https://www.scopus.com/inward/record.uri?eid=2-s2.0-84931380478&partnerID=40&md5=c2896d50234dec7af820132d18af9899

68. Martins J, Marques A, Sarmento H, Diniz J, C DCF. Prevalence and correlates of active commuting among Portuguese adolescents considering the home-to-school distance. Aten Primaria [Internet]. 2016;48(Supplement C):70. Available from: https://library-search.imperial.ac.uk/primo-explore/openurl?vid=ICL_VU1&institution=44IMP&?sid=OVID&isbn=&issn=1578-1275&volume=48&issue=Supplement+C&date=2016&title=Atencion+Primaria&atitle=Prevalence+and+correlates+of+active+commuting+among+Portuguese+adolescents+considering+the+home-to-school+distance&aulast=Martins+J.&spage=70

69. Mandic S, Hopkins D, García Bengoechea E, Moore A, Sandretto S, Coppell K, et al. Built environment changes and active transport to school among adolescents: BEATS Natural Experiment Study protocol. BMJ Open. 2020;10(3):e034899.

70. Lam HY, Jayasinghe S, Hills AP, Ahuja KDK. Risk versus fun: Perspectives of children and their parents on active school commuting in regional Australia and strategies on promotion. J Transp Health [Internet]. 2025;44. Available from: https://www.scopus.com/inward/record.uri?eid=2-s2.0-105007023087&doi=10.1016%2fj.jth.2025.102087&partnerID=40&md5=4e54b03ea96e8e9dfcfb30038c39546f

71. Jurak G, Sorić M, Ocvirk T, Potočnik ŽL, Meh K, Đurić S, et al. Barriers and determinants of active commuting to school in Slovenia. Sustainability (Switzerland) [Internet]. 2021;13(24). Available from: https://www.scopus.com/inward/record.uri?eid=2-s2.0-85121231515&doi=10.3390%2fsu132413808&partnerID=40&md5=9802cea578db74587bf98c58b74cf2ed

72. Ortiz-Hernández L, Vega-López A V, Ayala-Hilario C. Commuting to school among Mexican schoolchildren and adolescents. Bol Med Hosp Infant Mex. 2019;76(5):225–36.

73. Kotoula K, Botzoris G, Morfoulaki M, Aifandopoulou G. The existing school transportation framework in Greece - Barriers and problems comparing to other European countries. In: Transportation Research Procedia [Internet]. 2017. p. 385–92. Available from: https://www.scopus.com/inward/record.uri?eid=2-s2.0-85020291135&doi=10.1016%2fj.trpro.2017.05.096&partnerID=40&md5=e93cddfb8a97cc5eae19451635ced0aa

74. Stewarta O, Moudon A V, Claybrooke C. Common ground: Eight factors that influence walking and biking to school. Transp Policy (Oxf) [Internet]. 2012;24:240–8. Available from: https://www.scopus.com/inward/record.uri?eid=2-s2.0-84867122471&doi=10.1016%2fj.tranpol.2012.06.016&partnerID=40&md5=03f96f53992cd42b17257aedff4a24e8

75. Wasnyo Y, Tatah L, Mapa-Tassou C, Foley L, Mba C, Blanche N, et al. Individual and socio-environmental factors associated with active transport to school among adolescents in Yaoundé. J Transp Health [Internet]. 2025 Jun;42:101977. Available from: http://www.sciencedirect.com/science/article/pii/S2214140524002238

76. Osman N, Nasrudin N, Abdullah YA. SOCIOECONOMIC INFLUENCES AND PEDESTRIAN INFRASTRUCTURE IN PROMOTING ACTIVE TRAVEL TO SCHOOL AMONG PRIMARY SCHOOL CHILDREN. Planning Malaysia [Internet]. 2025;23(2):220–34. Available from: https://www.scopus.com/inward/record.uri?eid=2-s2.0-105005518858&doi=10.21837%2fpm.v23i36.1714&partnerID=40&md5=57e057e019d2eb8a9465cf59b6edfbd9

77. Agyeman S, Alimo PK. Promoting adolescent girls' active school travel in Ghana: Contextualizing health belief model and theory of planned behavior. Journal of Cycling and Micromobility Research [Internet]. 2024 Dec;2:100033. Available from: http://www.sciencedirect.com/science/article/pii/S295010592400024X

78. Lu W, McKyer ELJ, Lee C, Wang S, Goodson P, Ory MG. Active commuting to school: a test of a modified integrative model. Am J Health Behav [Internet]. 2014;38(6):900–13. Available from: https://library-search.imperial.ac.uk/primo-explore/openurl?vid=ICL_VU1&institution=44IMP&?sid=OVID&isbn=&issn=1087-3244&volume=38&issue=6&date=2014&title=American+Journal+of+Health+Behavior&atitle=Active+commuting+to+school%3A+a+test+of+a+modified+integrative+model.&aulast=Lu+W&spage=900

79. Pavelka J, Sigmundová D, Hamřík Z, Kalman M. Active transport among Czech school-aged children. Acta Universitatis Palackianae Olomucensis, Gymnica [Internet]. 2012;42(3):17–26. Available from: https://www.scopus.com/inward/record.uri?eid=2-s2.0-84878769999&doi=10.5507%2fag.2012.014&partnerID=40&md5=9a1cafc90afb310722aebd31da41d5c8

80. Souza S, Carvalho W, Matos AP, Silva A, Oleveira E, Soares I, et al. Modes of commuting to school among 5th and 6th grade schoolchildren. Revista Brasileira de Cineantropometria e Desempenho Humano [Internet]. 2019 Jan [cited 2025 Jul 20]; Available from: https://www.researchgate.net/publication/333680931_Modes_of_commuting_to_school_among_5th_and_6th_grade_schoolchildren

81. Chillon P, Villa-González E, Herrador-Colmenero M, Huertas-Delgado FJ. Parental Barriers And Active Commuting To School: An Association In Spanish Children And Adolescents. Med Sci Sports Exerc. 2017;49(5):893.

82. Hollein T, Vašíčková J, Bucksch J, Kalman M, Sigmundová D, Dijk JP V. School physical activity policies and active transport to school among pupils in the Czech Republic. J Transp Health [Internet]. 2017;6:306–12. Available from: https://www.scopus.com/inward/record.uri?eid=2-s2.0-85026442875&doi=10.1016%2fj.jth.2017.07.008&partnerID=40&md5=03479744cb58db9f0dd4a26b27a9f567

83. Marwa N, Muhajarine N. Understanding the effects of neighborhood design and built environment on active transportation among children in the city of saskatoon. Am J Epidemiol [Internet]. 2012;176(1):82. Available from: https://library-search.imperial.ac.uk/primo-explore/openurl?vid=ICL_VU1&institution=44IMP&?sid=OVID&isbn=&issn=0002-

9262&volume=176&issue=1&date=2012&title=American+Journal+of+Epidemiology&atitle=Understanding+the+effects+of+neighborhood+design+and+built+environment+on+active+transportation+among+children+in+the+city+of+saskatoon&aulast=Marwa+N.&spage=82

84. Paulusová I, Sharmeen F. School bikeability - what is it, and why is it important? An overview of key indicators and measurement. Transp Rev. 2025;45(1):49–76.

85. Chen S, Hong M, Wei W. A Study of the Influencing Mechanism of Travel Mode Choice for Primary School Students: A Case Study in Wuhan. Buildings [Internet]. 2025;15(5). Available from: https://www.scopus.com/inward/record.uri?eid=2-s2.0-86000583705&doi=10.3390%2fbuildings15050700&partnerID=40&md5=339dd0d7decb102c24bd474d42c139b9

86. Berasategi Sancho N, Legorburu Fernnadez I, Alonso Saez I, Idoiaga Mondragon N. Children's perceptions of safety in their school path: the case of the Basque Country. Child Geogr [Internet]. 2024;22(3):404–15. Available from: https://www.scopus.com/inward/record.uri?eid=2-s2.0-85180221632&doi=10.1080%2f14733285.2023.2292574&partnerID=40&md5=d0858ba2a95b3347ea66b9bf1395b324

87. Liu Y, Ao Y, Han L, Long Y. Rural parents' route choices under different modes for school children drop-off and pick-up: a stated choice experiment. Transportation Planning and Technology [Internet]. 2024; Available from: https://www.scopus.com/inward/record.uri?eid=2-s2.0-85212420998&doi=10.1080%2f03081060.2024.2436948&partnerID=40&md5=bb33209f622f77b6c630916e23a3f8d3

88. Smith M, Hawley G, Mackay L, Hosking J, Mackie H, Ikeda E, et al. Impact of changing road infrastructure on children's active travel: A multi-methods study from Auckland, New Zealand. J Transp Health. 2020;18.

89. Sun G, Han X, Sun S, Oreskovic N. Living in school catchment neighborhoods: Perceived built environments and active commuting behaviors of children in China. J Transp Health [Internet]. 2018;8:251–61. Available from: https://www.scopus.com/inward/record.uri?eid=2-s2.0-85039963384&doi=10.1016%2fj.jth.2017.12.009&partnerID=40&md5=f166583fb068959d74e1ebc9d8c415b3

90. Rodrigues D, Padez C, Machado-Rodrigues AM. Environmental and Socio-demographic Factors Associated with 6–10-Year-Old Children's School Travel in Urban and Non-urban Settings. Journal of Urban Health [Internet]. 2018;95(6):859–68. Available from: https://www.scopus.com/inward/record.uri?eid=2-s2.0-85050689104&doi=10.1007%2fs11524-018-0295-x&partnerID=40&md5=e78c2f858f90d4bfbd21d227f6222dcc

91. Huertas-Delgado FJ, Chillón P, Barranco-Ruiz Y, Herrador-Colmenero M, Rodríguez-Rodríguez F, Villa-González E. Which Are the Main Parental Barriers to Active Commuting to School in Ecuador? J Transp Health. 2017;5:S75–6.

92. Mehdizadeh M, Nordfjaern T, Mamdoohi AR, Shariat Mohaymany A. The role of parental risk judgements, transport safety attitudes, transport priorities and accident experiences on pupils' walking to school. Accid Anal Prev [Internet]. 2017;102:60–71. Available from: https://library-search.imperial.ac.uk/primo-explore/openurl?vid=ICL_VU1&institution=44IMP&?sid=OVID&isbn=&issn=0001-4575&volume=102&issue=&date=2017&title=Accident+Analysis+%26+Prevention&atitle=The+role+of+parental+risk+judgements%2C+transport+safety+attitudes%2C+transport+priorities+and+accident+experiences+on+pupils%27+walking+to+school.&aulast=Mehdizadeh+M&spage=60

93. Race DL, Sims-Gould J, Lee NC, Frazer AD, Voss C, Naylor PJ, et al. Urban and suburban children's experiences with school travel – A case study. J Transp Health [Internet]. 2017;4:305–15. Available from: https://www.scopus.com/inward/record.uri?eid=2-s2.0-85011628944&doi=10.1016%2fj.jth.2017.01.011&partnerID=40&md5=db25a83cb457708d0be5eb6b98b79893

94. Lee C, Yoon J, Zhu X. From sedentary to active school commute: Multi-level factors associated with travel mode shifts. Prev Med (Baltim) [Internet]. 2017;95 Suppl:S28–36. Available from: https://library-search.imperial.ac.uk/primo-

explore/openurl?vid=ICL_VU1&institution=44IMP&?sid=OVID&isbn=&issn=0091-7435&volume=95&issue=&date=2017&title=Preventive+Medicine&atitle=From+sedentary+to+active+school+commute%3A+Multi-level+factors+associated+with+travel+mode+shifts.&aulast=Lee+C&spage=S28

95. Fitch DT, Thigpen CG, Handy SL. Traffic stress and bicycling to elementary and junior high school: Evidence from Davis, California. J Transp Health. 2016;3(4):457–66.

96. Mandic S, Hodge C, Flaherty C, Horwood H, Mountfort A, Hopkins D, et al. Comparison of sociodemographic, family and environmental factors as motivations and barriers for walking to school among adolescents in Dunedin, New Zealand. Glob Heart [Internet]. 2014;9(1 SUPPL. 1):e299. Available from: https://library-search.imperial.ac.uk/primo-explore/openurl?vid=ICL_VU1&institution=44IMP&?sid=OVID&isbn=&issn=2211-8160&volume=9&issue=1+SUPPL.+1&date=2014&title=Global+Heart&atitle=Comparison+of+sociodemographic%2C+family+and+environmental+factors+as+motivations+and+barriers+for+walking+to+school+among+adolescents+in+Dunedin%2C+New+Zealand&aulast=Mandic+S.&spage=e299

97. Kamargianni M, Polydoropoulou A. Generation Y's Travel Behavior and Perceptions of Walkability Constraints. Transp Res Rec. 2014;(2430):59–71.

98. Lee C, Zhu X, Yoon J, Varni JW. Beyond distance: children's school travel mode choice. Ann Behav Med [Internet]. 2013;45 Suppl 1:S55-67. Available from: https://library-search.imperial.ac.uk/primo-explore/openurl?vid=ICL_VU1&institution=44IMP&?sid=OVID&isbn=&issn=0883-6612&volume=45&issue=1&date=2013&title=Annals+of+Behavioral+Medicine&atitle=Beyond+distance%3A+children%27s+school+travel+mode+choice.&aulast=Lee+C&spage=S55

99. Aarts MJ, Mathijssen JJP, van Oers JAM, Schuit AJ. Associations between environmental characteristics and active commuting to school among children: a cross-sectional study. Int J Behav Med [Internet]. 2013;20(4):538–55. Available from: https://library-search.imperial.ac.uk/primo-explore/openurl?vid=ICL_VU1&institution=44IMP&?sid=OVID&isbn=&issn=1070-5503&volume=20&issue=4&date=2013&title=International+Journal+of+Behavioral+Medicine&atitle=Associations+between+environmental+characteristics+and+active+commuting+to+school+among+children%3A+a+cross-sectional+study.&aulast=Aarts+MJ&spage=538

100. Pont K, Wadley D, Ziviani J, Khan A. The Influence of Urban Form and Family Decision Making on Children's Travel to School. J Urban Des (Abingdon) [Internet]. 2013;18(3):363–82. Available from: https://www.scopus.com/inward/record.uri?eid=2-s2.0-84879522072&doi=10.1080%2f13574809.2013.800452&partnerID=40&md5=f397980001da1bfebd89691e52352261

101. Xiao ZP, Lin TY, Liao JY, Lin YY. School Travel Inequity between Students from Public and Private Schools in the City of Shenzhen, China. J Adv Transp. 2021;2021.

102. da Silva Bezerra Cavalcanti, oel de Almeida Siqueira Junior, Francisco Timbó de Paiva Neto, Cassiano Ricardo Rech. ACTIVE SCHOOL COMMUTING: BARRIERS AND FACILITATORSIDENTIFIED IN THE SPEECHES OF CHILDREN IN A PUBLIC SCHOOL SYSTEM. J Phys Educ [Internet]. 2025 Apr 28 [cited 2025 Jul 22]; Available from: https://periodicos.uem.br/ojs/index.php/RevEducFis/article/view/73370

103. Ross A, Kurka JM. Predictors of Active Transportation Among Safe Routes to School Participants in Arizona: Impacts of Distance and Income. JOURNAL OF SCHOOL HEALTH. 2022;92(3):282–92.

104. Padilla JS, Cobos JMC, Sánchez EM, López JM, Quiñones IT. Benefits and barriers of active commuting to the school center: a systematic review. RETOS-NUEVAS TENDENCIAS EN EDUCACION FISICA DEPORTE Y RECREACION. 2021;(43):572–8.

105. Nunes de Oliveira B, Minatto G, Galdino da Costa BG, Veber Lopes M V, Samara da Silva K. Association between psychosocial factors and active commuting to school in Brazilian adolescents. J Transp Health [Internet]. 2020;19. Available from: https://www.scopus.com/inward/record.uri?eid=2-s2.0-85092635829&doi=10.1016%2fj.jth.2020.100964&partnerID=40&md5=006e6ce98c3a6acabb243d77b80b104a

106. Scharoun Benson SM, Bruner B, Mayer A. Encouraging active transportation to school: Lessons learned from implementing a walking school bus program in Northeastern Ontario. J Transp Health [Internet]. 2020;19. Available from: https://www.scopus.com/inward/record.uri?eid=2-s2.0-85090737129&doi=10.1016%2fj.jth.2020.100914&partnerID=40&md5=77e6c864ab71940a9da02a8a22a06308

107. Wang YN, Liu Y, Song SZ, Gittelsohn J, Ouellette M, Ma YA, et al. Individual, parental and built environmental features as influencing factors of active travel to school in northeast China: findings from a cross-sectional study. BMJ Open. 2022;12(1).

108. Omura JD, Hyde ET, Watson KB, Sliwa SA, Fulton JE, Carlson SA. Prevalence of children walking to school and related barriers—United States, 2017. Prev Med (Baltim) [Internet]. 2019;118:191–5. Available from: https://www.scopus.com/inward/record.uri?eid=2-s2.0-85056182469&doi=10.1016%2fj.ypmed.2018.10.016&partnerID=40&md5=1d14d3e1d0d80d8de5db85d9dc32222a

109. Eugenio Merellano-Navarro, John Alcaino-Pedreros, Omar Concha-López, Cristina Herrera-Pavez, Carlos Torres-Pérez, Andrés Godoy-Cumillaf. PERCEPTION OF THE ENVIRONMENT FOR ACTIVE TRANSPORTATIONIN HIGH SCHOOL STUDENTS IN CHILE. CULTURA, CIENCIA Y DEPORTE [Internet]. 2025 Jan 24 [cited 2025 Jul 22]; Available from: https://ccd.ucam.edu/index.php/revista/article/view/2215/1406

110. Gössling S, Kees J, Hologa R, Riach N, von Stülpnagel R. Children's safe routes to school: Real and perceived risks, and evidence of an incapacity-incapability space. Journal of Cycling and Micromobility Research [Internet]. 2024 Dec;2:100019. Available from: http://www.sciencedirect.com/science/article/pii/S295010592400010X

111. Etaati B, Jahangiri A, Fernandez G, Tsou MH, Machiani SG. Understanding Active Transportation to School Behavior in Socioeconomically Disadvantaged Communities: A Machine Learning and SHAP Analysis Approach. Sustainability. 2024;16(1).

112. Mihaere S, Smith M, Jones R. THE IMPORTANCE OF CULTURALLY SAFE ACTIVE SCHOOL TRAVEL OPTIONS TO ENABLE TAMARIKI MĀORI TO FLOURISH A narrative review and model. MAI Journal [Internet]. 2024;13(1):26–38. Available from: https://www.scopus.com/inward/record.uri?eid=2-s2.0-85191005495&doi=10.20507%2fMAIJournal.2024.13.1.3&partnerID=40&md5=7e4163dc7cddb94a73d0fe3da1c00150

113. Cadima C, von Schönfeld K, Ferreira A. Beyond Car-Centred Adultism? Exploring Parental Influences on Children's Mobility. Urban Plan [Internet]. 2024;9. Available from: https://www.scopus.com/inward/record.uri?eid=2-s2.0-85211358710&doi=10.17645%2fup.8643&partnerID=40&md5=c4cfabbf45f1c5393c7b5df36851d0a1

114. Katsavounidou G, Voutsa E, Sepetzi S. Active but not Independent: Children's School Travel Patterns in a Compact-City Environment in Greece. Urban Plan [Internet]. 2024;9. Available from: https://www.scopus.com/inward/record.uri?eid=2-s2.0-85211334256&doi=10.17645%2fup.8682&partnerID=40&md5=ae737a5533adf1b669cc72641b33d4c5

115. Tristram C, Marzi I, Beck F, Diehl K, Renninger D, Demetriou Y, et al. Parental and adolescents' perspectives on environmental predictors of active commuting to school – findings from the ARRIVE mixed-methods study. Front Built Environ [Internet]. 2024;10. Available from: https://www.scopus.com/inward/record.uri?eid=2-s2.0-85215693825&doi=10.3389%2ffbuil.2024.1494837&partnerID=40&md5=2c75f696dbbaaa8a73d7c0c588188d84

116. de Aguiar Greca JP, Korff T, Ryan J. The feasibility of cycling as a form of active commuting among children from a parental perspective: a qualitative study. Int J Health Promot Educ [Internet]. 2023;61(5):266–75. Available from: https://www.scopus.com/inward/record.uri?eid=2-s2.0-85158890064&doi=10.1080%2f14635240.2023.2207100&partnerID=40&md5=9893965f8427e59a496017799ded9829

117. Khalifa NA. Factors That Influence the Parents Perspective of Allowing Their Children to Walk to/From School- An Overview. JURNAL KEJURUTERAAN. 2023;35(4):789–802.

118. Wangzom D, White M, Paay J. Perceived Safety Influencing Active Travel to School—A Built Environment Perspective. Int J Environ Res Public Health [Internet]. 2023;20(2). Available from: https://www.scopus.com/inward/record.uri?eid=2-s2.0-85146598066&doi=10.3390%2fijerph20021026&partnerID=40&md5=4012d9e4e629a67aaec3014cf86fecf5

119. Kim YJ, Lee C. Built and Natural Environmental Correlates of Parental Safety Concerns for Children's Active Travel to School. Int J Environ Res Public Health. 2020;17(2).

120. Smith M, Hawley G, Mackay L, Hosking J, Mackie H, Ikeda E, et al. Impact of changing road infrastructure on children's active travel: A multi-methods study from Auckland, New Zealand: Impact of infrastructural changes on active school travel. J Transp Health [Internet]. 2020;18. Available from: https://www.scopus.com/inward/record.uri?eid=2-s2.0-85085744282&doi=10.1016%2fj.jth.2020.100868&partnerID=40&md5=a8c6e550644a4846ebde8b872d5bbcbd

121. Mandic S, Mountfort A, Hopkins D, Flaherty C, Williams J, Brook E, et al. Built Environment and Active Transport to School (BEATS) Study: Multidisciplinary and Multi-Sector Collaboration for Physical Activity Promotion. RETOS-NUEVAS TENDENCIAS EN EDUCACION FISICA DEPORTE Y RECREACION. 2015;(28):197–202.

122. Nasrudin N, Nor AbdRMd. Travelling to School: Transportation Selection by Parents and Awareness towards Sustainable Transportation. Procedia Environ Sci. 2013;17:392–400.

123. Aibar Solana A, Mandic S, Generelo Lanaspa E, Gallardo LO, Zaragoza Casterad J. Parental barriers to active commuting to school in children: does parental gender matter? J Transp Health [Internet]. 2018;9:141–9. Available from: https://www.scopus.com/inward/record.uri?eid=2-s2.0-85044274681&doi=10.1016%2fj.jth.2018.03.005&partnerID=40&md5=0a719bea09cf49dea84005e0276dce31

124. Yang Y, Markowitz E. Integrating parental attitudes in research on children's active school commuting. Transp Res Rec [Internet]. 2012;(2318):116–27. Available from: https://www.scopus.com/inward/record.uri?eid=2-s2.0-84874050523&doi=10.3141%2f2318-14&partnerID=40&md5=f9a6ade9e73e046182424b33a2ff1600

125. Van Dyck D, De Bourdeaudhuij I, Cardon G, Deforche B. Criterion distances and correlates of active transportation to school in Belgian older adolescents. INTERNATIONAL JOURNAL OF BEHAVIORAL NUTRITION AND PHYSICAL ACTIVITY. 2010;7.

126. Stewart OT. Safe Routes to School (SRTS). In: Children's Active Transportation [Internet]. 2018. p. 193–203. Available from: https://www.scopus.com/inward/record.uri?eid=2-s2.0-85081399785&doi=10.1016%2fB978-0-12-811931-0.00013-2&partnerID=40&md5=30850c6d877fca201ee0fa8ba20f6149

127. Ross A, Kwon JY, Kulinna PH, Searle M. Active transportation: The role of parent attitude, the physical environment, and social capital. J Phys Act Health [Internet]. 2019;16(1):60–7. Available from: https://www.scopus.com/inward/record.uri?eid=2-s2.0-85060010020&doi=10.1123%2fjpah.2017-0503&partnerID=40&md5=fafcc9399be494005db9a08787d02eba

128. Hecker V, Blanchette S, Faulkner GE, Riazi NA, Tremblay MS, Trudeau F, et al. Parental travel behaviours, accompaniment, and children's active transportation: A multi-site study. J Transp Health [Internet]. 2025 Mar;41:101988. Available from: http://www.sciencedirect.com/science/article/pii/S2214140525000088

129. Vaquero-Solís M, Tapia-Serrano MA, Prieto JP, Cerro-Herrero D, Sánchez-Miguel PA. ACTIVE COMMUTING IN ADOLESCENTS: IMPORTANCE OF PERCEIVED BARRIERS AND ACCOMPANIMENT. REVISTA INTERNACIONAL DE MEDICINA Y CIENCIAS DE LA ACTIVIDAD FISICA Y DEL DEPORTE. 2022;22(88):985–99.

130. Romero V. Children's experiences: Enjoyment and fun as additional encouragement for walking to school. J Transp Health. 2015;2(2):230–7.

131. Higgins R, Ahern A. Students' and Parents' Perceptions of Barriers to Cycling to School-An Analysis by Gender. Sustainability. 2021;13(23).

132. Frater J, Kingham S. Adolescents and bicycling to school: Does behaviour setting/place make a difference? J Transp Geogr. 2020;85.

133. Mandic S, Hopkins D, García Bengoechea E, Flaherty C, Williams J, Sloane L, et al. Adolescents' perceptions of cycling versus walking to school: Understanding the New Zealand context. J Transp Health [Internet]. 2017;4:294–304. Available from: https://www.scopus.com/inward/record.uri?eid=2-s2.0-85006036010&doi=10.1016%2fj.jth.2016.10.007&partnerID=40&md5=fe23a9e3eb191d32328ebca49dc2c984

134. Carlson JA, Steel C, Bejarano CM, Beauchamp MT, Davis AM, Sallis JF, et al. Walking School Bus Programs: Implementation Factors, Implementation Outcomes, and Student Outcomes, 2017–2018. Prev Chronic Dis [Internet]. 2020;17. Available from: https://www.scopus.com/inward/record.uri?eid=2-s2.0-85093493877&doi=10.5888%2fpcd17.200061&partnerID=40&md5=616d8f8488517b552b41211cfd5f8e36

135. McDonald NC, Steiner RL, Lee C, Smith TR, Zhu X, Yang Y. Impact of the safe routes to school program on walking and bicycling. Journal of the American Planning Association [Internet]. 2014;80(2):153–67. Available from: https://www.scopus.com/inward/record.uri?eid=2-s2.0-84907887466&doi=10.1080%2f01944363.2014.956654&partnerID=40&md5=e0f9c67b80ac4d7a3e0db2c22f91ec83

136. Koester MK, Bejarano CM, Davis AM, Brownson RC, Kerner J, Sallis JF, et al. Implementation contextual factors related to community-based active travel to school interventions: a mixed methods interview study. Implement Sci Commun [Internet]. 2021;2(1). Available from: https://www.scopus.com/inward/record.uri?eid=2-s2.0-85132063460&doi=10.1186%2fs43058-021-00198-7&partnerID=40&md5=51446befaf0306cf4cff1c83d170e8d9

137. Timperio A, Duncan S, Akram M, Molina-Garcia J, Van Dyck D, Barnett A, et al. Associations between parental perceptions of neighbourhood environments and active travel to school: IPEN Adolescent study. Int J Behav Nutr Phys Act [Internet]. 2025;22(1):55. Available from: https://library-search.imperial.ac.uk/primo-explore/openurl?vid=ICL_VU1&institution=44IMP&?sid=OVID&isbn=&issn=1479-5868&volume=22&issue=1&date=2025&title=International+Journal+of+Behavioral+Nutrition+%26+Physical+Activity&atitle=Associations+between+parental+perceptions+of+neighbourhood+environments+and+active+travel+to+school%3A+IPEN+Adolescent+study.&aulast=Timperio+A&spage=55

138. Puyol-Martín Y, Oliveras L, Marí-Dell'Olmo M, Bartoll X, Pérez K, Gómez-Gutiérrez A, et al. Social and urban determinants of school commuting in Barcelona and its relationship with air quality around schools. J Transp Health [Internet]. 2025;44. Available from: https://www.scopus.com/inward/record.uri?eid=2-s2.0-105006911469&doi=10.1016%2fj.jth.2025.102084&partnerID=40&md5=c566782840bab0a86efc23f93205da51

139. Alquhtani S. Built environment predictors of walking to school in a sprawled city. JOURNAL OF ASIAN ARCHITECTURE AND BUILDING ENGINEERING. 2024;

140. Herazo-Beltrán Y, Sánchez-Guette L, González SA, Pahuana-Escobar M, Berdejo-Sandoval V, Álvarez-González J, et al. Environmental and social barriers to active school transport in the Colombian Caribbean region. Revista de la Facultad de Medicina. 2024 Feb 1;72(1):e104638.

141. Smith M, Zhang YJ, Fainu HM, Cavadino A, Zhao JF, Morton S, et al. Socio-environmental factors associated with active school travel in children at ages 6 and 8 years. Transp Res Interdiscip Perspect. 2024;23.

142. Lutfur Rahman M, Moore AB, Mandic S. Adolescents' perceptions of school neighbourhood built environment for walking and cycling to school. Transp Res Part F Traffic Psychol Behav. 2022 Jul;88:111–21.

143. Mandic S, García Bengoechea E, Hopkins D, Coppell K, Spence JC. Adolescents' perceptions of walking and cycling to school differ based on how far they live from school. J Transp Health [Internet]. 2022;24. Available from: https://www.scopus.com/inward/record.uri?eid=2-s2.0-85120949317&doi=10.1016%2fj.jth.2021.101316&partnerID=40&md5=15c18fa51d1a3050461b6fe91d8cc2fd

144. Osman N, Nasrudin N, Abdullah YA, Khalid NS. Factors Influencing Active Travel Mode to School for Primary School Children. Environment-Behaviour Proceedings Journal. 2020 Mar 23;5(13):13.

145. Distefano N, Leonardi S, Pulvirenti G. Home-school travel: Analysis of factors affecting italian parents' mode choice. Civil Engineering and Architecture [Internet]. 2019;7(3):75–87. Available from: https://www.scopus.com/inward/record.uri?eid=2-s2.0-85068687953&doi=10.13189%2fcea.2019.070302&partnerID=40&md5=6275a8cf813c63ec24612225613fbd1a

146. Helbich M, Emmichoven MJZ van, Dijst MJ, Kwan MP, Pierik FH, Vries SI de. Natural and built environmental exposures on children's active school travel: A Dutch global positioning system-based cross-sectional study. Health Place [Internet]. 2016;39:101–9. Available from: https://library-search.imperial.ac.uk/primo-explore/openurl?vid=ICL_VU1&institution=44IMP&?sid=OVID&isbn=&issn=1353-8292&volume=39&issue=&date=2016&title=Health+%26+Place&atitle=Natural+and+built+environmental+exposures+on+children%27s+active+school+travel%3A+A+Dutch+global+positioning+system-based+cross-sectional+study.&aulast=Helbich+M&spage=101

147. Daniels N, Kelly C, Molcho M, Sixsmith J, Byrne M, Nic Gabhainn S. Investigating active travel to primary school in ireland. Health Educ [Internet]. 2014;114(6):501–15. Available from: https://www.scopus.com/inward/record.uri?eid=2-s2.0-84926352822&doi=10.1108%2fHE-08-2012-0045&partnerID=40&md5=a06150bd48380830a9a92eef3ccaa751

148. Shabus S, Tominitz M, Rigby J. The Influence of Distance and Neighbourhood Surroundings on Children's Travel Mode to School. In: GI_Forum 2013 – Creating the GISociety. Vienna: Austrian Academy of Sciences Press; 2013. p. 229–38.

149. Mackay L, Oliver M, Badland H, Mavoa S, Witten K, Hinckson E, et al. Parents' neighbourhood self-selection and children's active transport to school. J Sci Med Sport. 2012 Dec;15:S209–10.

150. McMinn D, Nelson NM, Murtagh S, Rowe DA. Perceived Barriers To Walking To School Among Parents Of 8-9 Year Old Children. Med Sci Sports Exerc. 2010;42(5):520.

151. Wang Y, Liu Y, Song S, Gittelsohn J, Ouellette M, Ma Y, et al. Individual, parental and built environmental features as influencing factors of active travel to school in northeast China: findings from a cross-sectional study. BMJ Open. 2022 Jan;12(1):e047816.

152. Larsen K, Buliung RN, Faulkner GEJ. Safety and School Travel: How Does the Environment Along the Route Relate to Safety and Mode Choice? Transp Res Rec. 2013;(2327):9–18.

153. Han L, Wang Y, Ao Y, Ding X, Li M, Wang T. The built environment impacts on route choice from home to school for rural students: A stated preference experiment. Front Public Health [Internet]. 2022;10:1087467. Available from: https://library-search.imperial.ac.uk/primo-explore/openurl?vid=ICL_VU1&institution=44IMP&?sid=OVID&isbn=&issn=2296-2565&volume=10&issue=&date=2022&title=Frontiers+in+Public+Health&atitle=The+built+environment+impacts+on+route+choice+from+home+to+school+for+rural+students%3A+A+stated+preference+experiment.&aulast=Han+L&spage=1087467

154. Xiong H, Ma L, Wei C, Yan X, Srinivasan S, Chen J. Exploring Behavioral Heterogeneities of Elementary School Students' Commute Mode Choices Through the Urban Travel Big Data of Beijing, China. IEEE Access [Internet]. 2019;7:22235–45. Available from: https://www.scopus.com/inward/record.uri?eid=2-s2.0-85062858431&doi=10.1109%2fACCESS.2019.2897890&partnerID=40&md5=c40b5ee6460d414e0c3ae92a857088af

155. Berasategi Sancho N, Eiguren Munitis A, Legorburu Fernandez IL, Idoiaga-Mondragon NI. Children's Active and Autonomous School Journeys and Their Implications: An Investigation Using Drawing Methodology. Child Youth Care Forum [Internet]. 2025; Available from: https://www.scopus.com/inward/record.uri?eid=2-s2.0-105005965708&doi=10.1007%2fs10566-025-09867-3&partnerID=40&md5=73ab0774ec6a5aeca947f0af80d6d957

156. Giles-Corti B, Wood G, Pikora T, Learnihan V, Bulsara M, Van Niel K, et al. School site and the potential to walk to school: The impact of street connectivity and traffic exposure in school neighborhoods. Health Place. 2011 Mar;17(2):545–50.

157. Kennedy J, Mammen G. Walking to and from School. In 2017. p. 99–111.

158. Ma L, Xiong H, Wang Z, Xie KQ. Impact of weather conditions on middle school students' commute mode choices: Empirical findings from Beijing, China. TRANSPORTATION RESEARCH PART D-TRANSPORT AND ENVIRONMENT. 2019;68:39–51.

159. Tetali S, Edwards P, Roberts GVSMI. How do children travel to school in urban India? A cross-sectional study of 5,842 children in Hyderabad. BMC Public Health [Internet]. 2016;16(1):1099. Available from: https://library-search.imperial.ac.uk/primo-explore/openurl?vid=ICL_VU1&institution=44IMP&?sid=OVID&isbn=&issn=1471-2458&volume=16&issue=1&date=2016&title=BMC+Public+Health&atitle=How+do+children+travel+to+school+in+urban+India%3F+A+cross-sectional+study+of+5%2C842+children+in+Hyderabad.&aulast=Tetali+S&spage=1099

160. Dalton MA, Longacre MR, Drake KM, Gibson L, Adachi-Mejia AM, Swain K, et al. Built environment predictors of active travel to school among rural adolescents. Am J Prev Med [Internet]. 2011;40(3):312–9. Available from: https://library-search.imperial.ac.uk/primo-explore/openurl?vid=ICL_VU1&institution=44IMP&?sid=OVID&isbn=&issn=0749-3797&volume=40&issue=3&date=2011&title=American+Journal+of+Preventive+Medicine&atitle=Built+environment+predictors+of+active+travel+to+school+among+rural+adolescents.&aulast=Dalton+MA&spage=312

161. Chen P, Jiao JF, Xu MY, Gao X, Bischak C. Promoting active student travel: A longitudinal study. J Transp Geogr. 2018;70:265–74.

162. Wu F, Li W, Qiu W. Examining non-linear relationship between streetscape features and propensity of walking to school in Hong Kong using machine learning techniques. J Transp Geogr [Internet]. 2023;113. Available from: https://www.scopus.com/inward/record.uri?eid=2-s2.0-85174358580&doi=10.1016%2fj.jtrangeo.2023.103698&partnerID=40&md5=bb366c39077df75774a39b2875699fb9

163. Ganzar LA, Burford K, Zhang Y, Gressett A, Kohl HW, Hoelscher DM. Association of Walking and Biking to School Policies and Active Commuting to School in Children. J Phys Act Health [Internet]. 2023;20(7):648–54. Available from: https://library-search.imperial.ac.uk/primo-explore/openurl?vid=ICL_VU1&institution=44IMP&?sid=OVID&isbn=&issn=1543-3080&volume=20&issue=7&date=2023&title=Journal+of+Physical+Activity+%26+Health&atitle=Association+of+Walking+and+Biking+to+School+Policies+and+Active+Commuting+to+School+in+Children.&aulast=Ganzar+LA&spage=648

164. Aranda-Balboa MJ, Huertas-Delgado FJ, Herrador-Colmenero M, Cardon G, Chillón P. Parental barriers to active transport to school: a systematic review. Int J Public Health. 2020 Jan 14;65(1):87–98.

165. An R, Shen J, Yang Q, Yang Y. Impact of built environment on physical activity and obesity among children and adolescents in China: A narrative systematic review. J Sport Health Sci. 2019 Mar;8(2):153–69.

166. Zhang Y, Koene M, Chen C, Wagenaar C, Reijneveld SA. Associations between the built environment and physical activity in children, adults and older people: A narrative review of reviews. Prev Med (Baltim). 2024 Mar;180:107856.

167. Buttazzoni A, Nelson Ferguson K, Gilliland J. Barriers to and facilitators of active travel from the youth perspective: A qualitative meta-synthesis. SSM Popul Health. 2023 Jun;22:101369.

168. Loitz C. C, Spencer-Cavaliere N. Exploring the Barriers and Facilitators to Children's Active Transportation to and From School From the Perspectives of Practitioners. J Phys Act Health [Internet]. 2013 [cited 2025 Aug 15]; Available from: https://2024.sci-hub.box/5970/7f945a32e14e11ff6105b56a6bfdc31e/10.1123@jpah.10.8.1128.pdf

169. Wilson Van Voorhis CR, Morgan BL. Understanding Power and Rules of Thumb for Determining Sample Sizes. Tutor Quant Methods Psychol. 2007 Sep 1;3(2):43–50.

170. Rothman L, Ling R, Hagel BE, Macarthur C, Macpherson AK, Buliung R, et al. Pilot study to evaluate school safety zone built environment interventions. Inj Prev [Internet]. 2022;28(3):243–8. Available from: https://library-search.imperial.ac.uk/primo-explore/openurl?vid=ICL_VU1&institution=44IMP&?sid=OVID&isbn=&issn=1353-8047&volume=28&issue=3&date=2022&title=Injury+Prevention&atitle=Pilot+study+to+evaluate+school+safety+zone+built+environment+interventions.&aulast=Rothman+L&spage=243

171. Triana CA, Sarmiento OL, Bravo-Balado A, González SA, Bolívar MA, Lemoine P, et al. Active streets for children: The case of the Bogotá Ciclovía. PLoS One. 2019 May 15;14(5):e0207791.

172. Kingham S, Ussher S. An assessment of the benefits of the walking school bus in Christchurch, New Zealand. Transp Res Part A Policy Pract. 2007 Jul;41(6):502–10.

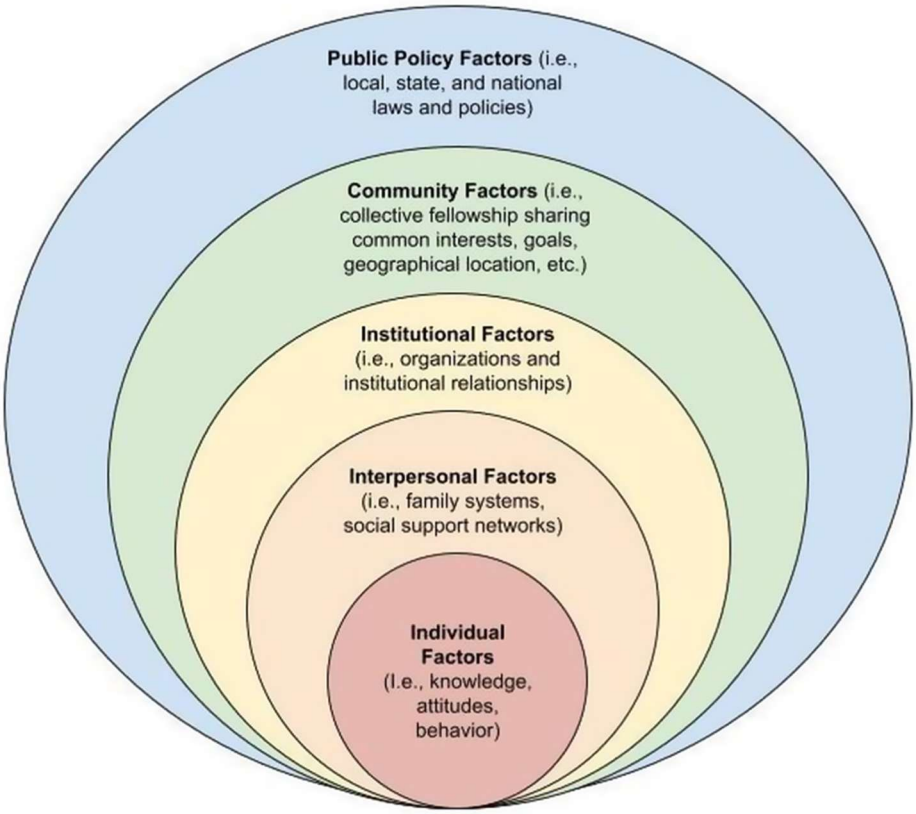
173. Fleming S. The Netherlands is Paying People to Cycle to Work. Intertraffic; 2019.

174. A ROLL-OUT PLAN FOR SHOVA KALULA BICYCLE PROJECT [Internet]. 2007 Aug [cited 2025 Aug 17]. Available from: http://www.kzntransport.gov.za/programmes/shova_kalula/Shova%20Rollout%20Plan%20Final%20Amendments%2006%20AUGUST%202007.pdf

Appendices

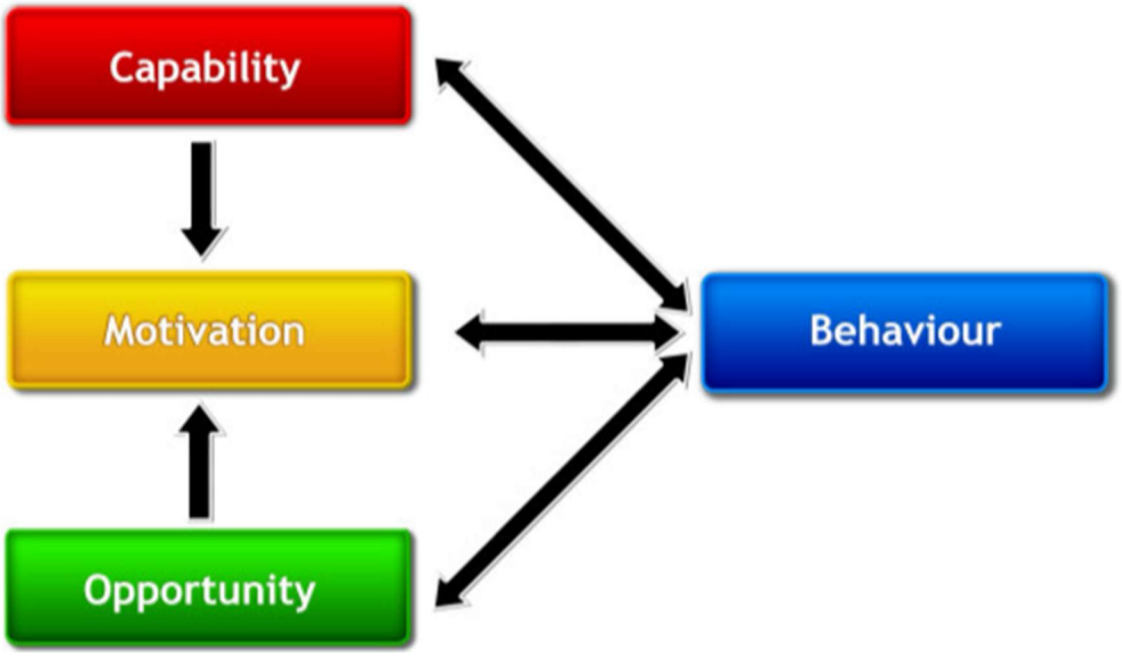
Appendix A

The five levels of the Socio-Ecological Model (SEM), adapted from McLeroy *et al.* (1988) by Breanne Prendeville. Originally created by psychologist Urie Bronfenrenner (1979) in *The Ecology of Human Development: Experiment by Nature and Design.*



Appendix B

The Capability, Opportunity, Motivation (COM-B) Behavioural Framework, created by Michie *et al.* (2011).



Appendix C

The Preferred Reporting items for Systematic reviews and Meta Analyses extension for Scoping Reviews (PRISMA-ScR), created by the Joanna Briggs Institute (2018).

Section	Item	PRISMA-ScR Checklist Item	Page Reported on
Title			
Title	1	Identify the report as a scoping review.	-
Abstract			
Structured summary	2	Provide a structured summary that includes (as applicable): background, objectives, eligibility criteria, sources of evidence, charting methods, results, and conclusions that relate to the review questions and objectives.	
Introduction			
Rationale	3	Describe the rationale for the review in the context of what is already known. Explain why the review questions/objectives lend themselves to a scoping review approach.	-
Objectives	4	Provide an explicit statement of the questions and objectives being addressed with reference to their key elements (e.g., population or participants, concepts, and context) or other relevant key elements used to conceptualize the review questions and/or objectives.	-
Methods			
Protocol and registration	5	Indicate whether a review protocol exists; state if and where it can be accessed (e.g., a Web address); and if available, provide registration information, including the registration number.	-
Eligibility criteria	6	Specify characteristics of the sources of evidence used as eligibility criteria (e.g., years considered, language, and publication status), and provide a rationale.	-
Information sources*	7	Describe all information sources in the search (e.g., databases with dates of coverage and contact with authors to identify additional sources), as well as the date the most recent search was executed.	-
Search	8	Present the full electronic search strategy for at least 1 database, including any limits used, such that it could be repeated.	-
Selection of sources of evidence†	9	State the process for selecting sources of evidence (i.e., screening and eligibility) included in the scoping review.	-
Data charting process‡	10	Describe the methods of charting data from the included sources of evidence (e.g., calibrated forms or forms that have been tested by the team before their use, and whether data charting was done independently or in duplicate) and any processes for obtaining and confirming data from investigators.	-
Data items	11	List and define all variables for which data were sought and any assumptions and simplifications made.	-
Critical appraisal of individual sources of evidence§	12	If done, provide a rationale for conducting a critical appraisal of included sources of evidence; describe the methods used and how this information was used in any data synthesis (if appropriate).	N/A
Synthesis of results	13	Describe the methods of handling and summarizing the data that were charted.	-

Appendix C (Continued)

Results			
Selection of sources of evidence	14	Give numbers of sources of evidence screened, assessed for eligibility, and included in the review, with reasons for exclusions at each stage, ideally using a flow diagram.	-
Characteristics of sources of evidence	15	For each source of evidence, present characteristics for which data were charted and provide the citations.	
Critical appraisal within sources of evidence	16	If done, present data on critical appraisal of included sources of evidence (see item 12).	N/A
Results of individual sources of evidence	17	For each included source of evidence, present the relevant data that were charted that relate to the review questions and objectives.	
Synthesis of results	18	Summarize and/or present the charting results as they relate to the review questions and objectives.	
Discussion			
Summary of evidence	19	Summarize the main results (including an overview of concepts, themes, and types of evidence available), link to the review questions and objectives, and consider the relevance to key groups.	
Limitations	20	Discuss the limitations of the scoping review process.	
Conclusions	21	Provide a general interpretation of the results with respect to the review questions and objectives, as well as potential implications and/or next steps.	
Funding			
Funding	22	Describe sources of funding for the included sources of evidence, as well as sources of funding for the scoping review. Describe the role of the funders of the scoping review.	

Appendix D

The complete search strategy across the six electronic databases (OVID Embase, OVID Medline, PubMed, Scopus, Web of Science (WoS), and Transport Research International Documentation (TRID)).

OVID Embase		
Population (P): School-aged children aged 5-19	Subject headings (Emtree)	Child/
		Adolescent/
		Minors/
		Juvenile/
		Teenager/
		exp Parent/
		exp Teacher/
	Keywords	("School-Age*" or "School-Going" or Child* or Schoolchild* or Adolescent* or Teen* or Kid or Kids or Minor or Minors or Youth* or Parent* or Guardian* or Teacher*).mp.
Context (C): Barriers and facilitators to AST across SEM levels.	Subject headings (Emtree)	Social determinants of health/
		Health behavior/
		Behavior change/
		Motivation/
		Extrinsic motivation/
		Intrinsic motivation/
		Incentive/
		Social incentive/
		Perverse incentive/
		Disincentive/
	Keywords	("Health Behavio?r" or "Mobility Behavio?r" or "Behavio?r Change" or "Health Literac*" or "Self-Efficac*" or "Peer-Pressure" or "Peer-Influence" or ((Cultural or Systemic or Underlying or Social) adj1 (Belief* or Determinant* or Barrier* or Facilitat* or Challenge* or Reason or Reasons)) or ((Positive or Good or Negative or Bad) adj1 (Experience* or Opinion* or Belief* or Perception*)) or Barrier* or Facilitat* or Reason or Reasons or Policy or Constrain* or Obstruct* or Restrain* or Restrict* or Enabler* or Obstacle* or Influence* or Determinant* or Motivator* or Limit* or Challenge or Attitude* or Belief* or Perception* or Motivation* or Stigma* or Trust* or Access* or Availab* or Guideline*).mp.
Concept (C): Non-motorised travel to and/or from school specifically.	Keywords	((("Non-Motori?ed Transport*" or "Activ* Travel*" or "Active Mobility" or "Physical Mobility" or "Soft Mobility" or "Cycle Path*" or "Cycle Zone*" or "Cycle Lane*" or "Cycling Path*" or "Cycling Zone*" or "Cycling Lane*" or "Bike Path*" or "Bike Zone*" or "Bike Lane*" or "Bicycle Path*" or "Bicycle Zone*" or "Bicycle Lane*" or "Walk* Path*" or "Walk* Zone*" or "Walk* Lane*" or "Foot Path*" or "Foot Zone*" or "Foot Lane*" or “Pedestrian Path*” or “Pedestrian Zone*” or “Pedestrian Lane*” or "Home-to-School Travel*" or “Home-from-School Travel” or “School-to-Home Travel” or “School-from-Home Travel” or “Home-to-and-from School Travel”) adj10 (School* or Highschool* or Kindergarten*)) or (("Non-Motori?ed Transport*" or "Activ* Travel*" or "Active Mobility" or "Physical Mobility" or "Soft Mobility" or "Cycle Path*" or "Cycle Zone*" or "Cycle Lane*" or "Cycling Path*" or "Cycling Zone*" or "Cycling Lane*" or "Bike Path*" or "Bike Zone*" or "Bike Lane*" or "Bicycle Path*" or "Bicycle Zone*" or "Bicycle Lane*" or "Walk* Path*" or "Walk* Zone*" or "Walk* Lane*" or "Foot Path*" or "Foot Zone*" or "Foot Lane*" or “Pedestrian Path*” or “Pedestrian Zone*” or “Pedestrian Lane*” or "Home-to-School Travel*" or “Home-from-School Travel” or “School-to-Home Travel” or “School-from-Home Travel” or “Home-to-and-from School Travel”) adj10 (“High School*” or "Secondary School*" or "Middle School*" or "Primary School*" or "Elementary School*")) or ((Commut* or Walk* or Sidewalk or Footpath or Bike or Bicycle or Cycle or Cycling or Trip or Trips or Travel* or Route* or Journey* or "Drop-off" or “Drop-offs” or "Pick-up" or “Pick-ups”) adj10 (School* or Highschool* or Kindergarten*)) or ((Commut* or Walk* or Sidewalk or Footpath or Bike or Bicycle or Cycle or Cycling or Trip or Trips or Travel* or Route* or Journey* or "Drop-off" or “Drop-offs” or "Pick-up" or “Pick-ups”) adj10 (“High School*” or "Secondary School*" or "Middle School*" or "Primary School*" or "Elementary School*"))).mp.

Appendix D (Continued)

OVID Medline		
Population	Subject headings (MeSH)	Child/
		Adolescent/
		exp Minors/
		Parents/
		Single Parent/
		Fathers/
		Mothers/
		Legal Guardians/
		School Teacher/
	Keywords	("School-Age*" or "School-Going" or Child* or Schoolchild* or Adolescent* or Teen* or Kid or Kids or Minor or Minors or Youth* or Parent* or Guardian* or Teacher*).mp.
Context	Subject headings (MeSH)	"Social determinants of health"/
		Health behavior/
		Motivation/
	Keywords	("Health Behavio?r" or "Mobility Behavio?r" or "Behavio?r Change" or "Health Literac*" or "Self-Efficac*" or "Peer-Pressure" or "Peer-Influence" or ((Cultural or Systemic or Underlying or Social) adj1 (Belief* or Determinant* or Barrier* or Facilitat* or Challenge* or Reason or Reasons)) or ((Positive or Good or Negative or Bad) adj1 (Experience* or Opinion* or Belief* or Perception*)) or Barrier* or Facilitat* or Reason or Reasons or Policy or Constrain* or Obstruct* or Restrain* or Restrict* or Enabler* or Obstacle* or Influence* or Determinant* or Motivator* or Limit* or Challenge or Attitude* or Belief* or Perception* or Motivation* or Stigma* or Trust* or Access* or Availab* or Guideline*).mp.
Concept	Keywords	((("Non-Motori?ed Transport*" or "Activ* Travel*" or "Active Mobility" or "Physical Mobility" or "Soft Mobility" or "Cycle Path*" or "Cycle Zone*" or "Cycle Lane*" or "Cycling Path*" or "Cycling Zone*" or "Cycling Lane*" or "Bike Path*" or "Bike Zone*" or "Bike Lane*" or "Bicycle Path*" or "Bicycle Zone*" or "Bicycle Lane*" or "Walk* Path*" or "Walk* Zone*" or "Walk* Lane*" or "Foot Path*" or "Foot Zone*" or "Foot Lane*" or "Pedestrian Path*" or "Pedestrian Zone*" or "Pedestrian Lane*" or "Home-to-School Travel*" or "Home-from-School Travel" or "School-to-Home Travel" or "School-from-Home Travel" or "Home-to-and-from School Travel") adj10 (School* or Highschool* or Kindergarten*)) or (("Non-Motori?ed Transport*" or "Activ* Travel*" or "Active Mobility" or "Physical Mobility" or "Soft Mobility" or "Cycle Path*" or "Cycle Zone*" or "Cycle Lane*" or "Cycling Path*" or "Cycling Zone*" or "Cycling Lane*" or "Bike Path*" or "Bike Zone*" or "Bike Lane*" or "Bicycle Path*" or "Bicycle Zone*" or "Bicycle Lane*" or "Walk* Path*" or "Walk* Zone*" or "Walk* Lane*" or "Foot Path*" or "Foot Zone*" or "Foot Lane*" or "Pedestrian Path*" or "Pedestrian Zone*" or "Pedestrian Lane*" or "Home-to-School Travel*" or "Home-from-School Travel" or "School-to-Home Travel" or "School-from-Home Travel" or "Home-to-and-from School Travel") adj10 ("High School*" or "Secondary School*" or "Middle School*" or "Primary School*" or "Elementary School*")) or ((Commut* or Walk* or Sidewalk or Footpath or Bike or Bicycle or Cycle or Cycling or Trip or Trips or Travel* or Route* or Journey* or "Drop-off" or "Drop-offs" or "Pick-up" or "Pick-ups") adj10 (School* or Highschool* or Kindergarten*)) or ((Commut* or Walk* or Sidewalk or Footpath or Bike or Bicycle or Cycle or Cycling or Trip or Trips or Travel* or Route* or Journey* or "Drop-off" or "Drop-offs" or "Pick-up" or "Pick-ups") adj10 ("High School*" or "Secondary School*" or "Middle School*" or "Primary School*" or "Elementary School*"))).mp.

Appendix D (Continued)

PubMed		
Population	Subject headings (MeSH) and keywords	*Child"[Mesh] OR "Adolescent"[Mesh] OR "Minors"[Mesh] OR "Parents"[Mesh:NoExp] OR "Fathers"[Mesh] OR "Mothers"[Mesh] OR "Legal Guardians"[Mesh:NoExp] OR "School Teachers"[Mesh] OR "School-Age*"[Title/Abstract] OR "School-Going"[Title/Abstract] OR Child*[Title/Abstract] OR Schoolchild*[Title/Abstract] OR Adolescent*[Title/Abstract] OR Teen*[Title/Abstract] OR Kid[Title/Abstract] OR Kids[Title/Abstract] OR Minor[Title/Abstract] OR Minors[Title/Abstract] OR Youth*[Title/Abstract] OR Parent*[Title/Abstract] OR Guardian*[Title/Abstract] OR Teacher*[Title/Abstract]
Context	Subject headings (MeSH) and keywords	"Social Determinants of Health"[Mesh] OR "Health Behavior"[Mesh:NoExp] OR "Motivation"[Mesh:NoExp] OR "Health Behavior"[Title/Abstract] OR "Health Behaviour"[Title/Abstract] "Mobility Behavior"[Title/Abstract] OR "Mobility Behaviour"[Title/Abstract] OR "Behavior Change"[Title/Abstract] OR "Behaviour Change"[Title/Abstract] OR "Health Literac*"[Title/Abstract] OR "Self-Efficac*"[Title/Abstract] OR "Peer-Pressure"[Title/Abstract] OR "Peer-Influence"[Title/Abstract] OR "Cultural Belief*"[Title/Abstract] OR "Systemic Belief*"[Title/Abstract] OR "Underlying Belief*"[Title/Abstract] OR "Social Belief*"[Title/Abstract] OR "Cultural Determinant*"[Title/Abstract] OR "Systemic Determinant*"[Title/Abstract] OR "Underlying Determinant*"[Title/Abstract] OR "Social Determinant*"[Title/Abstract] OR "Cultural Barrier*"[Title/Abstract] OR "Systemic Barrier*"[Title/Abstract] OR "Underlying Barrier*"[Title/Abstract] OR "Social Barrier*"[Title/Abstract] OR "Cultural Challenge*"[Title/Abstract] OR "Systemic Challenge*"[Title/Abstract] OR "Underlying Challenge*"[Title/Abstract] OR "Social Challenge*"[Title/Abstract] OR "Underlying Reason"[Title/Abstract] OR "Social Reason"[Title/Abstract] OR "Cultural Reasons"[Title/Abstract] OR "Systemic Reasons"[Title/Abstract] OR "Underlying Reasons"[Title/Abstract] OR "Social Reasons"[Title/Abstract] OR "Positive Experience*"[Title/Abstract] OR "Good Experience*"[Title/Abstract] OR "Negative Experience*"[Title/Abstract] OR "Bad Experience*"[Title/Abstract] OR "Positive Opinion*"[Title/Abstract] OR "Good Opinion*"[Title/Abstract] OR "Negative Opinion*"[Title/Abstract] OR "Bad Opinion*"[Title/Abstract] OR "Positive Belief*"[Title/Abstract] OR "Negative Belief*"[Title/Abstract] OR "Bad Belief*"[Title/Abstract] OR "Positive Perception*"[Title/Abstract] OR "Good Perception*"[Title/Abstract] OR "Negative Perception*"[Title/Abstract] OR "Bad Perception*"[Title/Abstract] OR Barrier*[Title/Abstract] OR Facilitat*[Title/Abstract] OR Reason[Title/Abstract] OR Reasons[Title/Abstract] OR Policy[Title/Abstract] OR Constrain*[Title/Abstract] OR Obstruct*[Title/Abstract] OR Restrain*[Title/Abstract] OR Restrict*[Title/Abstract] OR Enabler*[Title/Abstract] OR Obstacle*[Title/Abstract] OR Influence*[Title/Abstract] OR Determinant*[Title/Abstract] OR Motivator*[Title/Abstract] OR Limit*[Title/Abstract] OR Challenge[Title/Abstract] OR Attitude*[Title/Abstract] OR Belief*[Title/Abstract] OR Perception*[Title/Abstract] OR Motivation*[Title/Abstract] OR Stigma*[Title/Abstract] OR Trust*[Title/Abstract] OR Access*[Title/Abstract] OR Availab*[Title/Abstract] OR Guideline*[Title/Abstract]

Appendix D (Continued)

PubMed		
Concept	Keywords	((("Non-Motorised Transport*"[Title/Abstract] OR "Non-Motorized Transport*"[Title/Abstract] OR "Activ* Travel*"[Title/Abstract] OR "Active Mobility"[Title/Abstract] OR "Physical Mobility"[Title/Abstract] OR "Soft Mobility"[Title/Abstract] OR "Cycle Path*"[Title/Abstract] OR "Cycle Zone*"[Title/Abstract] OR "Cycle Lane*"[Title/Abstract] OR "Cycling Path*"[Title/Abstract] OR "Cycling Lane*"[Title/Abstract] OR "Bike Path*"[Title/Abstract] OR "Bike Lane*"[Title/Abstract] OR "Bicycle Path*"[Title/Abstract] OR "Bicycle Lane*"[Title/Abstract] OR "Walk* Path*"[Title/Abstract] OR "Walk* Lane*"[Title/Abstract] OR "Foot Path*"[Title/Abstract] OR "Foot Zone*"[Title/Abstract] OR "Pedestrian Path*"[Title/Abstract] OR "Pedestrian Zone*"[Title/Abstract] OR "Pedestrian Lane*"[Title/Abstract]) AND (School*[Title/Abstract] OR Highschool*[Title/Abstract] OR Kindergarten*[Title/Abstract])) OR (("Non-Motorised Transport*"[Title/Abstract] OR "Non-Motorized Transport*"[Title/Abstract] OR "Activ* Travel*"[Title/Abstract] OR "Active Mobility"[Title/Abstract] OR "Physical Mobility"[Title/Abstract] OR "Soft Mobility"[Title/Abstract] OR "Cycle Path*"[Title/Abstract] OR "Cycle Zone*"[Title/Abstract] OR "Cycle Lane*"[Title/Abstract] OR "Cycling Path*"[Title/Abstract] OR "Cycling Lane*"[Title/Abstract] OR "Bike Path*"[Title/Abstract] OR "Bike Lane*"[Title/Abstract] OR "Bicycle Path*"[Title/Abstract] OR "Bicycle Lane*"[Title/Abstract] OR "Walk* Path*"[Title/Abstract] OR "Walk* Lane*"[Title/Abstract] OR "Foot Path*"[Title/Abstract] OR "Foot Zone*"[Title/Abstract] OR "Pedestrian Path*"[Title/Abstract] OR "Pedestrian Zone*"[Title/Abstract] OR "Pedestrian Lane*"[Title/Abstract]) AND ("High School*"[Title/Abstract] OR "Secondary School*"[Title/Abstract] OR "Middle School*"[Title/Abstract] OR "Primary School*"[Title/Abstract] OR "Elementary School*"[Title/Abstract])) OR ((Commut*[Title/Abstract] OR Walk*[Title/Abstract] OR Sidewalk[Title/Abstract] OR Footpath[Title/Abstract] OR Bike[Title/Abstract] OR Bicycle[Title/Abstract] OR Cycle[Title/Abstract] OR Cycling[Title/Abstract] OR Trip[Title/Abstract] OR Trips[Title/Abstract] OR Travel*[Title/Abstract] OR Route*[Title/Abstract] OR Journey*[Title/Abstract] OR Drop-off[Title/Abstract] OR Drop-offs[Title/Abstract] OR Pick-up[Title/Abstract] OR Pick-ups[Title/Abstract]) AND (School*[Title/Abstract] OR Highschool*[Title/Abstract] OR Kindergarten*[Title/Abstract])) OR ((Commut*[Title/Abstract] OR Walk*[Title/Abstract] OR Sidewalk[Title/Abstract] OR Footpath[Title/Abstract] OR Bike[Title/Abstract] OR Bicycle[Title/Abstract] OR Cycle[Title/Abstract] OR Cycling[Title/Abstract] OR Trip[Title/Abstract] OR Trips[Title/Abstract] OR Travel*[Title/Abstract] OR Route*[Title/Abstract] OR Journey*[Title/Abstract] OR Drop-off[Title/Abstract] OR Drop-offs[Title/Abstract] OR Pick-up[Title/Abstract] OR Pick-ups[Title/Abstract]) AND ("High School*"[Title/Abstract] OR "Secondary School*"[Title/Abstract] OR "Middle School*"[Title/Abstract] OR "Primary School*"[Title/Abstract] OR "Elementary School*"[Title/Abstract])))

Appendix D (Continued)

Scopus		
Population	Keywords (TITLE-ABS-KEY(x))	School-Age* OR School-Going OR Child* OR Schoolchild* OR Adolescent* OR Teen* OR Kid OR Kids OR Minor OR Minors OR Youth* OR Parent* OR Guardian* OR Teacher*
Context	Keywords (TITLE-ABS-KEY(x))	(“Health Behavio*r" OR "Mobility Behavio*r" OR "Behavio*r Change" OR "Health Literac*" OR Self-Efficac* OR Peer-Pressure OR Peer-Influence OR ((Cultural OR Systemic OR Underlying OR Social) W/1 (Belief* OR Determinant* OR Barrier* OR Facilitat* OR Challenge* OR Reason OR Reasons)) OR ((Positive OR Good OR Negative OR Bad) W/1 (Experience* OR Opinion* OR Belief* OR Perception*)) OR Barrier* OR Facilitat* OR Reason OR Reasons OR Policy OR Constrain* OR Obstruct* OR Restrain* OR Restrict* OR Enabler* OR Obstacle* OR Influence* OR Determinant* OR Motivator* OR Limit* OR Challenge OR Attitude* OR Belief* OR Perception* OR Motivation* OR Stigma* OR Trust* OR Access* OR Availab* OR Guideline*)
Concept	Keywords (TITLE-ABS-KEY(x))	((("Non-Motori*ed Transport*" OR "Activ* Travel*" OR "Active Mobility" OR "Physical Mobility" OR "Soft Mobility" OR "Cycle Path*" OR "Cycle Zone*" OR "Cycle Lane*" OR "Cycling Path*" OR "Cycling Zone*" OR "Cycling Lane*" OR "Bike Path*" OR "Bike Zone*" OR "Bike Lane*" OR "Bicycle Path*" OR "Bicycle Zone*" OR "Bicycle Lane*" OR "Walk* Path*" OR "Walk* Zone*" OR "Walk* Lane*" OR "Foot Path*" OR "Foot Zone*" OR "Foot Lane*" OR "Pedestrian Path*" OR "Pedestrian Zone*" OR "Pedestrian Lane*" OR "Home-to-School Travel*" OR "Home-from-School Travel" OR "School-to-Home Travel" OR "School-from-Home Travel" OR "Home-to-and-from School Travel") W/10 (School* OR Highschool* OR Kindergarten*)) OR (("Non-Motori*ed Transport*" OR "Activ* Travel*" OR "Active Mobility" OR "Physical Mobility" OR "Soft Mobility" OR "Cycle Path*" OR "Cycle Zone*" OR "Cycle Lane*" OR "Cycling Path*" OR "Cycling Zone*" OR "Cycling Lane*" OR "Bike Path*" OR "Bike Zone*" OR "Bike Lane*" OR "Bicycle Path*" OR "Bicycle Zone*" OR "Bicycle Lane*" OR "Walk* Path*" OR "Walk* Zone*" OR "Walk* Lane*" OR "Foot Path*" OR "Foot Zone*" OR "Foot Lane*" OR "Pedestrian Path*" OR "Pedestrian Zone*" OR "Pedestrian Lane*" OR "Home-to-School Travel*" OR "Home-from-School Travel" OR "School-to-Home Travel" OR "School-from-Home Travel" OR "Home-to-and-from School Travel") W/10 ("High School*" OR "Secondary School*" OR "Middle School*" OR "Primary School*" OR "Elementary School*")) OR ((Commut* OR Walk* OR Sidewalk OR Footpath OR Bike OR Bicycle OR Cycle OR Cycling OR Trip OR Trips OR Travel* OR Route* OR Journey* OR Drop-off OR Drop-offs OR Pick-up OR Pick-ups) W/10 (School* OR Highschool* OR Kindergarten*)) OR ((Commut* OR Walk* OR Sidewalk OR Footpath OR Bike OR Bicycle OR Cycle OR Cycling OR Trip OR Trips OR Travel* OR Route* OR Journey* OR Drop-off OR Drop-offs OR Pick-up OR Pick-ups) W/10 ("High School*" OR "Secondary School*" OR "Middle School*" OR "Primary School*" OR "Elementary School*")))

Appendix D (Continued)

Web of Science		
Population	Keywords (TS=(x))	(School-Age* OR School-Going OR Child* OR Schoolchild* OR Adolescent* OR Teen* OR Kid OR Kids OR Minor OR Minors OR Youth* OR Parent* OR Guardian* OR Teacher*)
Context	Keywords (TS=)	("Health Behavio$r" OR "Mobility Behavio$r" OR "Behavio$r Change" OR "Health Literac*" OR Self-Efficac* OR Peer-Pressure OR Peer-Influence OR ((Cultural OR Systemic OR Underlying OR Social) NEAR/1 (Belief* OR Determinant* OR Barrier* OR Facilitat* OR Challenge* OR Reason OR Reasons)) OR ((Positive OR Good OR Negative OR Bad) NEAR/1 (Experience* OR Opinion* OR Belief* OR Perception*)) OR Barrier* OR Facilitat* OR Reason OR Reasons OR Policy OR Constrain* OR Obstruct* OR Restrain* OR Restrict* OR Enabler* OR Obstacle* OR Influence* OR Determinant* OR Motivator* OR Limit* OR Challenge OR Attitude* OR Belief* OR Perception* OR Motivation* OR Stigma* OR Trust* OR Access* OR Availab* OR Guideline*)
Concept	Keywords (TS=)	((("Non-Motori$ed Transport*" OR "Activ* Travel*" OR "Active Mobility" OR "Physical Mobility" OR "Soft Mobility" OR "Cycle Path*" OR "Cycle Zone*" OR "Cycle Lane*" OR "Cycling Path*" OR "Cycling Zone*" OR "Cycling Lane*" OR "Bike Path*" OR "Bike Zone*" OR "Bike Lane*" OR "Bicycle Path*" OR "Bicycle Zone*" OR "Bicycle Lane*" OR "Walk* Path*" OR "Walk* Zone*" OR "Walk* Lane*" OR "Foot Path*" OR "Foot Zone*" OR "Foot Lane*" OR "Pedestrian Path*" OR "Pedestrian Zone*" OR "Pedestrian Lane*" OR "Home-to-School Travel*" OR "Home-from-School Travel" OR "School-to-Home Travel" OR "School-from-Home Travel" OR "Home-to-and-from School Travel") NEAR/10 (School* OR Highschool* OR Kindergarten*)) OR (("Non-Motori$ed Transport*" OR "Activ* Travel*" OR "Active Mobility" OR "Physical Mobility" OR "Soft Mobility" OR "Cycle Path*" OR "Cycle Zone*" OR "Cycle Lane*" OR "Cycling Path*" OR "Cycling Zone*" OR "Cycling Lane*" OR "Bike Path*" OR "Bike Zone*" OR "Bike Lane*" OR "Bicycle Path*" OR "Bicycle Zone*" OR "Bicycle Lane*" OR "Walk* Path*" OR "Walk* Zone*" OR "Walk* Lane*" OR "Foot Path*" OR "Foot Zone*" OR "Foot Lane*" OR "Pedestrian Path*" OR "Pedestrian Zone*" OR "Pedestrian Lane*" OR "Home-to-School Travel*" OR "Home-from-School Travel" OR "School-to-Home Travel" OR "School-from-Home Travel" OR "Home-to-and-from School Travel") NEAR/10 ("High School*" OR "Secondary School*" OR "Middle School*" OR "Primary School*" OR "Elementary School*")) OR ((Commut* OR Walk* OR Sidewalk OR Footpath OR Bike OR Bicycle OR Cycle OR Cycling OR Trip OR Trips OR Travel* OR Route* OR Journey* OR Drop-off OR Drop-offs OR Pick-up OR Pick-ups) NEAR/10 (School* OR Highschool* OR Kindergarten*)) OR ((Commut* OR Walk* OR Sidewalk OR Footpath OR Bike OR Bicycle OR Cycle OR Cycling OR Trip OR Trips OR Travel* OR Route* OR Journey* OR Drop-off OR Drop-offs OR Pick-up OR Pick-ups) NEAR/10 ("High School*" OR "Secondary School*" OR "Middle School*" OR "Primary School*" OR "Elementary School*")))

Appendix D (Continued)

Transport Research International Documentation		
Population	Keywords	("School-Age*" OR "School-Going" OR Child* OR Schoolchild* OR Adolescent* OR Teen* OR Kid OR Kids OR Minor OR Minors OR Youth* OR Parent* OR Guardian* OR Teacher*)
Context	Keywords	("Health Behavior" OR "Health Behaviour" OR "Mobility Behavior" OR "Mobility Behaviour" OR "Behavior Change" OR "Behaviour Change" OR "Health Literac*" OR "Self-Efficac*" OR "Peer-Pressure" OR "Peer-Influence" OR ((Cultural OR Systemic OR Underlying OR Social) AND (Belief* OR Determinant* OR Barrier* OR Facilitat* OR Challenge* OR Reason OR Reasons)) OR ((Positive OR Good OR Negative OR Bad) AND (Experience* OR Opinion* OR Belief* OR Perception*)) OR Barrier* OR Facilitat* OR Reason OR Reasons OR Policy OR Constrain* OR Obstruct* OR Restrain* OR Restrict* OR Enabler* OR Obstacle* OR Influence* OR Determinant* OR Motivator* OR Limit* OR Challenge OR Attitude* OR Belief* OR Perception* OR Motivation* OR Stigma* OR Trust* OR Access* OR Availab* OR Guideline*)
Concept	Keywords	("Non-Motorised Transport*" OR "Non-Motorized Transport*" OR "Active Travel*" OR "Active Mobility" OR "Physical Mobility" OR "Soft Mobility" OR "Cycle Path*" OR "Cycle Zone*" OR "Cycle Lane*" OR "Cycling Path*" OR "Cycling Lane*" OR "Bike Path*" OR "Bike Lane*" OR "Bicycle Path*" OR "Bicycle Zone*" OR "Bicycle Lane*" OR "Walk Path*" OR "Walk Zone*" OR "Walking Zone*" OR "Walk Lane*" OR "Walking Lane*" OR "Foot Path" OR "Foot Zone*" OR "Foot Lane*" OR "Pedestrian Path*" OR "Pedestrian Zone*" OR "Pedestrian Lane*" OR Commut* OR Walk* OR Sidewalk OR Footpath OR Bike OR Bicycle OR Cycle OR Cycling OR Trip OR Trips OR Travel* OR Route* OR Journey*)
		(School* OR Highschool* OR Kindergarten* OR "High School*" OR "Secondary School*" OR "Middle School*" OR "Primary School*" OR "Elementary School*")

Appendix E

The Theoretical Domains Framework (TDF), created by Michie *et al.* (2005).

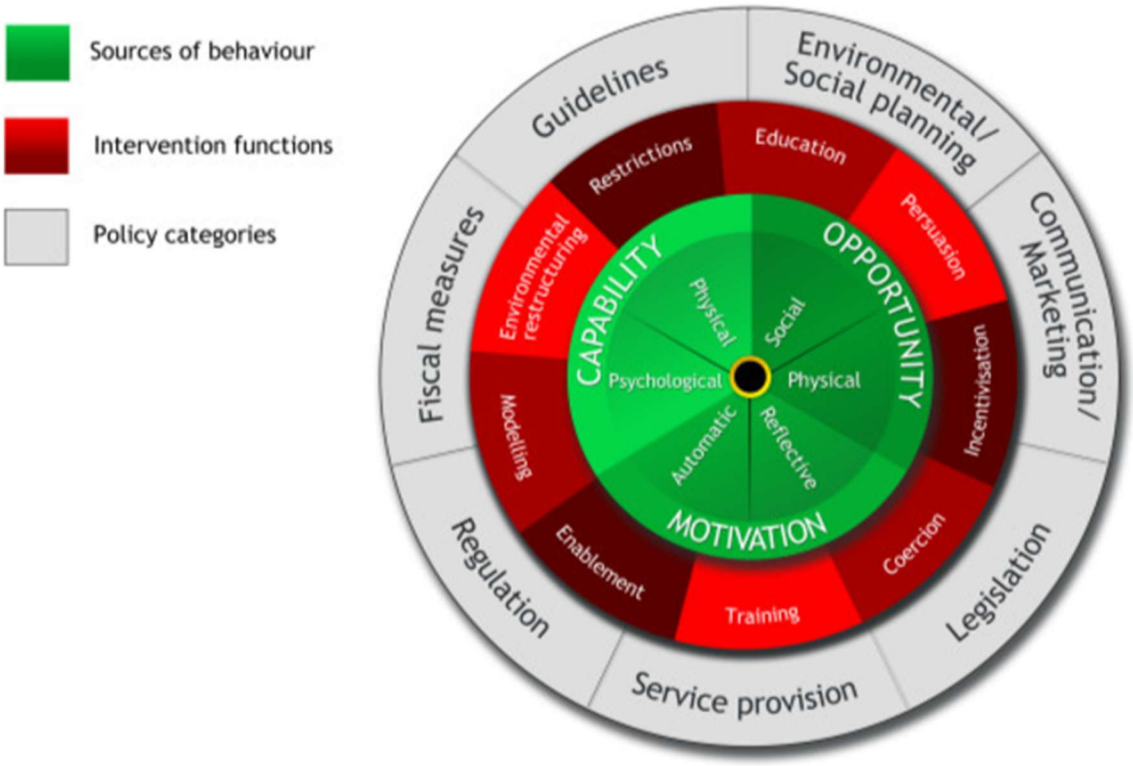
TDF Version 2	
Domain (definition)	Constructs
1. Knowledge (An awareness of the existence of something)	Knowledge (including knowledge of condition/scientific rationale); Procedural knowledge; Knowledge of task environment
2. Skills (An ability or proficiency acquired through practice)	Skills; Skills development; Competence; Ability; Interpersonal skills; Practice; Skill assessment
3. Social/professional role and identity (A coherent set of behaviours and displayed personal qualities of an individual in a social or work setting)	Professional identity; Professional role; Social identity; Identity; Professional boundaries; Professional confidence; Group identity; Leadership; Organisational commitment
4. Beliefs about capabilities (Acceptance of the truth, reality or validity about an ability, talent or facility that a person can put to constructive use)	Self-confidence; Perceived competence; Self-efficacy; Perceived behavioural control; Beliefs; Self-esteem; Empowerment; Professional confidence
5. Optimism (The confidence that things will happen for the best or that desired goals will be attained)	Optimism; Pessimism; Unrealistic optimism; Identity
6. Beliefs about Consequences (Acceptance of the truth, reality, or validity about outcomes of a behaviour in a given situation)	Beliefs; Outcome expectancies; Characteristics of outcome expectancies; Anticipated regret; Consequents
7. Reinforcement (Increasing the probability of a response by arranging a dependent relationship, or contingency, between the response and a given stimulus)	Rewards (proximal/distal, valued/not valued, probable/improbable); Incentives; Punishment; Consequents; Reinforcement; Contingencies; Sanctions
8. Intentions (A conscious decision to perform a behaviour or a resolve to act in a certain way)	Stability of intentions; Stages of change model; Transtheoretical model and stages of change

Appendix E (Continued)

9. Goals (Mental representations of outcomes or end states that an individual wants to achieve)	Goals (distal/proximal); Goal priority; Goal/target setting; Goals (autonomous/controlled); Action planning; Implementation intention
10. Memory, attention and decision processes (The ability to retain information, focus selectively on aspects of the environment and choose between two or more alternatives)	Memory; Attention; Attention control; Decision making; Cognitive overload/tiredness
11. Environmental context and resources (Any circumstance of a person's situation or environment that discourages or encourages the development of skills and abilities, independence, social competence and adaptive behaviour)	Environmental stressors; Resources/material resources; Organisational culture/climate; Salient events/critical incidents; Person × environment interaction; Barriers and facilitators
12. Social influences (Those interpersonal processes that can cause individuals to change their thoughts, feelings, or behaviours)	Social pressure; Social norms; Group conformity; Social comparisons; Group norms; Social support; Power; Intergroup conflict; Alienation; Group identity; Modelling
13. Emotion (A complex reaction pattern, involving experiential, behavioural, and physiological elements, by which the individual attempts to deal with a personally significant matter or event)	Fear; Anxiety; Affect; Stress; Depression; Positive/negative affect; Burn-out;
14. Behavioural regulation (Anything aimed at managing or changing objectively observed or measured actions)	Self-monitoring; Breaking habit; Action planning

Appendix F

The Behaviour Change Wheel (BCW), created by Michie *et al.* (2011).



Appendix G

The Full Table of Charted Items was uploaded to a new and anonymous Google Drive, as the extracted data was too large to be sufficiently accommodated within a Word document (21 columns and 123 rows).

https://drive.google.com/drive/folders/1dif7qwgOU1YNGUmg4jNpjvCtZpIcv7it?usp=sharing